

HCIA-Security Comprehensive Exam Question Bank

V3.0 Certification Preparation

Total Questions: 1070

Question Types: Single Choice, Multiple Choice, True/False

Coverage: All HCIA-Security V3.0 Exam Topics

Contents

I	Single-Answer Questions	5
1	Network Security Concepts	5
2	Network Security Standards	13
3	Network Fundamentals	19
4	OSI and TCP/IP Models	31
5	Network Protocols	32
6	Network Devices	33
7	Enterprise Security Threats	33
8	Communication Network Security	42
9	Zone Border Security	43
10	Computing Environment Security	43
11	Firewall Basics	43
12	Security Zones	51
13	Security Policies	52
14	Stateful Inspection	53
15	ASPF	53
16	NAT	54
17	NAT Basics	60
18	Source NAT	61
19	Destination NAT	61
20	NAT Server	61
21	NAT ALG	62
22	Firewall Hot Standby	62
23	VRRP	67
24	VGMP	68
25	HRP	68
26	Intrusion Prevention and Antivirus	68
27	Intrusion Overview	75
28	Intrusion Prevention	75
29	Antivirus	76

30 AAA and User Authentication	76
31 AAA Basics	82
32 Firewall User Authentication	83
33 Cryptography and PKI	84
34 Cryptography Basics	92
35 Hash Algorithms	93
36 Digital Signatures	94
37 PKI	94
38 VPN	95
39 VPN Overview	103
40 GRE VPN	104
41 IPsec VPN	104
42 L2TP VPN	105
43 SSL VPN	106
 II Multiple-Answer Questions	 106
44 Network Security Concepts	106
45 Network Security Standards	107
46 Network Fundamentals	109
47 OSI and TCP/IP Models	111
48 Network Protocols	111
49 Network Devices	112
50 Enterprise Security Threats	112
51 Zone Border Security	114
52 Firewall Basics	114
53 Security Zones	115
54 Security Policies	116
55 Stateful Inspection	116
56 NAT	116
57 NAT Basics	117
58 Source NAT	118
59 NAT ALG	118
60 Firewall Hot Standby	118

61 VRRP	120
62 HRP	120
63 Intrusion Prevention and Antivirus	120
64 Intrusion Prevention	122
65 Antivirus	122
66 AAA and User Authentication	122
67 AAA Basics	123
68 Firewall User Authentication	124
69 Cryptography and PKI	124
70 Cryptography Basics	126
71 PKI	126
72 VPN	126
73 VPN Overview	128
74 IPsec VPN	128
75 SSL VPN	129
 III True / False Questions	 129
76 Network Security Concepts	129
77 Network Security Standards	136
78 Network Fundamentals	141
79 OSI and TCP/IP Models	152
80 Network Protocols	153
81 Network Devices	153
82 Enterprise Security Threats	153
83 Communication Network Security	162
84 Zone Border Security	162
85 Firewall Basics	162
86 Security Zones	169
87 Security Policies	169
88 Stateful Inspection	169
89 ASPF	170
90 NAT	170
91 NAT Basics	175

92 Source NAT	176
93 Destination NAT	176
94 NAT Server	176
95 NAT ALG	176
96 Firewall Hot Standby	176
97 VRRP	181
98 VGMP	182
99 HRP	182
100 Intrusion Prevention and Antivirus	182
101 Intrusion Prevention	188
102 Antivirus	188
103 AAA and User Authentication	189
104 AAA Basics	194
105 Firewall User Authentication	194
106 Cryptography and PKI	194
107 Cryptography Basics	203
108 Hash Algorithms	204
109 PKI	204
110 VPN	204
111 GRE VPN	211
112 IPsec VPN	212
113 L2TP VPN	212
114 SSL VPN	212

PART I

Single-Answer Questions

1 Network Security Concepts

Question Question 1

Single Choice | Topic: Network Security Concepts

In a broad sense, network security refers to:

- A. Physical security of devices only
- B. Cybersecurity / cyberspace security at a national level
- C. Only firewall deployment
- D. Only antivirus protection

Correct Answer: B

Concept: Broadly, network security is synonymous with cybersecurity, covering national laws, regulations, and processes to protect cyberspace.

Question Question 2

Single Choice | Topic: Network Security Concepts

In a narrow sense, network security focuses on:

- A. National cyberspace policies only
- B. Vendor devices and solutions that ensure secure running of enterprise networks
- C. Only social engineering defense
- D. Only data-center cooling

Correct Answer: B

Concept: In a narrow sense, network security refers to measures such as firewalls, IPS, VPNs, and policies that protect enterprise networks.

Question Question 3

Single Choice | Topic: Network Security Concepts

Which period of security history mainly used stream ciphers and physical protection?

- A. Information security period
- B. Communication security period
- C. Cyberspace security period
- D. Information assurance period

Correct Answer: B

Concept: The communication security period (1940s) relied on physical security and cryptographic communication, mainly stream ciphers.

Question Question 4

Single Choice | Topic: Network Security Concepts

During the information security period, which three goals were the main focus?

- A. Confidentiality, integrity, availability
- B. Speed, bandwidth, latency
- C. Power, cooling, cabling
- D. Routing, switching, NAT

Correct Answer: A

Concept: The classic CIA triad — confidentiality, integrity, and availability — became the core objectives of information security.

Question Question 5

Single Choice | Topic: Network Security Concepts

Which two properties were added to the CIA triad during the information assurance period?

- A. Controllability and non-repudiation
- B. Speed and redundancy
- C. Encryption and hashing
- D. NAT and VLAN

Correct Answer: A

Concept: In the information assurance period, controllability (monitoring and control) and non-repudiation (proof of actions) joined the CIA triad.

Question Question 6

Single Choice | Topic: Network Security Concepts

The cyberspace security period aims to protect:

- A. Only web servers
- B. Only user passwords
- C. Facilities, data, users, and operations across cyberspace
- D. Only email gateways

Correct Answer: C

Concept: Cyberspace security expands protection to the entire digital ecosystem: facilities, data, users, and operations.

Question Question 7

Single Choice | Topic: Network Security Concepts

Which term describes the practice of protecting networks and data from unauthorized access?

- A. Cybersecurity
- B. Marketing
- C. Accounting
- D. Logistics

Correct Answer: A

Concept: Cybersecurity protects networks, devices, and data from unauthorized access or attack.

Question Question 8

Single Choice | Topic: Network Security Concepts

Which of the following best defines a vulnerability?

- A. A weakness that can be exploited
- B. A security patch
- C. A firewall rule
- D. A network diagram

Correct Answer: A**Concept:** A vulnerability is a weakness in a system that a threat actor can exploit.

Question Question 9

Single Choice | Topic: Network Security Concepts

What is an exploit?

- A. A software patch
- B. A method to take advantage of a vulnerability
- C. A type of firewall
- D. A routing protocol

Correct Answer: B**Concept:** An exploit is a technique or code that leverages a vulnerability.

Question Question 10

Single Choice | Topic: Network Security Concepts

Which term describes a potential danger to an asset?

- A. Risk
- B. Threat
- C. Vulnerability
- D. Control

Correct Answer: B**Concept:** A threat is any potential danger that could exploit a vulnerability.

Question Question 11

Single Choice | Topic: Network Security Concepts

Which term describes the likelihood and impact of a threat exploiting a vulnerability?

- A. Threat
- B. Risk
- C. Asset
- D. Control

Correct Answer: B**Concept:** Risk measures the probability and impact of a security incident.

Question Question 12

Single Choice | Topic: Network Security Concepts

Future security architecture is moving toward:

- A. Single perimeter defense
- B. Zero-trust and dynamic trust models
- C. Open anonymous access
- D. No logging

Correct Answer: B**Concept:** Zero trust assumes no implicit trust and verifies every access request dynamically.

Question Question 13

Single Choice | Topic: Network Security Concepts

Which technology is increasingly used to detect unknown threats?

- A. Signature-only antivirus
- B. Artificial intelligence and machine learning
- C. Hub-based networks
- D. Plaintext protocols

Correct Answer: B**Concept:** AI/ML helps analyze behavior and detect anomalies and unknown threats.

Question Question 14

Single Choice | Topic: Network Security Concepts

What does confidentiality ensure?

- A. Only speed
- B. Information is accessible only to authorized users
- C. Unlimited access
- D. Data duplication

Correct Answer: B**Concept:** Confidentiality ensures that information is disclosed only to authorized individuals.

Question Question 15

Single Choice | Topic: Network Security Concepts

What does integrity ensure?

- A. Fast transmission
- B. Information is not tampered with during transmission/storage
- C. Free access
- D. Unencrypted storage

Correct Answer: B**Concept:** Integrity ensures data is not altered or destroyed in an unauthorized manner.

Question Question 16

Single Choice | Topic: Network Security Concepts

What does availability ensure?

- A. Authorized users can access information when needed
- B. All users can access
- C. Systems are offline
- D. Data is hidden

Correct Answer: A

Concept: Availability guarantees reliable and timely access to information for authorized users.

Question Question 17

Single Choice | Topic: Network Security Concepts

What does controllability mean in information security?

- A. No monitoring
- B. Ability to monitor and control information/systems
- C. Unlimited downloads
- D. No access controls

Correct Answer: B

Concept: Controllability enables organizations to monitor, manage, and protect information and systems.

Question Question 18

Single Choice | Topic: Network Security Concepts

What does non-repudiation prevent?

- A. Fast data transfer
- B. Senders or receivers from denying their actions
- C. Encryption
- D. Backups

Correct Answer: B

Concept: Non-repudiation provides proof of origin and delivery so parties cannot deny participation.

Question Question 19

Single Choice | Topic: Network Security Concepts

The broad definition of network security is closest to:

- A. Cybersecurity
- B. Firewall configuration
- C. Antivirus deployment
- D. Physical security

Correct Answer: A

Concept: Broadly, network security equates to cybersecurity covering national and societal cyberspace protection.

Question Question 20

Single Choice | Topic: Network Security Concepts

The narrow definition of network security focuses on:

- A. National laws only
- B. Enterprise network protection solutions
- C. Social media policies
- D. Hardware warranties

Correct Answer: B

Concept: Narrowly, network security means vendor solutions and measures that keep enterprise networks secure.

Question Question 21

Single Choice | Topic: Network Security Concepts

During which period did communication cipher security emerge?

- A. Cyberspace security period
- B. Communication security period
- C. Information assurance period
- D. Cloud security period

Correct Answer: B

Concept: Stream ciphers and communication confidentiality appeared during the communication security period.

Question Question 22

Single Choice | Topic: Network Security Concepts

The information security period added which goal to the earlier communication security focus?

- A. Availability
- B. Confidentiality, integrity, and availability
- C. Non-repudiation only
- D. Controllability only

Correct Answer: B

Concept: The information security period formalized confidentiality, integrity, and availability as core goals.

Question Question 23

Single Choice | Topic: Network Security Concepts

Controllability and non-repudiation became prominent during:

- A. Communication security period
- B. Information security period
- C. Information assurance period
- D. Cloud computing period

Correct Answer: C

Concept: The information assurance period expanded security to include monitoring and proof of actions.

Question Question 24

Single Choice | Topic: Network Security Concepts

Which concept is described as: is synonymous with national cyberspace security?

- A. Cyberspace security
- B. Vulnerability
- C. Broad network security
- D. Integrity

Correct Answer: C

Concept: Broad network security refers to the concept where is synonymous with national cyberspace security.

Question Question 25

Single Choice | Topic: Network Security Concepts

Which concept is described as: focuses on enterprise network protection solutions?

- A. Narrow network security
- B. Broad network security
- C. Cyberspace security
- D. Controllability

Correct Answer: A

Concept: Narrow network security refers to the concept where focuses on enterprise network protection solutions.

Question Question 26

Single Choice | Topic: Network Security Concepts

Which concept is described as: consists of confidentiality, integrity, and availability?

- A. Controllability
- B. Availability
- C. Future security trend
- D. CIA triad

Correct Answer: D

Concept: CIA triad refers to the concept where consists of confidentiality, integrity, and availability.

Question Question 27

Single Choice | Topic: Network Security Concepts

Which concept is described as: added controllability and non-repudiation?

- A. Non-repudiation
- B. Information assurance period
- C. Future security trend
- D. AI and machine learning

Correct Answer: B

Concept: Information assurance period refers to the concept where added controllability and non-repudiation.

Question Question 28

Single Choice | Topic: Network Security Concepts

Which concept is described as: protects facilities, data, users, and operations?

- A. Future security trend
- B. CIA triad
- C. Narrow network security
- D. Cyberspace security

Correct Answer: D**Concept:** Cyberspace security refers to the concept where protects facilities, data, users, and operations.

Question Question 29

Single Choice | Topic: Network Security Concepts

Which concept is described as: ensures information is accessible only to authorized users?

- A. Confidentiality
- B. Broad network security
- C. AI and machine learning
- D. Integrity

Correct Answer: A**Concept:** Confidentiality refers to the concept where ensures information is accessible only to authorized users.

Question Question 30

Single Choice | Topic: Network Security Concepts

Which concept is described as: ensures data is not tampered with during transmission or storage?

- A. Narrow network security
- B. Availability
- C. Non-repudiation
- D. Integrity

Correct Answer: D**Concept:** Integrity refers to the concept where ensures data is not tampered with during transmission or storage.

Question Question 31

Single Choice | Topic: Network Security Concepts

Which concept is described as: ensures authorized users can access information when needed?

- A. Risk
- B. Availability
- C. Exploit
- D. Vulnerability

Correct Answer: B**Concept:** Availability refers to the concept where ensures authorized users can access information when needed.

Question Question 32

Single Choice | Topic: Network Security Concepts

Which concept is described as: enables monitoring and control of information and systems?

- A. Availability
- B. Controllability
- C. Future security trend
- D. Exploit

Correct Answer: B

Concept: Controllability refers to the concept where enables monitoring and control of information and systems.

Question Question 33

Single Choice | Topic: Network Security Concepts

Which concept is described as: prevents senders or receivers from denying actions?

- A. Future security trend
- B. Non-repudiation
- C. Exploit
- D. Broad network security

Correct Answer: B

Concept: Non-repudiation refers to the concept where prevents senders or receivers from denying actions.

Question Question 34

Single Choice | Topic: Network Security Concepts

Which concept is described as: is the combination of threat likelihood and impact?

- A. Threat
- B. Risk
- C. Integrity
- D. Confidentiality

Correct Answer: B

Concept: Risk refers to the concept where is the combination of threat likelihood and impact.

Question Question 35

Single Choice | Topic: Network Security Concepts

Which concept is described as: is a weakness that can be exploited by threats?

- A. Vulnerability
- B. Risk
- C. AI and machine learning
- D. Integrity

Correct Answer: A

Concept: Vulnerability refers to the concept where is a weakness that can be exploited by threats.

Question Question 36

Single Choice | Topic: Network Security Concepts

Which concept is described as: is a potential danger to an asset?

- A. Threat
- B. Confidentiality
- C. CIA triad
- D. Non-repudiation

Correct Answer: A**Concept:** Threat refers to the concept where is a potential danger to an asset.

Question Question 37

Single Choice | Topic: Network Security Concepts

Which concept is described as: is a method that takes advantage of a vulnerability?

- A. Availability
- B. Exploit
- C. Vulnerability
- D. Confidentiality

Correct Answer: B**Concept:** Exploit refers to the concept where is a method that takes advantage of a vulnerability.

Question Question 38

Single Choice | Topic: Network Security Concepts

Which concept is described as: includes zero-trust architecture?

- A. Confidentiality
- B. Cyberspace security
- C. Threat
- D. Future security trend

Correct Answer: D**Concept:** Future security trend refers to the concept where includes zero-trust architecture.

Question Question 39

Single Choice | Topic: Network Security Concepts

Which concept is described as: help detect unknown threats and anomalies?

- A. Narrow network security
- B. Integrity
- C. AI and machine learning
- D. Availability

Correct Answer: C**Concept:** AI and machine learning refers to the concept where help detect unknown threats and anomalies.

2 Network Security Standards

Question Question 1

Single Choice | Topic: Network Security Standards

ISO 27001 specifies requirements for:

- A. An information security management system (ISMS)
- B. A firewall hardware model
- C. A routing protocol
- D. A programming language

Correct Answer: A

Concept: ISO/IEC 27001 is the international standard for establishing, implementing, maintaining, and continually improving an ISMS.

Question Question 2

Single Choice | Topic: Network Security Standards

ISO 27002 is best described as:

- A. A certification standard
- B. A code of practice with security controls
- C. A firewall configuration guide
- D. A wireless protocol

Correct Answer: B

Concept: ISO/IEC 27002 provides best-practice security controls, while ISO 27001 is the certifiable requirements standard.

Question Question 3

Single Choice | Topic: Network Security Standards

Cybersecurity Classified Protection 2.0 is issued by:

- A. Huawei
- B. China (as a national standard/policy)
- C. IEEE
- D. IETF

Correct Answer: B

Concept: Classified Protection 2.0 is a Chinese national cybersecurity baseline and compliance framework.

Question Question 4

Single Choice | Topic: Network Security Standards

Classified Protection 2.0 uses a protection-level scale from:

- A. Level 1 to Level 5
- B. Level 0 to Level 4
- C. Level A to Level E
- D. Level I to Level V

Correct Answer: A

Concept: The Chinese classified protection system defines five levels, with Level 5 being the highest.

Question Question 5

Single Choice | Topic: Network Security Standards

Which of the following is NOT a typical Classified Protection 2.0 lifecycle phase?

- A. Classification
- B. Filing
- C. Construction and rectification
- D. Marketing and sales

Correct Answer: D

Concept: The Classified Protection lifecycle includes classification, filing, construction/rectification, level assessment, and supervision/inspection.

Question Question 6

Single Choice | Topic: Network Security Standards

Which ISO standard family addresses information security management?

- A. ISO 9000
- B. ISO 27000
- C. ISO 14000
- D. ISO 20000

Correct Answer: B

Concept: The ISO/IEC 27000 family covers information security management systems and controls.

Question Question 7

Single Choice | Topic: Network Security Standards

Which Chinese regulation is the basis for Classified Protection?

- A. Cybersecurity Law
- B. GDPR
- C. HIPAA
- D. PCI DSS

Correct Answer: A

Concept: China Cybersecurity Law underpins the Classified Protection compliance framework.

Question Question 8

Single Choice | Topic: Network Security Standards

ISO 27001 uses which methodology for risk management?

- A. Ignore risks
- B. Plan-Do-Check-Act (PDCA)
- C. Random patching
- D. No reviews

Correct Answer: B

Concept: ISO 27001 follows the PDCA cycle for continuous improvement of the ISMS.

Question Question 9

Single Choice | Topic: Network Security Standards

Which document in ISO 27000 family provides implementation guidance for controls?

- A. ISO 27001
- B. ISO 27002
- C. ISO 27003
- D. ISO 27005

Correct Answer: B**Concept:** ISO 27002 lists security controls and implementation guidance.

Question Question 10

Single Choice | Topic: Network Security Standards

Which is a key principle of Classified Protection 2.0?

- A. Security synchronization with construction
- B. Security as an afterthought
- C. No auditing
- D. Anonymous access

Correct Answer: A**Concept:** Classified Protection requires security to be planned, constructed, and used simultaneously with information systems.

Question Question 11

Single Choice | Topic: Network Security Standards

In Classified Protection 2.0, the security protection level is preliminarily determined by the network operator, but when the level reaches which level or above, expert review and approval are required?

- A. Level 1
- B. Level 2
- C. Level 3
- D. Level 4

Correct Answer: B**Concept:** For protection Level 2 and above, the preliminary determination must be reviewed and approved; Level 1 can be determined by the operator.

Question Question 12

Single Choice | Topic: Network Security Standards

Which standard is used to certify an ISMS?

- A. ISO 27001
- B. ISO 27002
- C. ISO 9001
- D. ISO 14001

Correct Answer: A**Concept:** ISO 27001 contains the certifiable requirements for an ISMS.

Question Question 13

Single Choice | Topic: Network Security Standards

ISO 27002 is primarily a:

- A. Certification standard
- B. Code of practice for security controls
- C. Risk assessment methodology
- D. Networking protocol

Correct Answer: B**Concept:** ISO 27002 provides best-practice controls but is not itself a certification standard.

Question Question 14

Single Choice | Topic: Network Security Standards

Classified Protection 2.0 compliance is primarily relevant in:

- A. China
- B. United States only
- C. European Union only
- D. Every country equally

Correct Answer: A**Concept:** Classified Protection 2.0 is a Chinese national cybersecurity compliance framework.

Question Question 15

Single Choice | Topic: Network Security Standards

Which is the highest protection level in Classified Protection 2.0?

- A. Level 1
- B. Level 3
- C. Level 5
- D. Level 10

Correct Answer: C**Concept:** Classified Protection defines five levels, with Level 5 being the most stringent.

Question Question 16

Single Choice | Topic: Network Security Standards

Expert review is required for Classified Protection levels starting at:

- A. Level 1
- B. Level 2
- C. Level 3
- D. Level 4

Correct Answer: B**Concept:** For Level 2 and above, expert review and approval are required.

Question Question 17

Single Choice | Topic: Network Security Standards

Which concept is described as: specifies certifiable ISMS requirements?

- A. Cybersecurity Law
- B. ISO 27001
- C. Level 2 and above
- D. ISO 27002

Correct Answer: B**Concept:** ISO 27001 refers to the concept where specifies certifiable ISMS requirements.

Question Question 18

Single Choice | Topic: Network Security Standards

Which concept is described as: provides a code of practice for security controls?

- A. Classified Protection 2.0
- B. ISO 27001:2022
- C. ISO 27001
- D. ISO 27002

Correct Answer: D**Concept:** ISO 27002 refers to the concept where provides a code of practice for security controls.

Question Question 19

Single Choice | Topic: Network Security Standards

Which concept is described as: is the continuous improvement methodology used in ISO 27001?

- A. ISO 27001
- B. Security synchronization
- C. PDCA
- D. ISO 27002

Correct Answer: C**Concept:** PDCA refers to the concept where is the continuous improvement methodology used in ISO 27001.

Question Question 20

Single Choice | Topic: Network Security Standards

Which concept is described as: is a Chinese national cybersecurity framework?

- A. Classified Protection 2.0
- B. Level 2 and above
- C. Classified Protection lifecycle
- D. PDCA

Correct Answer: A**Concept:** Classified Protection 2.0 refers to the concept where is a Chinese national cybersecurity framework.

Question Question 21

Single Choice | Topic: Network Security Standards

Which concept is described as: range from Level 1 to Level 5?

- A. PDCA
- B. Classified Protection lifecycle
- C. Classified Protection levels
- D. Cybersecurity Law

Correct Answer: C**Concept:** Classified Protection levels refers to the concept where range from Level 1 to Level 5.

Question Question 22

Single Choice | Topic: Network Security Standards

Which concept is described as: require expert review and approval?

- A. Cybersecurity Law
- B. PDCA
- C. ISO 27001
- D. Level 2 and above

Correct Answer: D**Concept:** Level 2 and above refers to the concept where require expert review and approval.

Question Question 23

Single Choice | Topic: Network Security Standards

Which concept is described as: includes classification, filing, construction, assessment, and supervision?

- A. Classified Protection lifecycle
- B. ISO 27001
- C. Security synchronization
- D. ISO 27001:2022

Correct Answer: A**Concept:** Classified Protection lifecycle refers to the concept where includes classification, filing, construction, assessment, and supervision.

Question Question 24

Single Choice | Topic: Network Security Standards

Which concept is described as: means planning security with information system construction?

- A. Classified Protection levels
- B. Cybersecurity Law
- C. Security synchronization
- D. Classified Protection 2.0

Correct Answer: C**Concept:** Security synchronization refers to the concept where means planning security with information system construction.

Question Question 25

Single Choice | Topic: Network Security Standards

Which concept is described as: organizes controls into four themes?

- A. PDCA
- B. ISO 27001:2022
- C. Cybersecurity Law
- D. ISO 27002

Correct Answer: B**Concept:** ISO 27001:2022 refers to the concept where organizes controls into four themes.

Question Question 26

Single Choice | Topic: Network Security Standards

Which concept is described as: underpins Classified Protection in China?

- A. Security synchronization
- B. Cybersecurity Law
- C. Classified Protection levels
- D. Level 2 and above

Correct Answer: B**Concept:** Cybersecurity Law refers to the concept where underpins Classified Protection in China.

3 Network Fundamentals

Question Question 1

Single Choice | Topic: Network Fundamentals

What is the default subnet mask for a Class C IPv4 address?

- A. 255.0.0.0
- B. 255.255.0.0
- C. 255.255.255.0
- D. 255.255.255.255

Correct Answer: C**Concept:** Class C networks use a default /24 subnet mask: 255.255.255.0.

Question Question 2

Single Choice | Topic: Network Fundamentals

Which protocol is used to automatically assign IPv6 addresses using MAC addresses?

- A. DHCPv6
- B. SLAAC
- C. ARP
- D. DNS

Correct Answer: B**Concept:** SLAAC can generate IPv6 addresses from MAC addresses.

Question Question 3

Single Choice | Topic: Network Fundamentals

Which layer-2 protocol prevents loops in redundant Ethernet topologies?

- A. VTP
- B. STP
- C. RSTP
- D. Both STP and RSTP

Correct Answer: D**Concept:** Spanning Tree Protocol and Rapid STP prevent Layer 2 loops.

Question Question 4

Single Choice | Topic: Network Fundamentals

What does VLAN stand for?

- A. Virtual Local Area Network
- B. Virtual Large Area Network
- C. Verified Local Access Node
- D. Virtual Link Aggregation Network

Correct Answer: A**Concept:** VLANs logically segment a switch into separate broadcast domains.

Question Question 5

Single Choice | Topic: Network Fundamentals

Which device interconnects VLANs at Layer 3?

- A. Layer 2 switch
- B. Router or Layer 3 switch
- C. Hub
- D. Bridge

Correct Answer: B**Concept:** Routers or Layer 3 switches perform inter-VLAN routing.

Question Question 6

Single Choice | Topic: Network Fundamentals

Which OSI layer is responsible for reliable data transfer between hosts?

- A. Network
- B. Transport
- C. Data link
- D. Physical

Correct Answer: B**Concept:** The transport layer manages end-to-end reliability, flow control, and error recovery.

Question Question 7

Single Choice | Topic: Network Fundamentals

In the TCP/IP model, which layer corresponds to the OSI network layer?

- A. Internet layer
- B. Network interface layer
- C. Transport layer
- D. Application layer

Correct Answer: A**Concept:** The TCP/IP internet layer maps to the OSI network layer and handles IP routing.

Question Question 8

Single Choice | Topic: Network Fundamentals

Which layer handles encryption and data format conversion?

- A. Application
- B. Presentation
- C. Session
- D. Transport

Correct Answer: B**Concept:** The presentation layer handles syntax, encryption, compression, and character encoding.

Question Question 9

Single Choice | Topic: Network Fundamentals

Which layer establishes, manages, and terminates application dialogs?

- A. Application
- B. Presentation
- C. Session
- D. Transport

Correct Answer: C**Concept:** The session layer controls dialog establishment, maintenance, and termination.

Question Question 10

Single Choice | Topic: Network Fundamentals

Which protocol is used to send email between servers?

- A. HTTP
- B. SMTP
- C. POP3
- D. IMAP

Correct Answer: B**Concept:** SMTP is used to send and relay email.

Question Question 11

Single Choice | Topic: Network Fundamentals

Which protocol is used by clients to retrieve email from a server while keeping messages on the server?

- A. POP3
- B. IMAP
- C. SMTP
- D. SNMP

Correct Answer: B**Concept:** IMAP allows clients to access and manage email while messages remain on the server.

Question Question 12

Single Choice | Topic: Network Fundamentals

Which protocol dynamically assigns IP addresses to hosts?

- A. DNS
- B. DHCP
- C. ARP
- D. FTP

Correct Answer: B**Concept:** DHCP dynamically assigns IP addresses and network configuration parameters.

Question Question 13

Single Choice | Topic: Network Fundamentals

Which protocol maps IP addresses to MAC addresses on a local network?

- A. DNS
- B. ARP
- C. DHCP
- D. ICMP

Correct Answer: B**Concept:** ARP resolves IP addresses to MAC addresses.

Question Question 14

Single Choice | Topic: Network Fundamentals

Which protocol is used for network management and monitoring?

- A. SNMP
- B. SMTP
- C. FTP
- D. Telnet

Correct Answer: A**Concept:** SNMP manages and monitors network devices.

Question Question 15

Single Choice | Topic: Network Fundamentals

What does ICMP primarily provide?

- A. File transfer
- B. Error reporting and diagnostics
- C. Email delivery
- D. Web browsing

Correct Answer: B**Concept:** ICMP reports errors and provides diagnostic functions such as ping and traceroute.

Question Question 16

Single Choice | Topic: Network Fundamentals

Which port does HTTP use by default?

- A. 20
- B. 21
- C. 80
- D. 443

Correct Answer: C**Concept:** HTTP uses TCP port 80 by default.

Question Question 17

Single Choice | Topic: Network Fundamentals

Which port does HTTPS use by default?

- A. 80
- B. 443
- C. 22
- D. 25

Correct Answer: B**Concept:** HTTPS uses TCP port 443 by default, providing encrypted web traffic.

Question Question 18

Single Choice | Topic: Network Fundamentals

Which port does SSH use by default?

- A. 21
- B. 22
- C. 23
- D. 25

Correct Answer: B**Concept:** SSH uses TCP port 22 for secure remote access.

Question Question 19

Single Choice | Topic: Network Fundamentals

Telnet uses which port and is considered insecure because it transmits data in plaintext?

- A. 21
- B. 22
- C. 23
- D. 25

Correct Answer: C**Concept:** Telnet uses TCP port 23 and sends data, including credentials, in plaintext.

Question Question 20

Single Choice | Topic: Network Fundamentals

Which address is a Layer 2 address?

- A. IP address
- B. MAC address
- C. Port number
- D. Domain name

Correct Answer: B**Concept:** A MAC address is a hardware address used at the data link layer.

Question Question 21

Single Choice | Topic: Network Fundamentals

A router forwards packets based on:

- A. MAC address
- B. IP address
- C. Port number
- D. VLAN ID

Correct Answer: B**Concept:** Routers use IP addresses to make forwarding decisions.

Question Question 22

Single Choice | Topic: Network Fundamentals

What is a subnet mask used for?

- A. MAC address
- B. IP address class
- C. Network and host portions of an IP address
- D. DNS server

Correct Answer: C**Concept:** A subnet mask divides an IP address into network and host portions.

Question Question 23

Single Choice | Topic: Network Fundamentals

The OSI model layer closest to the physical medium is:

- A. Data link
- B. Physical
- C. Network
- D. Transport

Correct Answer: B**Concept:** The physical layer deals with bits, cables, and signals.

Question Question 24

Single Choice | Topic: Network Fundamentals

TCP is considered reliable because it uses:

- A. Sequence numbers and acknowledgments
- B. Broadcasting
- C. No connection setup
- D. Fixed ports only

Correct Answer: A**Concept:** TCP uses sequence numbers, acknowledgments, and retransmissions for reliability.

Question Question 25

Single Choice | Topic: Network Fundamentals

UDP is preferred for:

- A. Reliable file transfer
- B. Real-time streaming and low-latency applications
- C. Email delivery
- D. Web browsing requiring guaranteed delivery

Correct Answer: B**Concept:** UDP low overhead and lack of retransmission make it suitable for real-time traffic.

Question Question 26

Single Choice | Topic: Network Fundamentals

DNS primarily uses:

- A. TCP port 53
- B. UDP port 53
- C. TCP port 80
- D. UDP port 161

Correct Answer: B**Concept:** DNS queries primarily use UDP port 53.

Question Question 27

Single Choice | Topic: Network Fundamentals

A Layer 3 switch combines functions of a:

- A. Hub and repeater
- B. Switch and router
- C. Firewall and IDS
- D. Modem and gateway

Correct Answer: B**Concept:** Layer 3 switches perform both Layer 2 switching and Layer 3 routing.

Question Question 28

Single Choice | Topic: Network Fundamentals

Which concept is described as: has seven layers?

- A. OSI model
- B. TCP/IP equivalent model
- C. Presentation layer
- D. SMTP

Correct Answer: A**Concept:** OSI model refers to the concept where has seven layers.

Question Question 29

Single Choice | Topic: Network Fundamentals

Which concept is described as: has four layers?

- A. TCP/IP standard model
- B. TCP
- C. ICMP
- D. DNS

Correct Answer: A**Concept:** TCP/IP standard model refers to the concept where has four layers.

Question Question 30

Single Choice | Topic: Network Fundamentals

Which concept is described as: has five layers?

- A. Physical layer
- B. VLAN
- C. TCP/IP equivalent model
- D. Data link layer

Correct Answer: C**Concept:** TCP/IP equivalent model refers to the concept where has five layers.

Question Question 31

Single Choice | Topic: Network Fundamentals

Which concept is described as: transmits raw bits over a medium?

- A. Hub
- B. Physical layer
- C. Application layer
- D. SMTP

Correct Answer: B**Concept:** Physical layer refers to the concept where transmits raw bits over a medium.

Question Question 32

Single Choice | Topic: Network Fundamentals

Which concept is described as: handles framing and MAC addressing?

- A. TCP/IP standard model
- B. DNS
- C. Data link layer
- D. FTP

Correct Answer: C**Concept:** Data link layer refers to the concept where handles framing and MAC addressing.

Question Question 33

Single Choice | Topic: Network Fundamentals

Which concept is described as: handles logical addressing and routing?

- A. FTP
- B. DHCP
- C. Network layer
- D. UDP

Correct Answer: C**Concept:** Network layer refers to the concept where handles logical addressing and routing.

Question Question 34

Single Choice | Topic: Network Fundamentals

Which concept is described as: provides end-to-end communication and reliability?

- A. Data link layer
- B. Session layer
- C. TCP/IP equivalent model
- D. Transport layer

Correct Answer: D**Concept:** Transport layer refers to the concept where provides end-to-end communication and reliability.

Question Question 35

Single Choice | Topic: Network Fundamentals

Which concept is described as: manages application dialogs?

- A. UDP
- B. Session layer
- C. Transport layer
- D. Presentation layer

Correct Answer: B**Concept:** Session layer refers to the concept where manages application dialogs.

Question Question 36

Single Choice | Topic: Network Fundamentals

Which concept is described as: handles encryption and data format conversion?

- A. Presentation layer
- B. Hub
- C. ARP
- D. IPv4 address

Correct Answer: A**Concept:** Presentation layer refers to the concept where handles encryption and data format conversion.

Question Question 37

Single Choice | Topic: Network Fundamentals

Which concept is described as: provides network services to applications?

- A. FTP
- B. Application layer
- C. SSH
- D. Data link layer

Correct Answer: B**Concept:** Application layer refers to the concept where provides network services to applications.

Question Question 38

Single Choice | Topic: Network Fundamentals

Which concept is described as: is connection-oriented and reliable?

- A. Telnet
- B. FTP
- C. HTTP
- D. TCP

Correct Answer: D**Concept:** TCP refers to the concept where is connection-oriented and reliable.

Question Question 39

Single Choice | Topic: Network Fundamentals

Which concept is described as: is connectionless and has low overhead?

- A. RFC 1918
- B. ICMP
- C. UDP
- D. Session layer

Correct Answer: C**Concept:** UDP refers to the concept where is connectionless and has low overhead.

Question Question 40

Single Choice | Topic: Network Fundamentals

Which concept is described as: uses TCP ports 20 and 21?

- A. FTP
- B. IPv4 address
- C. ICMP
- D. UDP

Correct Answer: A**Concept:** FTP refers to the concept where uses TCP ports 20 and 21.

Question Question 41

Single Choice | Topic: Network Fundamentals

Which concept is described as: uses TCP port 80?

- A. Telnet
- B. HTTP
- C. IPv4 address
- D. SMTP

Correct Answer: B**Concept:** HTTP refers to the concept where uses TCP port 80.

Question Question 42

Single Choice | Topic: Network Fundamentals

Which concept is described as: uses TCP port 443?

- A. ICMP
- B. TCP/IP standard model
- C. HTTPS
- D. TCP/IP equivalent model

Correct Answer: C**Concept:** HTTPS refers to the concept where uses TCP port 443.

Question Question 43

Single Choice | Topic: Network Fundamentals

Which concept is described as: uses TCP port 22?

- A. SSH
- B. Network layer
- C. Router
- D. Transport layer

Correct Answer: A**Concept:** SSH refers to the concept where uses TCP port 22.

Question Question 44

Single Choice | Topic: Network Fundamentals

Which concept is described as: uses TCP port 23 and is insecure?

- A. Telnet
- B. TCP/IP equivalent model
- C. FTP
- D. Transport layer

Correct Answer: A**Concept:** Telnet refers to the concept where uses TCP port 23 and is insecure.

Question Question 45

Single Choice | Topic: Network Fundamentals

Which concept is described as: primarily uses UDP port 53?

- A. Network layer
- B. Application layer
- C. DNS
- D. ARP

Correct Answer: C**Concept:** DNS refers to the concept where primarily uses UDP port 53.

Question Question 46

Single Choice | Topic: Network Fundamentals

Which concept is described as: uses TCP port 25?

- A. Switch
- B. SMTP
- C. SSH
- D. Hub

Correct Answer: B**Concept:** SMTP refers to the concept where uses TCP port 25.

Question Question 47

Single Choice | Topic: Network Fundamentals

Which concept is described as: dynamically assigns IP addresses?

- A. Presentation layer
- B. HTTP
- C. DHCP
- D. SSH

Correct Answer: C**Concept:** DHCP refers to the concept where dynamically assigns IP addresses.

Question Question 48

Single Choice | Topic: Network Fundamentals

Which concept is described as: maps IP addresses to MAC addresses?

- A. ICMP
- B. TCP
- C. Telnet
- D. ARP

Correct Answer: D**Concept:** ARP refers to the concept where maps IP addresses to MAC addresses.

Question Question 49

Single Choice | Topic: Network Fundamentals

Which concept is described as: provides error reporting and diagnostics?

- A. TCP
- B. Network layer
- C. IPv6 address
- D. ICMP

Correct Answer: D**Concept:** ICMP refers to the concept where provides error reporting and diagnostics.

Question Question 50

Single Choice | Topic: Network Fundamentals

Which concept is described as: is 32 bits long?

- A. Switch
- B. TCP/IP standard model
- C. IPv4 address
- D. HTTP

Correct Answer: C**Concept:** IPv4 address refers to the concept where is 32 bits long.

Question Question 51

Single Choice | Topic: Network Fundamentals

Which concept is described as: is 128 bits long?

- A. RFC 1918
- B. IPv4 address
- C. STP
- D. IPv6 address

Correct Answer: D**Concept:** IPv6 address refers to the concept where is 128 bits long.

Question Question 52

Single Choice | Topic: Network Fundamentals

Which concept is described as: is a Layer 2 hardware address?

- A. MAC address
- B. TCP/IP equivalent model
- C. VLAN
- D. Telnet

Correct Answer: A**Concept:** MAC address refers to the concept where is a Layer 2 hardware address.

Question Question 53

Single Choice | Topic: Network Fundamentals

Which concept is described as: operates at Layer 2 and forwards frames by MAC?

- A. Data link layer
- B. Switch
- C. DHCP
- D. DNS

Correct Answer: B**Concept:** Switch refers to the concept where operates at Layer 2 and forwards frames by MAC.

Question Question 54

Single Choice | Topic: Network Fundamentals

Which concept is described as: operates at Layer 3 and forwards packets by IP?

- A. VLAN
- B. DHCP
- C. SMTP
- D. Router

Correct Answer: D**Concept:** Router refers to the concept where operates at Layer 3 and forwards packets by IP.

Question Question 55

Single Choice | Topic: Network Fundamentals

Which concept is described as: operates at Layer 1 and repeats signals?

- A. TCP/IP equivalent model
- B. MAC address
- C. Session layer
- D. Hub

Correct Answer: D**Concept:** Hub refers to the concept where operates at Layer 1 and repeats signals.

Question Question 56

Single Choice | Topic: Network Fundamentals

Which concept is described as: logically segments a network into broadcast domains?

- A. VLAN
- B. TCP
- C. SSH
- D. DHCP

Correct Answer: A**Concept:** VLAN refers to the concept where logically segments a network into broadcast domains.

Question Question 57

Single Choice | Topic: Network Fundamentals

Which concept is described as: prevents Layer 2 loops?

- A. Application layer
- B. DNS
- C. HTTP
- D. STP

Correct Answer: D**Concept:** STP refers to the concept where prevents Layer 2 loops.

Question Question 58

Single Choice | Topic: Network Fundamentals

Which concept is described as: defines private IPv4 address ranges?

- A. MAC address
- B. Data link layer
- C. DNS
- D. RFC 1918

Correct Answer: D**Concept:** RFC 1918 refers to the concept where defines private IPv4 address ranges.

4 OSI and TCP/IP Models

Question Question 1

Single Choice | Topic: OSI and TCP/IP Models

How many layers does the OSI reference model have?

- A. 4
- B. 5
- C. 7
- D. 9

Correct Answer: C

Concept: The OSI model has seven layers: physical, data link, network, transport, session, presentation, and application.

Question Question 2

Single Choice | Topic: OSI and TCP/IP Models

How many layers does the TCP/IP standard model have?

- A. 4
- B. 5
- C. 6
- D. 7

Correct Answer: A

Concept: The TCP/IP standard model has four layers: network interface, internet, transport, and application.

Question Question 3

Single Choice | Topic: OSI and TCP/IP Models

Which layer is responsible for end-to-end communication and flow control?

- A. Network layer
- B. Transport layer
- C. Data link layer
- D. Physical layer

Correct Answer: B

Concept: The transport layer provides end-to-end communication, reliability, and flow control (e.g., TCP).

Question Question 4

Single Choice | Topic: OSI and TCP/IP Models

Which layer handles logical addressing and routing?

- A. Transport layer
- B. Network layer
- C. Session layer
- D. Presentation layer

Correct Answer: B

Concept: The network layer uses logical addresses (IP) and routes packets between networks.

Question Question 5

Single Choice | Topic: OSI and TCP/IP Models

Which layer is responsible for framing, MAC addressing, and error detection on a local link?

- A. Network layer
- B. Transport layer
- C. Data link layer
- D. Application layer

Correct Answer: C**Concept:** The data link layer handles framing, MAC addresses, and local link error detection.

5 Network Protocols

Question Question 1

Single Choice | Topic: Network Protocols

Which protocol uses ports 20 and 21 for file transfer?

- A. HTTP
- B. FTP
- C. SMTP
- D. DNS

Correct Answer: B**Concept:** FTP uses TCP port 21 for control and port 20 for data transfer.

Question Question 2

Single Choice | Topic: Network Protocols

Which protocol resolves domain names to IP addresses?

- A. DHCP
- B. DNS
- C. SNMP
- D. NTP

Correct Answer: B**Concept:** DNS (Domain Name System) maps human-readable domain names to IP addresses.

Question Question 3

Single Choice | Topic: Network Protocols

Which transport protocol is connection-oriented and reliable?

- A. UDP
- B. TCP
- C. ICMP
- D. IP

Correct Answer: B**Concept:** TCP provides connection-oriented, reliable, ordered delivery with flow and congestion control.

Question Question 4

Single Choice | Topic: Network Protocols

Which transport protocol is connectionless and has low overhead?

- A. TCP
- B. UDP
- C. HTTP
- D. FTP

Correct Answer: B**Concept:** UDP is connectionless, offers low latency and overhead, but does not guarantee delivery.

6 Network Devices

Question Question 1

Single Choice | Topic: Network Devices

Which device operates at Layer 2 and forwards frames based on MAC addresses?

- A. Router
- B. Switch
- C. Hub
- D. Firewall

Correct Answer: B**Concept:** A switch operates at the data link layer and forwards frames using MAC address tables.

Question Question 2

Single Choice | Topic: Network Devices

Which device operates at Layer 3 and forwards packets based on IP addresses?

- A. Switch
- B. Router
- C. Hub
- D. Bridge

Correct Answer: B**Concept:** A router operates at the network layer and makes forwarding decisions based on IP routing tables.

Question Question 3

Single Choice | Topic: Network Devices

Which device simply repeats signals to all connected ports?

- A. Switch
- B. Router
- C. Hub
- D. Firewall

Correct Answer: C**Concept:** A hub is a Layer 1 device that broadcasts signals to all connected devices.

7 Enterprise Security Threats

Question Question 1

Single Choice | Topic: Enterprise Security Threats

Which attack sends an overwhelming amount of traffic from many sources?

- A. DoS
- B. DDoS
- C. Phishing
- D. Spoofing

Correct Answer: B**Concept:** DDoS uses multiple distributed sources to flood a target.

Question Question 2

Single Choice | Topic: Enterprise Security Threats

Which attack falsifies the source address of packets?

- A. Phishing
- B. Spoofing
- C. Sniffing
- D. Spamming

Correct Answer: B**Concept:** Spoofing impersonates a trusted source by falsifying addresses or identities.

Question Question 3

Single Choice | Topic: Enterprise Security Threats

Which type of malware replicates itself across networks without user action?

- A. Virus
- B. Worm
- C. Trojan
- D. Spyware

Correct Answer: B**Concept:** Worms can self-replicate and spread across networks, often without user interaction.

Question Question 4

Single Choice | Topic: Enterprise Security Threats

Which malware disguises itself as legitimate software?

- A. Virus
- B. Worm
- C. Trojan
- D. Adware

Correct Answer: C**Concept:** A Trojan horse masquerades as legitimate software to trick users into executing it.

Question Question 5

Single Choice | Topic: Enterprise Security Threats

Ransomware primarily does what?

- A. Steals credentials
- B. Encrypts data and demands payment
- C. Deletes logs
- D. Monitors traffic

Correct Answer: B

Concept: Ransomware encrypts files and demands a ransom for decryption.

Question Question 6

Single Choice | Topic: Enterprise Security Threats

Which attack intercepts communications between two parties without their knowledge?

- A. Man-in-the-middle
- B. DoS
- C. Phishing
- D. Tailgating

Correct Answer: A

Concept: A man-in-the-middle attack intercepts and possibly alters communication between parties.

Question Question 7

Single Choice | Topic: Enterprise Security Threats

Which protocol can help prevent man-in-the-middle attacks by verifying server identity?

- A. HTTP
- B. HTTPS
- C. FTP
- D. Telnet

Correct Answer: B

Concept: HTTPS uses TLS/SSL to encrypt traffic and authenticate the server.

Question Question 8

Single Choice | Topic: Enterprise Security Threats

Which wireless security protocol is more secure than WEP?

- A. WPA2/WPA3
- B. WEP
- C. WPA (TKIP only)
- D. Open network

Correct Answer: A

Concept: WPA2 and WPA3 provide stronger encryption and authentication than WEP.

Question Question 9

Single Choice | Topic: Enterprise Security Threats

What is a DMZ typically used for?

- A. Hosting internal databases only
- B. Hosting public-facing servers
- C. User workstations
- D. Printer sharing

Correct Answer: B

Concept: A DMZ hosts public-facing services between the Internet and internal network.

Question Question 10

Single Choice | Topic: Enterprise Security Threats

Which security zone usually has the highest priority in Huawei firewalls?

- A. Untrust
- B. Trust
- C. DMZ
- D. Local

Correct Answer: D

Concept: The Local zone (firewall itself) has priority 100, the highest default priority.

Question Question 11

Single Choice | Topic: Enterprise Security Threats

Which practice limits the damage from compromised accounts?

- A. Least privilege
- B. Sharing admin passwords
- C. Disabling logs
- D. Open shares

Correct Answer: A

Concept: Least privilege gives users only the access they need, limiting blast radius.

Question Question 12

Single Choice | Topic: Enterprise Security Threats

What is the purpose of host-based intrusion prevention?

- A. Route packets
- B. Detect and block malicious host activity
- C. Assign IP addresses
- D. Synchronize time

Correct Answer: B

Concept: Host-based IPS monitors and blocks suspicious activity on an endpoint.

Question Question 13

Single Choice | Topic: Enterprise Security Threats

Which technology collects and analyzes security logs centrally?

- A. SIEM
- B. SNMP
- C. DHCP
- D. ARP

Correct Answer: A**Concept:** SIEM centralizes log collection and analysis.

Question Question 14

Single Choice | Topic: Enterprise Security Threats

What is a key benefit of centralized security management?

- A. Reduced visibility
- B. Unified policy and monitoring
- C. More complex passwords
- D. Faster malware spread

Correct Answer: B**Concept:** Centralized management provides unified visibility, policy control, and monitoring.

Question Question 15

Single Choice | Topic: Enterprise Security Threats

What is an insider threat?

- A. An attack from outside the network
- B. A security risk originating from within the organization
- C. A hardware vendor
- D. A natural disaster

Correct Answer: B**Concept:** Insider threats come from employees, contractors, or partners with internal access.

Question Question 16

Single Choice | Topic: Enterprise Security Threats

Which attack sends oversized ICMP packets to crash a target?

- A. Ping of death
- B. Ping sweep
- C. ARP spoofing
- D. DNS hijacking

Correct Answer: A**Concept:** Ping of death sends ICMP packets larger than the maximum allowed size.

Question Question 17

Single Choice | Topic: Enterprise Security Threats

Which attack uses broadcast pings to amplify traffic?

- A. Smurf attack
- B. SYN flood
- C. Fraggle
- D. Session hijacking

Correct Answer: A**Concept:** Smurf attacks send ICMP echo requests to broadcast addresses with a spoofed source.

Question Question 18

Single Choice | Topic: Enterprise Security Threats

Which attack intercepts ARP messages to redirect traffic?

- A. ARP spoofing
- B. DNS spoofing
- C. IP spoofing
- D. MAC flooding

Correct Answer: A**Concept:** ARP spoofing sends forged ARP replies to associate an attacker MAC with another IP.

Question Question 19

Single Choice | Topic: Enterprise Security Threats

A SYN flood is a type of:

- A. Denial of Service attack
- B. Phishing attack
- C. Data exfiltration
- D. Physical attack

Correct Answer: A**Concept:** SYN floods exhaust server connection resources with half-open TCP connections.

Question Question 20

Single Choice | Topic: Enterprise Security Threats

ARP spoofing is used to:

- A. Intercept traffic on a local network
- B. Encrypt files
- C. Speed up routing
- D. Authenticate users

Correct Answer: A**Concept:** Attackers use forged ARP messages to redirect local traffic through themselves.

Question Question 21

Single Choice | Topic: Enterprise Security Threats

The principle of least privilege aims to:

- A. Give everyone admin rights
- B. Limit user access to the minimum necessary
- C. Disable all logging
- D. Remove firewalls

Correct Answer: B**Concept:** Least privilege reduces risk by restricting access to only what is needed.

Question Question 22

Single Choice | Topic: Enterprise Security Threats

A SIEM system is primarily used for:

- A. Centralized log analysis and threat detection
- B. Assigning IP addresses
- C. Switching frames
- D. Manufacturing cables

Correct Answer: A**Concept:** SIEM aggregates and analyzes security events and logs.

Question Question 23

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: monitors traffic without altering it?

- A. Ransomware
- B. Passive attack
- C. Ping of death
- D. DoS attack

Correct Answer: B**Concept:** Passive attack refers to the concept where monitors traffic without altering it.

Question Question 24

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: modifies or disrupts systems or data?

- A. Active attack
- B. Ping of death
- C. Man-in-the-middle attack
- D. DDoS attack

Correct Answer: A**Concept:** Active attack refers to the concept where modifies or disrupts systems or data.

Question Question 25

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: includes viruses, worms, Trojans, and ransomware?

- A. Malware
- B. Worm
- C. Virus
- D. Social engineering

Correct Answer: A**Concept:** Malware refers to the concept where includes viruses, worms, Trojans, and ransomware.

Question Question 26

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: requires a host program to spread?

- A. Active attack
- B. Man-in-the-middle attack
- C. Virus
- D. DMZ

Correct Answer: C**Concept:** Virus refers to the concept where requires a host program to spread.

Question Question 27

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: can self-replicate across networks?

- A. DDoS attack
- B. DoS attack
- C. Worm
- D. ARP spoofing

Correct Answer: C**Concept:** Worm refers to the concept where can self-replicate across networks.

Question Question 28

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: disguises itself as legitimate software?

- A. Phishing
- B. Trojan
- C. Active attack
- D. Worm

Correct Answer: B**Concept:** Trojan refers to the concept where disguises itself as legitimate software.

Question Question 29

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: encrypts data and demands payment?

- A. ARP spoofing
- B. Ransomware
- C. DoS attack
- D. Worm

Correct Answer: B**Concept:** Ransomware refers to the concept where encrypts data and demands payment.

Question Question 30

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: uses deception to steal information?

- A. SIEM
- B. SYN flood
- C. Phishing
- D. Trojan

Correct Answer: C**Concept:** Phishing refers to the concept where uses deception to steal information.

Question Question 31

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: manipulates people into breaking security rules?

- A. Smurf attack
- B. Man-in-the-middle attack
- C. Social engineering
- D. Worm

Correct Answer: C**Concept:** Social engineering refers to the concept where manipulates people into breaking security rules.

Question Question 32

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: aims to make a service unavailable?

- A. Smurf attack
- B. Virus
- C. DoS attack
- D. Patch management

Correct Answer: C**Concept:** DoS attack refers to the concept where aims to make a service unavailable.

Question Question 33

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: uses many distributed sources to flood a target?

- A. Passive attack
- B. DDoS attack
- C. Worm
- D. Active attack

Correct Answer: B**Concept:** DDoS attack refers to the concept where uses many distributed sources to flood a target.

Question Question 34

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: intercepts communication between two parties?

- A. Man-in-the-middle attack
- B. Passive attack
- C. Malware
- D. Patch management

Correct Answer: A**Concept:** Man-in-the-middle attack refers to the concept where intercepts communication between two parties.

Question Question 35

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: forges ARP replies to redirect traffic?

- A. ARP spoofing
- B. Patch management
- C. Man-in-the-middle attack
- D. Social engineering

Correct Answer: A**Concept:** ARP spoofing refers to the concept where forges ARP replies to redirect traffic.

Question Question 36

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: sends many half-open TCP connections?

- A. Phishing
- B. SYN flood
- C. DMZ
- D. Passive attack

Correct Answer: B**Concept:** SYN flood refers to the concept where sends many half-open TCP connections.

Question Question 37

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: sends oversized ICMP packets?

- A. Smurf attack
- B. DMZ
- C. Ping of death
- D. Ransomware

Correct Answer: C**Concept:** Ping of death refers to the concept where sends oversized ICMP packets.

Question Question 38

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: uses broadcast pings to amplify traffic?

- A. SIEM
- B. Patch management
- C. Active attack
- D. Smurf attack

Correct Answer: D**Concept:** Smurf attack refers to the concept where uses broadcast pings to amplify traffic.

Question Question 39

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: limits user access to the minimum necessary?

- A. Ransomware
- B. Worm
- C. Patch management
- D. Least privilege

Correct Answer: D**Concept:** Least privilege refers to the concept where limits user access to the minimum necessary.

Question Question 40

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: eliminates known vulnerabilities?

- A. Patch management
- B. Active attack
- C. DoS attack
- D. Phishing

Correct Answer: A**Concept:** Patch management refers to the concept where eliminates known vulnerabilities.

Question Question 41

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: centralizes log collection and analysis?

- A. Smurf attack
- B. Trojan
- C. Phishing
- D. SIEM

Correct Answer: D**Concept:** SIEM refers to the concept where centralizes log collection and analysis.

Question Question 42

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: hosts public-facing servers between Internet and internal network?

- A. ARP spoofing
- B. DoS attack
- C. Man-in-the-middle attack
- D. DMZ

Correct Answer: D**Concept:** DMZ refers to the concept where hosts public-facing servers between Internet and internal network.

Question Question 43

Single Choice | Topic: Enterprise Security Threats

Which concept is described as: originates from within the organization?

- A. Trojan
- B. Man-in-the-middle attack
- C. Insider threat
- D. SIEM

Correct Answer: C**Concept:** Insider threat refers to the concept where originates from within the organization.

8 Communication Network Security

Question Question 1

Single Choice | Topic: Communication Network Security

Which threat involves intercepting data traveling across a network?

- A. Eavesdropping
- B. DoS
- C. Phishing
- D. Tailgating

Correct Answer: A**Concept:** Eavesdropping intercepts network communications to steal information.

9 Zone Border Security

Question Question 1

Single Choice | Topic: Zone Border Security

What is the main purpose of a security zone?

- A. Increase Internet speed
- B. Group interfaces with similar security levels
- C. Replace antivirus
- D. Disable routing

Correct Answer: B**Concept:** Security zones group network interfaces with similar trust levels to enforce inter-zone policies.

Question Question 2

Single Choice | Topic: Zone Border Security

Which device is typically deployed at a zone border to enforce access control?

- A. Hub
- B. Firewall
- C. Repeater
- D. Bridge

Correct Answer: B**Concept:** Firewalls are deployed at network boundaries to enforce security policies between zones.

10 Computing Environment Security

Question Question 1

Single Choice | Topic: Computing Environment Security

Which control helps protect endpoints from malware?

- A. Antivirus/EDR
- B. Physical locks only
- C. Hub replacement
- D. Disabling DNS

Correct Answer: A**Concept:** Antivirus and endpoint detection and response (EDR) protect hosts from malware and suspicious behavior.

11 Firewall Basics

Question Question 1

Single Choice | Topic: Firewall Basics

Which firewall feature inspects application-layer data to detect attacks?

- A. Packet filtering
- B. Application inspection
- C. Hub forwarding
- D. NAT

Correct Answer: B**Concept:** Application inspection examines payload content for malicious patterns.

Question Question 2

Single Choice | Topic: Firewall Basics

Which Huawei firewall command mode is used for system-level configuration?

- A. User view
- B. System view
- C. Interface view
- D. Policy view

Correct Answer: B**Concept:** System view allows global-level configuration on Huawei devices.

Question Question 3

Single Choice | Topic: Firewall Basics

Which view is entered after the system-view command?

- A. User view
- B. System view
- C. Interface view
- D. None

Correct Answer: B**Concept:** The system-view command enters system view from user view.

Question Question 4

Single Choice | Topic: Firewall Basics

What is the purpose of a security profile in NGFW?

- A. Physical installation
- B. Applying advanced security features such as IPS/AV/URL filtering
- C. Layer 1 forwarding
- D. DHCP relay

Correct Answer: B**Concept:** Security profiles group advanced inspection features applied to policies.

Question Question 5

Single Choice | Topic: Firewall Basics

Which firewall generation combines traditional firewall functions with application awareness and intrusion prevention?

- A. Packet-filtering firewall
- B. Next-generation firewall
- C. Stateful firewall
- D. Proxy firewall

Correct Answer: B**Concept:** Next-generation firewalls integrate application awareness, IPS, antivirus, and user identity.

Question Question 6

Single Choice | Topic: Firewall Basics

What is the purpose of a default security policy deny action?

- A. Improve performance
- B. Ensure least-privilege access
- C. Enable logging
- D. Allow management

Correct Answer: B**Concept:** Default deny enforces least privilege by blocking unspecified traffic.

Question Question 7

Single Choice | Topic: Firewall Basics

Which zone would typically contain public web servers?

- A. Trust
- B. Untrust
- C. DMZ
- D. Local

Correct Answer: C**Concept:** Public-facing servers are placed in the DMZ for isolation.

Question Question 8

Single Choice | Topic: Firewall Basics

What does policy hit counting help with?

- A. Routing optimization
- B. Verifying policy usage and troubleshooting
- C. MAC learning
- D. DHCP assignment

Correct Answer: B**Concept:** Hit counts show which policies are triggered, aiding auditing and troubleshooting.

Question Question 9

Single Choice | Topic: Firewall Basics

Which of the following is NOT a firewall deployment mode?

- A. Routing mode
- B. Transparent mode
- C. Hybrid mode
- D. Broadcast mode

Correct Answer: D

Concept: Firewalls deploy in routing, transparent (bridge), or hybrid modes; broadcast mode is not standard.

Question Question 10

Single Choice | Topic: Firewall Basics

What happens when traffic moves from a higher-priority zone to a lower-priority zone?

- A. Inbound traffic
- B. Outbound traffic
- C. No traffic allowed
- D. Always dropped

Correct Answer: B

Concept: Traffic from a higher-priority zone to a lower-priority zone is considered outbound.

Question Question 11

Single Choice | Topic: Firewall Basics

What is the priority range for custom security zones on Huawei firewalls?

- A. 0-50
- B. 1-100
- C. 1-99 excluding defaults
- D. 0-100

Correct Answer: C

Concept: Custom zone priorities range from 1 to 99, avoiding reserved default values.

Question Question 12

Single Choice | Topic: Firewall Basics

Which action in a security policy can log permitted or denied traffic?

- A. Permit
- B. Deny
- C. Both permit and deny
- D. Neither

Correct Answer: C

Concept: Security policies can enable logging for both permit and deny actions.

Question Question 13

Single Choice | Topic: Firewall Basics

Which security policy action is typically applied to suspicious traffic for further inspection?

- A. Permit
- B. Deny
- C. Block
- D. Redirect

Correct Answer: B**Concept:** Suspicious traffic is commonly denied or blocked by policy.

Question Question 14

Single Choice | Topic: Firewall Basics

When a return packet arrives, what does a stateful firewall check first?

- A. Routing table
- B. Session table
- C. MAC table
- D. ARP table

Correct Answer: B**Concept:** A stateful firewall first checks the session table to find a matching connection.

Question Question 15

Single Choice | Topic: Firewall Basics

What happens if a return packet does not match any session table entry?

- A. Forwarded normally
- B. Dropped or inspected against policies
- C. Always permitted
- D. Sent to DMZ

Correct Answer: B**Concept:** Without a matching session, the firewall treats the packet as a new connection and applies policies.

Question Question 16

Single Choice | Topic: Firewall Basics

Which protocol commonly requires ASPF support due to dynamic port negotiation?

- A. HTTP
- B. FTP
- C. DNS
- D. NTP

Correct Answer: B**Concept:** FTP negotiates dynamic data ports in its control channel, requiring ASPF.

Question Question 17

Single Choice | Topic: Firewall Basics

ASPF primarily inspects traffic at which layer?

- A. Network layer only
- B. Application layer
- C. Physical layer
- D. Data link layer

Correct Answer: B**Concept:** ASPF performs application-layer inspection to identify negotiated connections.

Question Question 18

Single Choice | Topic: Firewall Basics

In Huawei firewalls, the default priority of Trust is:

- A. 5
- B. 50
- C. 85
- D. 100

Correct Answer: C**Concept:** Trust zone default priority is 85.

Question Question 19

Single Choice | Topic: Firewall Basics

Traffic from a low-priority zone to a high-priority zone is considered:

- A. Outbound
- B. Inbound
- C. Blocked by default
- D. Always permitted

Correct Answer: B**Concept:** Traffic entering a higher-priority zone from a lower-priority zone is inbound.

Question Question 20

Single Choice | Topic: Firewall Basics

The default security policy action is:

- A. Permit
- B. Deny
- C. Log
- D. Redirect

Correct Answer: B**Concept:** Huawei firewalls deny traffic by default unless explicitly permitted.

Question Question 21

Single Choice | Topic: Firewall Basics

Stateful inspection relies on:

- A. Session tables
- B. MAC tables only
- C. DNS records
- D. DHCP leases

Correct Answer: A

Concept: Session tables track connection state for stateful inspection.

Question Question 22

Single Choice | Topic: Firewall Basics

ASPF helps firewalls handle:

- A. Multi-channel protocols
- B. Static routing
- C. MAC learning
- D. DHCP relay

Correct Answer: A

Concept: ASPF inspects protocols that negotiate dynamic secondary channels.

Question Question 23

Single Choice | Topic: Firewall Basics

Which concept is described as: is 85 by default?

- A. Security policy logging
- B. Security zones
- C. Trust zone priority
- D. Hybrid mode

Correct Answer: C

Concept: Trust zone priority refers to the concept where is 85 by default.

Question Question 24

Single Choice | Topic: Firewall Basics

Which concept is described as: is 50 by default?

- A. DMZ zone priority
- B. Hybrid mode
- C. Intra-zone traffic
- D. Routing mode

Correct Answer: A

Concept: DMZ zone priority refers to the concept where is 50 by default.

Question Question 25

Single Choice | Topic: Firewall Basics

Which concept is described as: is 5 by default?

- A. Untrust zone priority
- B. DMZ zone priority
- C. ASPF
- D. Policy hit count

Correct Answer: A

Concept: Untrust zone priority refers to the concept where is 5 by default.

Question Question 26

Single Choice | Topic: Firewall Basics

Which concept is described as: is 100 by default?

- A. ASPF
- B. Hybrid mode
- C. Policy hit count
- D. Local zone priority

Correct Answer: D

Concept: Local zone priority refers to the concept where is 100 by default.

Question Question 27

Single Choice | Topic: Firewall Basics

Which concept is described as: is deny?

- A. Trust zone priority
- B. Routing mode
- C. Default security policy action
- D. Transparent mode

Correct Answer: C

Concept: Default security policy action refers to the concept where is deny.

Question Question 28

Single Choice | Topic: Firewall Basics

Which concept is described as: tracks connections in a session table?

- A. Stateful firewall
- B. Untrust zone priority
- C. Routing mode
- D. DMZ zone priority

Correct Answer: A

Concept: Stateful firewall refers to the concept where tracks connections in a session table.

Question Question 29

Single Choice | Topic: Firewall Basics

Which concept is described as: inspects packets individually without state?

- A. Trust zone priority
- B. Routing mode
- C. Security zones
- D. Packet-filtering firewall

Correct Answer: D

Concept: Packet-filtering firewall refers to the concept where inspects packets individually without state.

Question Question 30

Single Choice | Topic: Firewall Basics

Which concept is described as: inspects multi-channel protocols like FTP?

- A. ASPF
- B. Local zone priority
- C. Stateful firewall
- D. Security policy logging

Correct Answer: A

Concept: ASPF refers to the concept where inspects multi-channel protocols like FTP.

Question Question 31

Single Choice | Topic: Firewall Basics

Which concept is described as: includes application awareness and IPS?

- A. Local zone priority
- B. Next-generation firewall
- C. Security policy logging
- D. Untrust zone priority

Correct Answer: B

Concept: Next-generation firewall refers to the concept where includes application awareness and IPS.

Question Question 32

Single Choice | Topic: Firewall Basics

Which concept is described as: group interfaces with similar trust levels?

- A. Security policy logging
- B. Next-generation firewall
- C. Untrust zone priority
- D. Security zones

Correct Answer: D

Concept: Security zones refers to the concept where group interfaces with similar trust levels.

Question Question 33

Single Choice | Topic: Firewall Basics

Which concept is described as: can be controlled by policies?

- A. Trust zone priority
- B. Next-generation firewall
- C. Packet-filtering firewall
- D. Intra-zone traffic

Correct Answer: D**Concept:** Intra-zone traffic refers to the concept where can be controlled by policies.

Question Question 34

Single Choice | Topic: Firewall Basics

Which concept is described as: helps verify which rules are used?

- A. Untrust zone priority
- B. Intra-zone traffic
- C. Policy hit count
- D. DMZ zone priority

Correct Answer: C**Concept:** Policy hit count refers to the concept where helps verify which rules are used.

Question Question 35

Single Choice | Topic: Firewall Basics

Which concept is described as: can be enabled for permit and deny actions?

- A. Routing mode
- B. Security policy logging
- C. ASPF
- D. Security zones

Correct Answer: B**Concept:** Security policy logging refers to the concept where can be enabled for permit and deny actions.

Question Question 36

Single Choice | Topic: Firewall Basics

Which concept is described as: is a firewall deployment mode?

- A. Security zones
- B. Policy hit count
- C. Routing mode
- D. Packet-filtering firewall

Correct Answer: C**Concept:** Routing mode refers to the concept where is a firewall deployment mode.

Question Question 37

Single Choice | Topic: Firewall Basics

Which concept is described as: is a firewall deployment mode?

- A. Trust zone priority
- B. Security zones
- C. Transparent mode
- D. Stateful firewall

Correct Answer: C**Concept:** Transparent mode refers to the concept where is a firewall deployment mode.

Question Question 38

Single Choice | Topic: Firewall Basics

Which concept is described as: is a firewall deployment mode?

- A. Next-generation firewall
- B. Intra-zone traffic
- C. Hybrid mode
- D. Packet-filtering firewall

Correct Answer: C**Concept:** Hybrid mode refers to the concept where is a firewall deployment mode.

12 Security Zones

Question Question 1

Single Choice | Topic: Security Zones

What is the default priority of the Trust zone on a Huawei firewall?

- A. 5
- B. 50
- C. 85
- D. 100

Correct Answer: C**Concept:** The Trust zone has a default priority of 85.

Question Question 2

Single Choice | Topic: Security Zones

What is the default priority of the DMZ zone on a Huawei firewall?

- A. 5
- B. 50
- C. 85
- D. 100

Correct Answer: B**Concept:** The DMZ zone has a default priority of 50.

Question Question 3

Single Choice | Topic: Security Zones

What is the default priority of the Untrust zone on a Huawei firewall?

- A. 5
- B. 50
- C. 85
- D. 100

Correct Answer: A**Concept:** The Untrust zone has a default priority of 5.

Question Question 4

Single Choice | Topic: Security Zones

What is the default priority of the Local zone on a Huawei firewall?

- A. 5
- B. 50
- C. 85
- D. 100

Correct Answer: D**Concept:** The Local zone represents the firewall itself and has the highest default priority of 100.

13 Security Policies

Question Question 1

Single Choice | Topic: Security Policies

What is the default action of a Huawei firewall security policy if no rule matches?

- A. Permit
- B. Deny
- C. Log only
- D. Redirect

Correct Answer: B**Concept:** By default, unmatched traffic is denied.

Question Question 2

Single Choice | Topic: Security Policies

Which components can be used as matching conditions in a security policy? (single best answer)

- A. Source/destination zone, address, service, application, user, time
- B. Only source IP
- C. Only destination port
- D. Only protocol

Correct Answer: A**Concept:** Security policies can match on zones, addresses, users, services, applications, and time.

14 Stateful Inspection

Question Question 1

Single Choice | Topic: Stateful Inspection

What is a session table used for in a stateful firewall?

- A. Storing user passwords
- B. Tracking established connections
- C. Routing packets
- D. Encrypting traffic

Correct Answer: B

Concept: The session table tracks connection state so return traffic can be permitted without a new policy lookup.

Question Question 2

Single Choice | Topic: Stateful Inspection

Which type of firewall inspects packets individually without connection state?

- A. Stateful firewall
- B. Packet-filtering firewall
- C. Proxy firewall
- D. Next-generation firewall

Correct Answer: B

Concept: Packet-filtering firewalls inspect each packet independently based on header information.

15 ASPF

Question Question 1

Single Choice | Topic: ASPF

What does ASPF stand for?

- A. Advanced Stateful Packet Filtering
- B. Application Specific Packet Filtering
- C. Automatic Security Policy Forwarding
- D. Advanced Stateful Policy Firewall

Correct Answer: B

Concept: ASPF is Application Specific Packet Filtering, used to inspect multi-channel protocols.

Question Question 2

Single Choice | Topic: ASPF

Why is ASPF needed for protocols like FTP?

- A. FTP only uses one port
- B. FTP opens dynamic data channels
- C. FTP encrypts all traffic
- D. FTP does not use TCP

Correct Answer: B

Concept: FTP opens a dynamic data channel, and ASPF inspects the control channel to permit the negotiated data channel.

16 NAT

Question Question 1

Single Choice | Topic: NAT

Which NAT type is also called many-to-one NAT?

- A. Static NAT
- B. Dynamic NAT
- C. NAT
- D. NAT Server

Correct Answer: C

Concept: NAT maps multiple private addresses to one public IP using port numbers.

Question Question 2

Single Choice | Topic: NAT

Which NAT type preserves a one-to-one mapping between public and private addresses?

- A. Static NAT
- B. NAT
- C. Easy-IP
- D. Dynamic PAT

Correct Answer: A

Concept: Static NAT maps one public IP to one private IP consistently.

Question Question 3

Single Choice | Topic: NAT

Which NAT type uses the outbound interface IP as the public address?

- A. NAT
- B. NAT No-PAT
- C. Easy-IP
- D. Destination NAT

Correct Answer: C

Concept: Easy-IP uses the firewall egress interface IP for translation.

Question Question 4

Single Choice | Topic: NAT

Which source NAT mode maps one private address to one public address without port translation?

- A. NAT
- B. NAT No-PAT
- C. Easy-IP
- D. NAT Server

Correct Answer: B

Concept: NAT No-PAT performs one-to-one address translation without changing port numbers.

Question Question 5

Single Choice | Topic: NAT

When many internal hosts access the Internet using one public IP, which NAT is used?

- A. NAT No-PAT
- B. NAT
- C. NAT Server
- D. Static NAT

Correct Answer: B**Concept:** NAT allows many-to-one address translation using port numbers.

Question Question 6

Single Choice | Topic: NAT

Destination NAT is also known as:

- A. Static NAT
- B. Port forwarding / virtual IP
- C. PAT
- D. Proxy ARP

Correct Answer: B**Concept:** Destination NAT maps external destinations to internal hosts, commonly called port forwarding or virtual IP.

Question Question 7

Single Choice | Topic: NAT

Which NAT type maps a public IP/port to a private server IP/port?

- A. Source NAT
- B. NAT Server
- C. NAT
- D. Bidirectional NAT

Correct Answer: B**Concept:** NAT Server publishes internal services to external networks.

Question Question 8

Single Choice | Topic: NAT

Bidirectional NAT is useful when:

- A. Only source address translation is needed
- B. Both source and destination must be rewritten
- C. No NAT is needed
- D. Only DNS traffic is involved

Correct Answer: B**Concept:** It translates both source and destination addresses in a single flow.

Question Question 9

Single Choice | Topic: NAT

Which problem can occur if NAT ALG is disabled for FTP?

- A. FTP control channel fails
- B. FTP data channel may fail because dynamic ports are not opened
- C. FTP becomes encrypted
- D. FTP speeds up

Correct Answer: B

Concept: Without ALG, the firewall cannot open the negotiated FTP data channel.

Question Question 10

Single Choice | Topic: NAT

Which address type is preserved by NAT No-PAT?

- A. Source port
- B. Destination port
- C. Both source and destination ports
- D. No ports

Correct Answer: C

Concept: NAT No-PAT translates only IP addresses, preserving port numbers.

Question Question 11

Single Choice | Topic: NAT

What is a major benefit of NAT?

- A. One-to-one address mapping
- B. Conserves public IP addresses
- C. Full protocol transparency
- D. No ALG needed

Correct Answer: B

Concept: NAT conserves public IPs by sharing one address across many internal hosts.

Question Question 12

Single Choice | Topic: NAT

A NAT Server entry is typically:

- A. Bidirectional and persistent
- B. Temporary and random
- C. Encrypted
- D. Only for outbound traffic

Correct Answer: A

Concept: NAT Server entries are configured statically and persist for inbound access.

Question Question 13

Single Choice | Topic: NAT

NAT ALG is often enabled together with:

- A. ASPF
- B. Routing
- C. DHCP
- D. DNS

Correct Answer: A

Concept: ALG and ASPF work together to inspect dynamic application channels.

Question Question 14

Single Choice | Topic: NAT

Which RFC defines private IPv4 address space?

- A. RFC 1918
- B. RFC 2328
- C. RFC 793
- D. RFC 959

Correct Answer: A

Concept: RFC 1918 defines the private IPv4 address ranges.

Question Question 15

Single Choice | Topic: NAT

Hairpin NAT allows internal users to access an internal server using:

- A. The server private IP only
- B. The server public NAT address
- C. A VPN tunnel
- D. Broadcast address

Correct Answer: B

Concept: Hairpin NAT lets internal hosts reach an internal server via its public NAT address.

Question Question 16

Single Choice | Topic: NAT

NAPT is also known as:

- A. PAT
- B. Static NAT
- C. NAT Server
- D. Bridge NAT

Correct Answer: A

Concept: Port Address Translation (PAT) is another name for NAPT.

Question Question 17

Single Choice | Topic: NAT

Static NAT is most often used when:

- A. A server must always use the same public IP
- B. Many hosts share one IP
- C. Ports must be preserved
- D. Outbound only traffic exists

Correct Answer: A

Concept: Static NAT provides a consistent one-to-one public-to-private mapping.

Question Question 18

Single Choice | Topic: NAT

Destination NAT is commonly used to:

- A. Allow internal hosts to browse the web
- B. Publish internal services to external users
- C. Encrypt VPN traffic
- D. Assign IP addresses

Correct Answer: B

Concept: Destination NAT maps external destinations to internal servers.

Question Question 19

Single Choice | Topic: NAT

Which feature rewrites IP addresses inside application payloads?

- A. NAT ALG
- B. NAPT
- C. Static NAT
- D. Route lookup

Correct Answer: A

Concept: NAT ALG handles protocols that embed address information in payloads.

Question Question 20

Single Choice | Topic: NAT

Which concept is described as: translates IP addresses between private and public?

- A. NAT
- B. NAPT
- C. Easy-IP
- D. NAT Server

Correct Answer: A

Concept: NAT refers to the concept where translates IP addresses between private and public.

Question Question 21

Single Choice | Topic: NAT

Which concept is described as: translates the source IP of outbound packets?

- A. RFC 1918
- B. NAT Server
- C. Static NAT
- D. Source NAT

Correct Answer: D**Concept:** Source NAT refers to the concept where translates the source IP of outbound packets.

Question Question 22

Single Choice | Topic: NAT

Which concept is described as: translates the destination IP of inbound packets?

- A. NAPT
- B. NAT No-PAT
- C. Static NAT
- D. Destination NAT

Correct Answer: D**Concept:** Destination NAT refers to the concept where translates the destination IP of inbound packets.

Question Question 23

Single Choice | Topic: NAT

Which concept is described as: publishes internal services to the Internet?

- A. NAT Server
- B. Destination NAT
- C. RFC 1918
- D. Static NAT

Correct Answer: A**Concept:** NAT Server refers to the concept where publishes internal services to the Internet.

Question Question 24

Single Choice | Topic: NAT

Which concept is described as: translates both source and destination addresses?

- A. Static NAT
- B. NAT
- C. Bidirectional NAT
- D. Destination NAT

Correct Answer: C**Concept:** Bidirectional NAT refers to the concept where translates both source and destination addresses.

Question Question 25

Single Choice | Topic: NAT

Which concept is described as: translates many private addresses to one public IP using ports?

- A. Source NAT
- B. NAT
- C. NAT
- D. NAT ALG

Correct Answer: B

Concept: NAT refers to the concept where translates many private addresses to one public IP using ports.

Question Question 26

Single Choice | Topic: NAT

Which concept is described as: performs one-to-one address translation preserving ports?

- A. Destination NAT
- B. Easy-IP
- C. RFC 1918
- D. NAT No-PAT

Correct Answer: D

Concept: NAT No-PAT refers to the concept where performs one-to-one address translation preserving ports.

Question Question 27

Single Choice | Topic: NAT

Which concept is described as: uses the outbound interface IP for translation?

- A. Source NAT
- B. NAT
- C. Easy-IP
- D. Hairpin NAT

Correct Answer: C

Concept: Easy-IP refers to the concept where uses the outbound interface IP for translation.

Question Question 28

Single Choice | Topic: NAT

Which concept is described as: rewrites addresses inside application payloads?

- A. Source NAT
- B. NAT ALG
- C. NAT
- D. Static NAT

Correct Answer: B

Concept: NAT ALG refers to the concept where rewrites addresses inside application payloads.

Question Question 29

Single Choice | Topic: NAT

Which concept is described as: defines private IPv4 address ranges?

- A. Destination NAT
- B. Static NAT
- C. NAT ALG
- D. RFC 1918

Correct Answer: D**Concept:** RFC 1918 refers to the concept where defines private IPv4 address ranges.

Question Question 30

Single Choice | Topic: NAT

Which concept is described as: lets internal users access servers via public NAT address?

- A. NAT
- B. Hairpin NAT
- C. NAT No-PAT
- D. Destination NAT

Correct Answer: B**Concept:** Hairpin NAT refers to the concept where lets internal users access servers via public NAT address.

Question Question 31

Single Choice | Topic: NAT

Which concept is described as: provides a persistent one-to-one mapping?

- A. Static NAT
- B. NAT
- C. RFC 1918
- D. Bidirectional NAT

Correct Answer: A**Concept:** Static NAT refers to the concept where provides a persistent one-to-one mapping.

17 NAT Basics

Question Question 1

Single Choice | Topic: NAT Basics

What does NAT primarily translate?

- A. MAC addresses
- B. IP addresses
- C. Port numbers only
- D. Domain names

Correct Answer: B**Concept:** NAT translates IP addresses, typically between private and public addresses.

18 Source NAT

Question Question 1

Single Choice | Topic: Source NAT

Which NAT type translates the source IP address of outgoing packets?

- A. Source NAT
- B. Destination NAT
- C. Bidirectional NAT
- D. NAT Server

Correct Answer: A

Concept: Source NAT changes the source IP so internal hosts can use public addresses to access the Internet.

Question Question 2

Single Choice | Topic: Source NAT

What is the main purpose of NAT?

- A. Translate many private addresses to one public IP using port multiplexing
- B. Hide the destination IP
- C. Encrypt traffic
- D. Provide VPN access

Correct Answer: A

Concept: NAT (Network Address and Port Translation) maps multiple private addresses to a single public IP using different ports.

19 Destination NAT

Question Question 1

Single Choice | Topic: Destination NAT

Which NAT type translates the destination IP address of incoming packets?

- A. Source NAT
- B. Destination NAT
- C. Bidirectional NAT
- D. NAT Server

Correct Answer: B

Concept: Destination NAT rewrites the destination IP, often to expose internal servers to external users.

20 NAT Server

Question Question 1

Single Choice | Topic: NAT Server

What is a NAT Server used for?

- A. Hiding internal server IP addresses
- B. Publishing internal services to the Internet
- C. Encrypting server traffic
- D. Load balancing only

Correct Answer: B

Concept: NAT Server maps public IPs/ports to private server IPs/ports so external users can access internal services.

21 NAT ALG

Question Question 1

Single Choice | Topic: NAT ALG

What is the purpose of NAT ALG?

- A. Encrypt NAT traffic
- B. Translate application-layer addresses embedded in protocol payloads
- C. Replace firewalls
- D. Assign IP addresses

Correct Answer: B

Concept: NAT Application Level Gateway inspects and rewrites IP addresses inside application-layer protocols (e.g., FTP, SIP).

22 Firewall Hot Standby

Question Question 1

Single Choice | Topic: Firewall Hot Standby

In active/standby mode, how many firewalls forward traffic at the same time?

- A. One
- B. Two
- C. All
- D. None

Correct Answer: A

Concept: In active/standby mode, one firewall actively forwards while the other remains standby.

Question Question 2

Single Choice | Topic: Firewall Hot Standby

In load-sharing mode, how many firewalls forward traffic?

- A. One
- B. Two or more
- C. None
- D. Only the backup

Correct Answer: B**Concept:** Load-sharing mode allows multiple firewalls to forward traffic simultaneously.

Question Question 3

Single Choice | Topic: Firewall Hot Standby

What is the default VRRP advertisement interval?

- A. 1 second
- B. 3 seconds
- C. 10 seconds
- D. 30 seconds

Correct Answer: A**Concept:** VRRP master sends advertisements approximately every 1 second by default.

Question Question 4

Single Choice | Topic: Firewall Hot Standby

Which VRRP priority value is highest configurable?

- A. 1
- B. 100
- C. 254
- D. 255

Correct Answer: C**Concept:** VRRP priority ranges from 1 to 254 configurable; 255 is reserved for the IP owner.

Question Question 5

Single Choice | Topic: Firewall Hot Standby

What happens if a backup router stops receiving VRRP advertisements?

- A. It remains backup
- B. It preempts and becomes master
- C. It shuts down
- D. It sends ARP requests

Correct Answer: B**Concept:** Missing advertisements trigger a backup to assume the master role.

Question Question 6

Single Choice | Topic: Firewall Hot Standby

What does VGMP group priority determine?

- A. NAT pool size
- B. Which firewall becomes active
- C. Number of interfaces
- D. DHCP lease time

Correct Answer: B**Concept:** VGMP priority determines the active/standby role among redundant firewalls.

Question Question 7

Single Choice | Topic: Firewall Hot Standby

Which HRP state indicates the firewall is currently forwarding traffic?

- A. Standby
- B. Active
- C. Initialize
- D. Backup

Correct Answer: B**Concept:** The active HRP peer forwards traffic and synchronizes state to the standby.

Question Question 8

Single Choice | Topic: Firewall Hot Standby

Which HRP command enables automatic configuration synchronization?

- A. hrp enable
- B. hrp auto-sync config
- C. hrp mirror session
- D. hrp vrrp enable

Correct Answer: B**Concept:** hrp auto-sync config enables automatic configuration synchronization between peers.

Question Question 9

Single Choice | Topic: Firewall Hot Standby

Which deployment places both firewalls inline with the same traffic path?

- A. Dual-hot standby
- B. Active/standby inline
- C. Route backup only
- D. Transparent bridge

Correct Answer: B**Concept:** Inline active/standby deployment places firewalls in the same forwarding path.

Question Question 10

Single Choice | Topic: Firewall Hot Standby

What is a common requirement for HRP heartbeat links?

- A. Use the same interface as user traffic
- B. Use a dedicated high-reliability link
- C. Use wireless link only
- D. No link required

Correct Answer: B**Concept:** Heartbeat links should be dedicated and reliable to avoid false failovers.

Question Question 11

Single Choice | Topic: Firewall Hot Standby

Which mechanism reduces unnecessary VRRP state changes when interface flaps briefly?

- A. Preempt delay
- B. Track interface
- C. Authentication
- D. Priority 255

Correct Answer: A**Concept:** Preempt delay prevents rapid master/backup flapping.

Question Question 12

Single Choice | Topic: Firewall Hot Standby

Which HRP feature ensures sessions survive a failover?

- A. Session backup
- B. Route backup
- C. NAT ALG
- D. URL filtering

Correct Answer: A**Concept:** HRP session backup keeps session state synchronized so connections continue after failover.

Question Question 13

Single Choice | Topic: Firewall Hot Standby

VRRP master election is based primarily on:

- A. Priority value
- B. Router hostname
- C. Number of interfaces
- D. Uptime only

Correct Answer: A**Concept:** The router with the highest priority becomes the VRRP master.

Question Question 14

Single Choice | Topic: Firewall Hot Standby

VGMP group management prevents:

- A. Split-brain VRRP states
- B. NAT exhaustion
- C. DNS failures
- D. DHCP conflicts

Correct Answer: A**Concept:** VGMP ensures VRRP groups fail over consistently.

Question Question 15

Single Choice | Topic: Firewall Hot Standby

HRP is used to synchronize:

- A. Session and configuration data
- B. Email messages
- C. Web pages
- D. Printer drivers

Correct Answer: A**Concept:** HRP synchronizes sessions, NAT, server-map entries, and configurations.

Question Question 16

Single Choice | Topic: Firewall Hot Standby

In active/standby mode, the standby firewall:

- A. Does not forward traffic until failover
- B. Forwards half the traffic
- C. Processes all traffic
- D. Acts as a switch

Correct Answer: A**Concept:** The standby firewall remains ready but inactive until failover.

Question Question 17

Single Choice | Topic: Firewall Hot Standby

Which concept is described as: provides gateway redundancy with a virtual IP?

- A. VRRP
- B. Load-sharing mode
- C. Active/standby mode
- D. VRRP master

Correct Answer: A**Concept:** VRRP refers to the concept where provides gateway redundancy with a virtual IP.

Question Question 18

Single Choice | Topic: Firewall Hot Standby

Which concept is described as: forwards traffic for the virtual IP?

- A. VRRP
- B. VRRP backup
- C. HRP session backup
- D. VRRP master

Correct Answer: D**Concept:** VRRP master refers to the concept where forwards traffic for the virtual IP.

Question Question 19

Single Choice | Topic: Firewall Hot Standby

Which concept is described as: takes over if the master fails?

- A. VRRP master
- B. VRRP backup
- C. Load-sharing mode
- D. Active/standby mode

Correct Answer: B**Concept:** VRRP backup refers to the concept where takes over if the master fails.

Question Question 20

Single Choice | Topic: Firewall Hot Standby

Which concept is described as: manages VRRP group states consistently?

- A. Load-sharing mode
- B. Preempt delay
- C. VRRP
- D. VGMP

Correct Answer: D**Concept:** VGMP refers to the concept where manages VRRP group states consistently.

Question Question 21

Single Choice | Topic: Firewall Hot Standby

Which concept is described as: synchronizes sessions and configurations?

- A. Active/standby mode
- B. HRP
- C. Heartbeat link
- D. VRRP master

Correct Answer: B**Concept:** HRP refers to the concept where synchronizes sessions and configurations.

Question Question 22

Single Choice | Topic: Firewall Hot Standby

Which concept is described as: has one firewall forwarding traffic?

- A. Preempt delay
- B. Active/standby mode
- C. Load-sharing mode
- D. VGMP

Correct Answer: B**Concept:** Active/standby mode refers to the concept where has one firewall forwarding traffic.

Question Question 23

Single Choice | Topic: Firewall Hot Standby

Which concept is described as: has multiple firewalls forwarding traffic?

- A. VRRP backup
- B. Load-sharing mode
- C. Heartbeat link
- D. VGMP

Correct Answer: B**Concept:** Load-sharing mode refers to the concept where has multiple firewalls forwarding traffic.

Question Question 24

Single Choice | Topic: Firewall Hot Standby

Which concept is described as: should be dedicated and reliable?

- A. VRRP master
- B. HRP
- C. VGMP
- D. Heartbeat link

Correct Answer: D**Concept:** Heartbeat link refers to the concept where should be dedicated and reliable.

Question Question 25

Single Choice | Topic: Firewall Hot Standby

Which concept is described as: ensures connections survive failover?

- A. HRP session backup
- B. VGMP
- C. Preempt delay
- D. Active/standby mode

Correct Answer: A**Concept:** HRP session backup refers to the concept where ensures connections survive failover.

Question Question 26

Single Choice | Topic: Firewall Hot Standby

Which concept is described as: reduces rapid master-backup flapping?

- A. Load-sharing mode
- B. HRP session backup
- C. VRRP backup
- D. Preempt delay

Correct Answer: D**Concept:** Preempt delay refers to the concept where reduces rapid master-backup flapping.

23 VRRP

Question Question 1

Single Choice | Topic: VRRP

What does VRRP stand for?

- A. Virtual Router Redundancy Protocol
- B. Virtual Routing Resolution Protocol
- C. Virtual Redundant Routing Path
- D. Virtual Routing Relay Protocol

Correct Answer: A**Concept:** VRRP provides gateway redundancy by allowing multiple routers to share a virtual IP address.

Question Question 2

Single Choice | Topic: VRRP

In VRRP, which router forwards traffic for the virtual IP?

- A. Backup router
- B. Master router
- C. All routers simultaneously
- D. Only the physical IP owner

Correct Answer: B**Concept:** The VRRP master router owns the virtual IP and forwards traffic; backups take over if it fails.

24 VGMP

Question Question 1

Single Choice | Topic: VGMP

What does VGMP manage on Huawei firewalls?

- A. VRRP groups
- B. NAT pools
- C. Routing tables
- D. DHCP scopes

Correct Answer: A

Concept: VGMP (VRRP Group Management Protocol) manages the state of multiple VRRP groups consistently.

25 HRP

Question Question 1

Single Choice | Topic: HRP

What is the primary purpose of HRP?

- A. Encrypt traffic
- B. Synchronize sessions and configurations between firewalls
- C. Assign IP addresses
- D. Filter URLs

Correct Answer: B

Concept: HRP (Huawei Redundancy Protocol) synchronizes sessions, configurations, and state between hot-standby firewalls.

26 Intrusion Prevention and Antivirus

Question Question 1

Single Choice | Topic: Intrusion Prevention and Antivirus

Which phase of an attack involves identifying vulnerable services?

- A. Exploitation
- B. Reconnaissance
- C. Installation
- D. Exfiltration

Correct Answer: B

Concept: Reconnaissance gathers information about targets before exploitation.

Question Question 2

Single Choice | Topic: Intrusion Prevention and Antivirus

Which phase involves gaining unauthorized access?

- A. Reconnaissance
- B. Exploitation
- C. Covering tracks
- D. Reporting

Correct Answer: B**Concept:** Exploitation uses vulnerabilities to gain access to the target.

Question Question 3

Single Choice | Topic: Intrusion Prevention and Antivirus

Which deployment mode allows IDS to monitor traffic without affecting it?

- A. Inline
- B. Tap/SPAN
- C. Bridge
- D. Gateway

Correct Answer: B**Concept:** IDS commonly connects to a tap or switch SPAN port for passive monitoring.

Question Question 4

Single Choice | Topic: Intrusion Prevention and Antivirus

Which signature action drops the offending packet?

- A. Alert
- B. Block
- C. Permit
- D. Mirror

Correct Answer: B**Concept:** Block actions discard malicious packets.

Question Question 5

Single Choice | Topic: Intrusion Prevention and Antivirus

What is a false positive in IPS?

- A. Attack missed
- B. Legitimate traffic blocked as malicious
- C. Correct detection
- D. Signature update failure

Correct Answer: B**Concept:** False positives incorrectly classify legitimate traffic as malicious.

Question Question 6

Single Choice | Topic: Intrusion Prevention and Antivirus

What is a false negative in IPS?

- A. Legitimate traffic blocked
- B. Attack missed by IPS
- C. Correct block
- D. Logging error

Correct Answer: B**Concept:** False negatives occur when an actual attack is not detected.

Question Question 7

Single Choice | Topic: Intrusion Prevention and Antivirus

Which malware type encrypts user files and demands ransom?

- A. Virus
- B. Worm
- C. Trojan
- D. Ransomware

Correct Answer: D**Concept:** Ransomware encrypts files and demands a ransom for decryption.

Question Question 8

Single Choice | Topic: Intrusion Prevention and Antivirus

Which approach only allows known-good applications to run?

- A. Blacklisting
- B. Whitelisting
- C. Heuristics
- D. Sandboxing

Correct Answer: B**Concept:** Whitelisting permits only approved applications.

Question Question 9

Single Choice | Topic: Intrusion Prevention and Antivirus

Which technique detects malware by examining code behavior rather than signatures?

- A. Signature scanning
- B. Heuristic analysis
- C. Whitelisting
- D. Patching

Correct Answer: B**Concept:** Heuristics analyze code behavior to identify potentially malicious activity.

Question Question 10

Single Choice | Topic: Intrusion Prevention and Antivirus

Which protocol is commonly targeted by brute-force attacks against remote access?

- A. HTTP
- B. SSH
- C. DNS
- D. NTP

Correct Answer: B**Concept:** SSH and similar remote-access protocols are frequent brute-force targets.

Question Question 11

Single Choice | Topic: Intrusion Prevention and Antivirus

What does an IPS signature database contain?

- A. User passwords
- B. Known attack patterns
- C. Routing tables
- D. DNS records

Correct Answer: B**Concept:** Signature databases contain patterns used to identify known attacks.

Question Question 12

Single Choice | Topic: Intrusion Prevention and Antivirus

What is the role of the antivirus engine on a firewall?

- A. Route packets
- B. Scan files and traffic for malware
- C. Assign IP addresses
- D. Authenticate users

Correct Answer: B**Concept:** The antivirus engine inspects files and traffic to detect and block malware.

Question Question 13

Single Choice | Topic: Intrusion Prevention and Antivirus

An IPS differs from an IDS mainly because an IPS can:

- A. Block traffic
- B. Only log events
- C. Operate out-of-band only
- D. Never generate alerts

Correct Answer: A**Concept:** IPS can take active blocking actions, while IDS primarily detects.

Question Question 14

Single Choice | Topic: Intrusion Prevention and Antivirus

Signature-based detection cannot easily detect:

- A. Known attacks
- B. Zero-day attacks
- C. Old malware
- D. Common exploits

Correct Answer: B**Concept:** Zero-day attacks have no known signatures.

Question Question 15

Single Choice | Topic: Intrusion Prevention and Antivirus

Sandboxing helps detect malware by:

- A. Observing behavior in an isolated environment
- B. Matching signatures
- C. Comparing file sizes
- D. Checking file names

Correct Answer: A**Concept:** Sandboxing executes suspicious files safely to observe malicious behavior.

Question Question 16

Single Choice | Topic: Intrusion Prevention and Antivirus

Antivirus engines require regular updates to:

- A. Recognize new malware variants
- B. Improve routing
- C. Increase bandwidth
- D. Disable firewalls

Correct Answer: A**Concept:** Signature and detection databases must be updated to detect emerging threats.

Question Question 17

Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: detects intrusions and generates alerts?

- A. IDS
- B. False negative
- C. Reconnaissance
- D. Exploitation

Correct Answer: A**Concept:** IDS refers to the concept where detects intrusions and generates alerts.

Question Question 18

Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: detects and blocks intrusions inline?

- A. IPS
- B. Anomaly detection
- C. Exploitation
- D. IDS

Correct Answer: A

Concept: IPS refers to the concept where detects and blocks intrusions inline.

Question Question 19

Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: matches known attack patterns?

- A. Signature detection
- B. Heuristic analysis
- C. Antivirus signatures
- D. Exploitation

Correct Answer: A

Concept: Signature detection refers to the concept where matches known attack patterns.

Question Question 20

Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: flags deviations from normal behavior?

- A. IPS
- B. Anomaly detection
- C. Reconnaissance
- D. Antivirus signatures

Correct Answer: B

Concept: Anomaly detection refers to the concept where flags deviations from normal behavior.

Question Question 21

Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: incorrectly blocks legitimate traffic?

- A. Heuristic analysis
- B. Reconnaissance
- C. False positive
- D. Exploitation

Correct Answer: C

Concept: False positive refers to the concept where incorrectly blocks legitimate traffic.

Question Question 22

Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: misses an actual attack?

- A. Signature detection
- B. False negative
- C. Exploitation
- D. False positive

Correct Answer: B**Concept:** False negative refers to the concept where misses an actual attack.

Question Question 23

Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: observes file behavior in an isolated environment?

- A. Antivirus signatures
- B. HIDS
- C. Sandboxing
- D. Whitelisting

Correct Answer: C**Concept:** Sandboxing refers to the concept where observes file behavior in an isolated environment.

Question Question 24

Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: detects malware by examining behavior?

- A. Sandboxing
- B. HIDS
- C. Heuristic analysis
- D. IDS

Correct Answer: C**Concept:** Heuristic analysis refers to the concept where detects malware by examining behavior.

Question Question 25

Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: allows only approved applications?

- A. Whitelisting
- B. Reconnaissance
- C. Signature detection
- D. False positive

Correct Answer: A**Concept:** Whitelisting refers to the concept where allows only approved applications.

Question Question 26

Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: must be updated regularly?

- A. Sandboxing
- B. IDS
- C. Antivirus signatures
- D. IPS

Correct Answer: C**Concept:** Antivirus signatures refers to the concept where must be updated regularly.

Question Question 27

Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: monitors network traffic for intrusions?

- A. Antivirus signatures
- B. NIDS
- C. IPS
- D. Exploitation

Correct Answer: B**Concept:** NIDS refers to the concept where monitors network traffic for intrusions.

Question Question 28

Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: monitors host activity for intrusions?

- A. IDS
- B. False positive
- C. Heuristic analysis
- D. HIDS

Correct Answer: D**Concept:** HIDS refers to the concept where monitors host activity for intrusions.

Question Question 29

Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: is an early attack phase?

- A. HIDS
- B. IDS
- C. Anomaly detection
- D. Reconnaissance

Correct Answer: D**Concept:** Reconnaissance refers to the concept where is an early attack phase.

Question Question 30

Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: gains unauthorized access using vulnerabilities?

- A. Exploitation
- B. Antivirus signatures
- C. False positive
- D. IPS

Correct Answer: A**Concept:** Exploitation refers to the concept where gains unauthorized access using vulnerabilities.

27 Intrusion Overview

Question Question 1

Single Choice | Topic: Intrusion Overview

What is an intrusion?

- A. Authorized access
- B. Unauthorized access or attack on a network/system
- C. A software patch
- D. A backup operation

Correct Answer: B**Concept:** An intrusion is any unauthorized attempt to access, manipulate, or harm a system or network.

28 Intrusion Prevention

Question Question 1

Single Choice | Topic: Intrusion Prevention

What is the main difference between IDS and IPS?

- A. IDS only detects; IPS detects and blocks
- B. IDS is faster
- C. IPS cannot log
- D. IDS operates inline

Correct Answer: A**Concept:** IDS detects intrusions and alerts; IPS can also take action to block malicious traffic.

Question Question 2

Single Choice | Topic: Intrusion Prevention

Which detection method compares traffic against known attack signatures?

- A. Anomaly detection
- B. Signature detection
- C. Behavioral analysis
- D. Heuristic detection

Correct Answer: B**Concept:** Signature-based detection matches traffic patterns against a database of known attack signatures.

Question Question 3

Single Choice | Topic: Intrusion Prevention

Which detection method builds a baseline of normal behavior and flags deviations?

- A. Signature detection
- B. Anomaly detection
- C. Static analysis
- D. Port scanning

Correct Answer: B**Concept:** Anomaly detection identifies traffic that deviates from a learned baseline.

Question Question 4

Single Choice | Topic: Intrusion Prevention

Where is an IPS typically deployed?

- A. Out-of-band only
- B. Inline in the traffic path
- C. On user desktops only
- D. Inside a printer

Correct Answer: B**Concept:** IPS is deployed inline so it can block malicious traffic in real time.

29 Antivirus

Question Question 1

Single Choice | Topic: Antivirus

What does antivirus software primarily detect and remove?

- A. Routing loops
- B. Malware
- C. Cable faults
- D. DNS errors

Correct Answer: B**Concept:** Antivirus detects and removes malware such as viruses, worms, and Trojans.

Question Question 2

Single Choice | Topic: Antivirus

Which antivirus technique looks for known byte sequences of malware?

- A. Heuristic analysis
- B. Signature scanning
- C. Sandboxing
- D. Whitelisting

Correct Answer: B**Concept:** Signature scanning matches files/traffic against known malware signatures.

Question Question 3

Single Choice | Topic: Antivirus

Which method executes suspicious files in an isolated environment to observe behavior?

- A. Signature scanning
- B. Heuristic analysis
- C. Sandboxing
- D. Whitelisting

Correct Answer: C**Concept:** Sandboxing runs files in an isolated environment to detect malicious behavior.

30 AAA and User Authentication

Question Question 1

Single Choice | Topic: AAA and User Authentication

Which AAA protocol commonly uses UDP ports 1812 and 1813?

- A. TACACS+
- B. RADIUS
- C. LDAP
- D. Kerberos

Correct Answer: B**Concept:** RADIUS commonly uses UDP ports 1812/1813 (or 1645/1646 legacy).

Question Question 2

Single Choice | Topic: AAA and User Authentication

Which AAA protocol uses TCP port 49 by default?

- A. RADIUS
- B. TACACS+
- C. LDAP
- D. DHCP

Correct Answer: B**Concept:** TACACS+ uses TCP port 49 by default.

Question Question 3

Single Choice | Topic: AAA and User Authentication

In AAA, which process maps users to privileges?

- A. Authentication
- B. Authorization
- C. Accounting
- D. Admission

Correct Answer: B**Concept:** Authorization maps authenticated identities to permitted actions/resources.

Question Question 4

Single Choice | Topic: AAA and User Authentication

Which authentication type identifies users based on source IP?

- A. User authentication
- B. Access user authentication
- C. Device authentication
- D. IP/MAC binding

Correct Answer: B**Concept:** Access user authentication can identify users by IP/MAC for transparent access control.

Question Question 5

Single Choice | Topic: AAA and User Authentication

Which user type is created directly on the firewall?

- A. RADIUS user
- B. LDAP user
- C. Local user
- D. Domain user

Correct Answer: C**Concept:** Local users are defined directly in the firewall local user database.

Question Question 6

Single Choice | Topic: AAA and User Authentication

Which component is queried when using LDAP authentication?

- A. DNS server
- B. LDAP directory server
- C. NTP server
- D. SNMP server

Correct Answer: B**Concept:** The firewall queries an LDAP directory server to validate credentials.

Question Question 7

Single Choice | Topic: AAA and User Authentication

What is the purpose of a user group?

- A. Increase Internet speed
- B. Apply policies to multiple users collectively
- C. Replace firewalls
- D. Encrypt traffic

Correct Answer: B**Concept:** User groups simplify policy management by grouping users with similar access needs.

Question Question 8

Single Choice | Topic: AAA and User Authentication

Which authentication method is often used for wired/wireless network access control?

- A. Portal
- B. 802.1X
- C. Local
- D. RADIUS accounting

Correct Answer: B**Concept:** 802.1X is commonly used for port-based network access control on wired and wireless networks.

Question Question 9

Single Choice | Topic: AAA and User Authentication

What does single sign-on (SSO) provide?

- A. Multiple passwords
- B. One authentication for multiple services
- C. No authentication
- D. Local only access

Correct Answer: B**Concept:** SSO allows users to authenticate once and access multiple services.

Question Question 10

Single Choice | Topic: AAA and User Authentication

Which user authentication policy condition can match traffic?

- A. Source zone
- B. Time range
- C. Source/destination address
- D. All of the above

Correct Answer: D**Concept:** Authentication policies can match zones, addresses, time ranges, and other conditions.

Question Question 11

Single Choice | Topic: AAA and User Authentication

The first A in AAA stands for:

- A. Authentication
- B. Authorization
- C. Accounting
- D. Auditing

Correct Answer: A**Concept:** AAA stands for Authentication, Authorization, and Accounting.

Question Question 12

Single Choice | Topic: AAA and User Authentication

RADIUS commonly runs over:

- A. UDP
- B. TCP
- C. SCTP
- D. ICMP

Correct Answer: A

Concept: RADIUS typically uses UDP ports 1812/1813.

Question Question 13

Single Choice | Topic: AAA and User Authentication

TACACS+ commonly runs over:

- A. UDP
- B. TCP
- C. HTTP
- D. FTP

Correct Answer: B

Concept: TACACS+ uses TCP port 49.

Question Question 14

Single Choice | Topic: AAA and User Authentication

Portal authentication is typically used for:

- A. Web-based guest access
- B. Cable modem registration
- C. Switch MAC learning
- D. DNS resolution

Correct Answer: A

Concept: Portal authentication presents a web login page.

Question Question 15

Single Choice | Topic: AAA and User Authentication

802.1X authentication occurs at:

- A. Layer 2 port level
- B. Application layer only
- C. Physical layer only
- D. Transport layer only

Correct Answer: A

Concept: 802.1X provides port-based network access control at Layer 2.

Question Question 16

Single Choice | Topic: AAA and User Authentication

Which concept is described as: verifies user identity?

- A. SSO
- B. LDAP
- C. Authentication
- D. Accounting

Correct Answer: C**Concept:** Authentication refers to the concept where verifies user identity.

Question Question 17

Single Choice | Topic: AAA and User Authentication

Which concept is described as: determines what a user can access?

- A. Authentication
- B. Authorization
- C. LDAP
- D. Portal authentication

Correct Answer: B**Concept:** Authorization refers to the concept where determines what a user can access.

Question Question 18

Single Choice | Topic: AAA and User Authentication

Which concept is described as: records user activities and resource usage?

- A. Accounting
- B. Local authentication
- C. SSO
- D. 802.1X

Correct Answer: A**Concept:** Accounting refers to the concept where records user activities and resource usage.

Question Question 19

Single Choice | Topic: AAA and User Authentication

Which concept is described as: uses UDP ports 1812 and 1813?

- A. Local authentication
- B. RADIUS
- C. TACACS+
- D. LDAP

Correct Answer: B**Concept:** RADIUS refers to the concept where uses UDP ports 1812 and 1813.

Question Question 20

Single Choice | Topic: AAA and User Authentication

Which concept is described as: uses TCP port 49?

- A. User group
- B. 802.1X
- C. LDAP
- D. TACACS+

Correct Answer: D**Concept:** TACACS+ refers to the concept where uses TCP port 49.

Question Question 21

Single Choice | Topic: AAA and User Authentication

Which concept is described as: uses a web login page?

- A. Portal authentication
- B. Local authentication
- C. Authorization
- D. TACACS+

Correct Answer: A**Concept:** Portal authentication refers to the concept where uses a web login page.

Question Question 22

Single Choice | Topic: AAA and User Authentication

Which concept is described as: provides port-based network access control?

- A. RADIUS
- B. Authorization
- C. 802.1X
- D. Local authentication

Correct Answer: C**Concept:** 802.1X refers to the concept where provides port-based network access control.

Question Question 23

Single Choice | Topic: AAA and User Authentication

Which concept is described as: uses the device user database?

- A. SSO
- B. Accounting
- C. User group
- D. Local authentication

Correct Answer: D**Concept:** Local authentication refers to the concept where uses the device user database.

Question Question 24

Single Choice | Topic: AAA and User Authentication

Which concept is described as: queries a directory server for authentication?

- A. Local authentication
- B. LDAP
- C. User group
- D. Authentication

Correct Answer: B**Concept:** LDAP refers to the concept where queries a directory server for authentication.

Question Question 25

Single Choice | Topic: AAA and User Authentication

Which concept is described as: applies policies to multiple users collectively?

- A. RADIUS
- B. 802.1X
- C. Portal authentication
- D. User group

Correct Answer: D**Concept:** User group refers to the concept where applies policies to multiple users collectively.

Question Question 26

Single Choice | Topic: AAA and User Authentication

Which concept is described as: allows one authentication for multiple services?

- A. Authentication
- B. Local authentication
- C. 802.1X
- D. SSO

Correct Answer: D**Concept:** SSO refers to the concept where allows one authentication for multiple services.

31 AAA Basics

Question Question 1

Single Choice | Topic: AAA Basics

What does AAA stand for?

- A. Authentication, Authorization, Accounting
- B. Access, Audit, Alert
- C. Attack, Analysis, Action
- D. Application, Authorization, Audit

Correct Answer: A**Concept:** AAA provides authentication (who are you), authorization (what can you do), and accounting (what did you do).

Question Question 2

Single Choice | Topic: AAA Basics

Which AAA component verifies a user identity?

- A. Authentication
- B. Authorization
- C. Accounting
- D. Auditing

Correct Answer: A**Concept:** Authentication verifies identity, for example, by username/password or certificate.

Question Question 3

Single Choice | Topic: AAA Basics

Which AAA component determines what resources a user can access?

- A. Authentication
- B. Authorization
- C. Accounting
- D. Auditing

Correct Answer: B**Concept:** Authorization defines permissions and access rights after authentication.

Question Question 4

Single Choice | Topic: AAA Basics

Which AAA component records user activities for billing or auditing?

- A. Authentication
- B. Authorization
- C. Accounting
- D. Auditing

Correct Answer: C**Concept:** Accounting logs user activities such as login time, data usage, and commands executed.

32 Firewall User Authentication

Question Question 1

Single Choice | Topic: Firewall User Authentication

Which authentication method redirects a user to a web portal for login?

- A. Local authentication
- B. RADIUS authentication
- C. Portal authentication
- D. LDAP authentication

Correct Answer: C**Concept:** Portal authentication presents a web page to users for credential entry.

Question Question 2

Single Choice | Topic: Firewall User Authentication

Which authentication method uses a client-side agent to authenticate users before network access?

- A. Portal authentication
- B. 802.1X authentication
- C. Local authentication
- D. Certificate authentication

Correct Answer: B**Concept:** 802.1X uses a supplicant client to authenticate devices before granting network access.

Question Question 3

Single Choice | Topic: Firewall User Authentication

Which protocol is commonly used for centralized authentication and accounting?

- A. RADIUS
- B. SMTP
- C. SNMP
- D. NTP

Correct Answer: A**Concept:** RADIUS is a widely used AAA protocol for centralized authentication, authorization, and accounting.

Question Question 4

Single Choice | Topic: Firewall User Authentication

Which protocol provides authentication with stronger security and uses TCP?

- A. RADIUS
- B. TACACS+
- C. LDAP
- D. DHCP

Correct Answer: B**Concept:** TACACS+ uses TCP and separates authentication, authorization, and accounting functions.

33 Cryptography and PKI

Question Question 1

Single Choice | Topic: Cryptography and PKI

Which symmetric algorithm uses variable key lengths of 128, 192, or 256 bits?

- A. DES
- B. AES
- C. RSA
- D. MD5

Correct Answer: B**Concept:** AES supports 128, 192, and 256-bit keys.

Question Question 2

Single Choice | Topic: Cryptography and PKI

How many bits is a DES key?

- A. 56
- B. 64
- C. 128
- D. 256

Correct Answer: A**Concept:** DES uses a 56-bit effective key length.

Question Question 3

Single Choice | Topic: Cryptography and PKI

Which asymmetric algorithm is commonly used for key exchange in TLS?

- A. AES
- B. RSA
- C. SHA-256
- D. HMAC

Correct Answer: B**Concept:** RSA or Diffie-Hellman are commonly used for TLS key exchange.

Question Question 4

Single Choice | Topic: Cryptography and PKI

What is the main disadvantage of symmetric encryption?

- A. Slow speed
- B. Key distribution challenge
- C. Large key sizes
- D. No confidentiality

Correct Answer: B**Concept:** Distributing the shared secret key securely is the main challenge.

Question Question 5

Single Choice | Topic: Cryptography and PKI

Which property ensures a small input change produces a very different hash?

- A. Determinism
- B. Avalanche effect
- C. Fixed output
- D. Reversibility

Correct Answer: B**Concept:** The avalanche effect means tiny input changes cause large output changes.

Question Question 6

Single Choice | Topic: Cryptography and PKI

Which hash algorithm produces a 128-bit digest?

- A. MD5
- B. SHA-1
- C. SHA-256
- D. SHA-512

Correct Answer: A**Concept:** MD5 produces a 128-bit digest.

Question Question 7

Single Choice | Topic: Cryptography and PKI

What is HMAC?

- A. A hash-based message authentication code
- B. A symmetric cipher
- C. A key exchange protocol
- D. A firewall feature

Correct Answer: A**Concept:** HMAC combines a hash function with a secret key to provide message authentication.

Question Question 8

Single Choice | Topic: Cryptography and PKI

Which key is used to create a digital signature?

- A. Public key
- B. Private key
- C. Shared secret
- D. Session key

Correct Answer: B**Concept:** The signer uses their private key to create a digital signature.

Question Question 9

Single Choice | Topic: Cryptography and PKI

Which key is used to verify a digital signature?

- A. Public key
- B. Private key
- C. Shared secret
- D. Master key

Correct Answer: A**Concept:** The verifier uses the signer public key to verify the signature.

Question Question 10

Single Choice | Topic: Cryptography and PKI

Which standard defines the format of digital certificates?

- A. X.509
- B. X.500
- C. LDAP
- D. DNS

Correct Answer: A**Concept:** X.509 is the standard format for public key certificates.

Question Question 11

Single Choice | Topic: Cryptography and PKI

What does OCSP provide?

- A. Online certificate status checking
- B. Certificate issuance
- C. Key generation
- D. Email encryption

Correct Answer: A**Concept:** OCSP checks certificate revocation status in real time.

Question Question 12

Single Choice | Topic: Cryptography and PKI

In a PKI hierarchy, which certificate is self-signed?

- A. End-entity certificate
- B. Root CA certificate
- C. Intermediate CA certificate
- D. Server certificate

Correct Answer: B**Concept:** Root CA certificates are self-signed because they are the trust anchor.

Question Question 13

Single Choice | Topic: Cryptography and PKI

Which entity stores issued certificates for retrieval?

- A. CA
- B. RA
- C. Repository
- D. CRL

Correct Answer: C**Concept:** A repository distributes certificates and CRLs to relying parties.

Question Question 14

Single Choice | Topic: Cryptography and PKI

Which cipher mode provides both confidentiality and authentication?

- A. ECB
- B. CBC
- C. GCM
- D. CTR

Correct Answer: C**Concept:** Galois/Counter Mode (GCM) provides authenticated encryption.

Question Question 15

Single Choice | Topic: Cryptography and PKI

Which mode should be avoided because identical plaintext blocks produce identical ciphertext?

- A. ECB
- B. CBC
- C. GCM
- D. CFB

Correct Answer: A**Concept:** ECB reveals patterns in plaintext and is generally insecure.

Question Question 16

Single Choice | Topic: Cryptography and PKI

AES is an example of:

- A. Symmetric encryption
- B. Asymmetric encryption
- C. Hashing
- D. Steganography

Correct Answer: A**Concept:** AES uses a single shared key for encryption and decryption.

Question Question 17

Single Choice | Topic: Cryptography and PKI

RSA is an example of:

- A. Asymmetric encryption
- B. Symmetric encryption
- C. Hash algorithm
- D. Compression algorithm

Correct Answer: A**Concept:** RSA uses public and private key pairs.

Question Question 18

Single Choice | Topic: Cryptography and PKI

A hash function produces:

- A. A fixed-length digest
- B. The original input
- C. An encrypted file
- D. A random key

Correct Answer: A

Concept: Hash functions map variable input to a fixed-size digest.

Question Question 19

Single Choice | Topic: Cryptography and PKI

MD5 is considered insecure because:

- A. Collision vulnerabilities exist
- B. It is too slow
- C. It produces too long output
- D. It requires a public key

Correct Answer: A

Concept: MD5 collision attacks make it unsuitable for security purposes.

Question Question 20

Single Choice | Topic: Cryptography and PKI

A digital certificate binds a public key to:

- A. An identity
- B. A password
- C. A MAC address
- D. A VLAN

Correct Answer: A

Concept: Certificates associate a public key with an entity identity.

Question Question 21

Single Choice | Topic: Cryptography and PKI

Certificate revocation status can be checked using:

- A. CRL or OCSP
- B. DNS only
- C. DHCP only
- D. SNMP only

Correct Answer: A

Concept: CRLs and OCSP provide revocation status information.

Question Question 22

Single Choice | Topic: Cryptography and PKI

Which concept is described as: uses the same key for encryption and decryption?

- A. RSA
- B. DES
- C. PKI
- D. Symmetric encryption

Correct Answer: D

Concept: Symmetric encryption refers to the concept where uses the same key for encryption and decryption.

Question Question 23

Single Choice | Topic: Cryptography and PKI

Which concept is described as: uses public and private key pairs?

- A. Hash function
- B. Digital signature
- C. Asymmetric encryption
- D. CRL

Correct Answer: C

Concept: Asymmetric encryption refers to the concept where uses public and private key pairs.

Question Question 24

Single Choice | Topic: Cryptography and PKI

Which concept is described as: is a symmetric encryption standard?

- A. DES
- B. GCM
- C. Symmetric encryption
- D. AES

Correct Answer: D

Concept: AES refers to the concept where is a symmetric encryption standard.

Question Question 25

Single Choice | Topic: Cryptography and PKI

Which concept is described as: is an asymmetric encryption algorithm?

- A. Digital signature
- B. RSA
- C. ECB
- D. Root CA certificate

Correct Answer: B

Concept: RSA refers to the concept where is an asymmetric encryption algorithm.

Question Question 26

Single Choice | Topic: Cryptography and PKI

Which concept is described as: uses a 56-bit effective key?

- A. CRL
- B. GCM
- C. AES
- D. DES

Correct Answer: D**Concept:** DES refers to the concept where uses a 56-bit effective key.

Question Question 27

Single Choice | Topic: Cryptography and PKI

Which concept is described as: produces a fixed-length digest?

- A. PKI
- B. RSA
- C. CA
- D. Hash function

Correct Answer: D**Concept:** Hash function refers to the concept where produces a fixed-length digest.

Question Question 28

Single Choice | Topic: Cryptography and PKI

Which concept is described as: is considered broken due to collision attacks?

- A. MD5
- B. PKI
- C. Asymmetric encryption
- D. Public key

Correct Answer: A**Concept:** MD5 refers to the concept where is considered broken due to collision attacks.

Question Question 29

Single Choice | Topic: Cryptography and PKI

Which concept is described as: produces a 256-bit digest?

- A. Digital signature
- B. X.509
- C. RA
- D. SHA-256

Correct Answer: D**Concept:** SHA-256 refers to the concept where produces a 256-bit digest.

Question Question 30

Single Choice | Topic: Cryptography and PKI

Which concept is described as: provides message authentication using a hash and key?

- A. HMAC
- B. AES
- C. MD5
- D. DES

Correct Answer: A**Concept:** HMAC refers to the concept where provides message authentication using a hash and key.

Question Question 31

Single Choice | Topic: Cryptography and PKI

Which concept is described as: verifies authenticity and integrity?

- A. HMAC
- B. CRL
- C. Digital signature
- D. RA

Correct Answer: C**Concept:** Digital signature refers to the concept where verifies authenticity and integrity.

Question Question 32

Single Choice | Topic: Cryptography and PKI

Which concept is described as: is used to verify a digital signature?

- A. Hash function
- B. Public key
- C. Asymmetric encryption
- D. GCM

Correct Answer: B**Concept:** Public key refers to the concept where is used to verify a digital signature.

Question Question 33

Single Choice | Topic: Cryptography and PKI

Which concept is described as: is used to create a digital signature?

- A. Asymmetric encryption
- B. Private key
- C. CA
- D. RSA

Correct Answer: B**Concept:** Private key refers to the concept where is used to create a digital signature.

Question Question 34

Single Choice | Topic: Cryptography and PKI

Which concept is described as: manages public keys and certificates?

- A. Hash function
- B. PKI
- C. Asymmetric encryption
- D. X.509

Correct Answer: B**Concept:** PKI refers to the concept where manages public keys and certificates.

Question Question 35

Single Choice | Topic: Cryptography and PKI

Which concept is described as: issues and signs digital certificates?

- A. SHA-256
- B. CA
- C. Asymmetric encryption
- D. MD5

Correct Answer: B**Concept:** CA refers to the concept where issues and signs digital certificates.

Question Question 36

Single Choice | Topic: Cryptography and PKI

Which concept is described as: validates identity before certificate issuance?

- A. PKI
- B. RA
- C. Asymmetric encryption
- D. CRL

Correct Answer: B**Concept:** RA refers to the concept where validates identity before certificate issuance.

Question Question 37

Single Choice | Topic: Cryptography and PKI

Which concept is described as: lists revoked certificates?

- A. RA
- B. Symmetric encryption
- C. Root CA certificate
- D. CRL

Correct Answer: D**Concept:** CRL refers to the concept where lists revoked certificates.

Question Question 38

Single Choice | Topic: Cryptography and PKI

Which concept is described as: provides online certificate status checking?

- A. Root CA certificate
- B. SHA-256
- C. RSA
- D. OCSP

Correct Answer: D**Concept:** OCSP refers to the concept where provides online certificate status checking.

Question Question 39

Single Choice | Topic: Cryptography and PKI

Which concept is described as: is the standard certificate format?

- A. CRL
- B. X.509
- C. OCSP
- D. AES

Correct Answer: B**Concept:** X.509 refers to the concept where is the standard certificate format.

Question Question 40

Single Choice | Topic: Cryptography and PKI

Which concept is described as: is self-signed and acts as a trust anchor?

- A. Hash function
- B. Root CA certificate
- C. Asymmetric encryption
- D. Symmetric encryption

Correct Answer: B**Concept:** Root CA certificate refers to the concept where is self-signed and acts as a trust anchor.

Question Question 41

Single Choice | Topic: Cryptography and PKI

Which concept is described as: provides authenticated encryption?

- A. DES
- B. AES
- C. GCM
- D. SHA-256

Correct Answer: C**Concept:** GCM refers to the concept where provides authenticated encryption.

Question Question 42

Single Choice | Topic: Cryptography and PKI

Which concept is described as: should be avoided because it reveals plaintext patterns?

- A. Public key
- B. MD5
- C. ECB
- D. Hash function

Correct Answer: C**Concept:** ECB refers to the concept where should be avoided because it reveals plaintext patterns.

34 Cryptography Basics

Question Question 1

Single Choice | Topic: Cryptography Basics

Which encryption type uses the same key for encryption and decryption?

- A. Symmetric encryption
- B. Asymmetric encryption
- C. Hashing
- D. Steganography

Correct Answer: A**Concept:** Symmetric encryption uses a single shared key for both encryption and decryption.

Question Question 2

Single Choice | Topic: Cryptography Basics

Which encryption type uses a key pair: public key and private key?

- A. Symmetric encryption
- B. Asymmetric encryption
- C. Hashing
- D. Encoding

Correct Answer: B**Concept:** Asymmetric encryption uses a public key to encrypt and a private key to decrypt (or vice versa).

Question Question 3

Single Choice | Topic: Cryptography Basics

Which algorithm is a symmetric encryption standard?

- A. RSA
- B. AES
- C. ECC
- D. DSA

Correct Answer: B**Concept:** AES (Advanced Encryption Standard) is a widely used symmetric encryption algorithm.

Question Question 4

Single Choice | Topic: Cryptography Basics

Which algorithm is an asymmetric encryption algorithm?

- A. AES
- B. DES
- C. RSA
- D. 3DES

Correct Answer: C**Concept:** RSA is an asymmetric algorithm used for encryption, digital signatures, and key exchange.

35 Hash Algorithms

Question Question 1

Single Choice | Topic: Hash Algorithms

What is a key property of a cryptographic hash function?

- A. Reversible output
- B. Fixed-length output regardless of input size
- C. Requires a public key
- D. Uses the same key for hashing and verification

Correct Answer: B**Concept:** Hash functions produce a fixed-length digest for any input size.

Question Question 2

Single Choice | Topic: Hash Algorithms

Which hash algorithm produces a 256-bit digest?

- A. MD5
- B. SHA-1
- C. SHA-256
- D. SHA-512

Correct Answer: C**Concept:** SHA-256 produces a 256-bit hash digest.

Question Question 3

Single Choice | Topic: Hash Algorithms

Which hash algorithm is considered broken and unsuitable for security use?

- A. SHA-256
- B. MD5
- C. SHA-3
- D. BLAKE2

Correct Answer: B**Concept:** MD5 is vulnerable to collision attacks and should not be used for security-sensitive applications.

36 Digital Signatures

Question Question 1

Single Choice | Topic: Digital Signatures

What is the purpose of a digital signature?

- A. Encrypt all data
- B. Verify authenticity and integrity
- C. Compress files
- D. Hide sender identity

Correct Answer: B

Concept: Digital signatures verify the sender identity and ensure data has not been altered.

37 PKI

Question Question 1

Single Choice | Topic: PKI

What does PKI stand for?

- A. Public Key Infrastructure
- B. Private Key Interface
- C. Packet Knowledge Index
- D. Policy Key Identifier

Correct Answer: A

Concept: PKI provides a framework for managing public keys, certificates, and trust relationships.

Question Question 2

Single Choice | Topic: PKI

Which entity issues digital certificates in a PKI?

- A. Registration Authority
- B. Certificate Authority
- C. End user
- D. Repository

Correct Answer: B

Concept: A Certificate Authority (CA) issues and signs digital certificates.

Question Question 3

Single Choice | Topic: PKI

Which PKI component validates an applicant identity before certificate issuance?

- A. Certificate Authority
- B. Registration Authority
- C. Repository
- D. CRL

Correct Answer: B

Concept: A Registration Authority (RA) verifies identities and forwards requests to the CA.

Question Question 4

Single Choice | Topic: PKI

What is a CRL used for?

- A. Storing public keys
- B. Listing revoked certificates
- C. Encrypting emails
- D. Generating hash values

Correct Answer: B

Concept: A Certificate Revocation List (CRL) contains certificates that have been revoked before expiration.

38 VPN

Question Question 1

Single Choice | Topic: VPN

Which VPN deployment connects two fixed sites?

- A. Remote access VPN
- B. Site-to-site VPN
- C. Clientless VPN
- D. Host-to-host VPN

Correct Answer: B

Concept: Site-to-site VPNs connect entire networks at different locations.

Question Question 2

Single Choice | Topic: VPN

What is a key advantage of GRE?

- A. Built-in encryption
- B. Can encapsulate multicast and non-IP protocols
- C. Uses TCP
- D. Automatic key exchange

Correct Answer: B

Concept: GRE can tunnel multicast and non-IP traffic, making it useful for routing protocols.

Question Question 3

Single Choice | Topic: VPN

Which protocol number is assigned to GRE?

- A. 47
- B. 50
- C. 51
- D. 89

Correct Answer: A

Concept: GRE is IP protocol 47.

Question Question 4

Single Choice | Topic: VPN

What does IKE Phase 1 establish?

- A. IPsec SA
- B. IKE SA
- C. Routing table
- D. DHCP lease

Correct Answer: B

Concept: IKE Phase 1 establishes a secure IKE SA for protected negotiation.

Question Question 5

Single Choice | Topic: VPN

What does IKE Phase 2 establish?

- A. IKE SA
- B. IPsec SA
- C. SSL session
- D. ARP entry

Correct Answer: B

Concept: IKE Phase 2 negotiates the IPsec SA that protects actual data traffic.

Question Question 6

Single Choice | Topic: VPN

Which authentication method does IKE support? (single best answer)

- A. Pre-shared key
- B. Digital signatures
- C. Both pre-shared key and digital signatures
- D. None

Correct Answer: C

Concept: IKE supports both pre-shared key and digital signature authentication.

Question Question 7

Single Choice | Topic: VPN

Which Diffie-Hellman group provides stronger key exchange?

- A. Group 1
- B. Group 2
- C. Group 5
- D. Group 14 or higher

Correct Answer: D

Concept: Higher Diffie-Hellman groups such as 14 or above provide stronger security.

Question Question 8

Single Choice | Topic: VPN

Which IKE version provides faster SA setup and improved reliability?

- A. IKEv1
- B. IKEv2
- C. IKEv3
- D. ISAKMP

Correct Answer: B**Concept:** IKEv2 simplifies negotiation and improves resilience compared to IKEv1.

Question Question 9

Single Choice | Topic: VPN

Which UDP ports does L2TP use?

- A. 500 and 4500
- B. 1701
- C. 443
- D. 50 and 51

Correct Answer: B**Concept:** L2TP uses UDP port 1701.

Question Question 10

Single Choice | Topic: VPN

Why is L2TP often combined with IPsec?

- A. To provide encryption
- B. To increase speed
- C. To reduce overhead
- D. To avoid tunnels

Correct Answer: A**Concept:** L2TP lacks encryption, so IPsec is added for confidentiality and integrity.

Question Question 11

Single Choice | Topic: VPN

Which SSL VPN mode grants the remote host a virtual IP on the internal network?

- A. Web proxy
- B. Network extension
- C. File sharing
- D. Port forwarding

Correct Answer: B**Concept:** Network extension mode assigns a virtual IP and routes traffic into the corporate network.

Question Question 12

Single Choice | Topic: VPN

Which SSL VPN mode is best for accessing internal web apps without installing a client?

- A. Web proxy
- B. Network extension
- C. IPsec
- D. L2TP

Correct Answer: A**Concept:** Web proxy mode requires only a browser.

Question Question 13

Single Choice | Topic: VPN

Which encapsulation mode is typically used for site-to-site VPNs?

- A. Transport mode
- B. Tunnel mode
- C. Bridge mode
- D. Trunk mode

Correct Answer: B**Concept:** Site-to-site VPNs typically use tunnel mode to protect the entire original packet.

Question Question 14

Single Choice | Topic: VPN

Which tunneling protocol is considered a Layer 2 VPN technology?

- A. GRE
- B. IPsec
- C. L2TP
- D. SSL VPN

Correct Answer: C**Concept:** L2TP operates at Layer 2, while GRE, IPsec, and SSL VPN operate at higher layers.

Question Question 15

Single Choice | Topic: VPN

Which component defines the security parameters for an IPsec connection?

- A. SPD
- B. SA
- C. NAT
- D. ARP

Correct Answer: B**Concept:** A Security Association (SA) defines algorithms, keys, and lifetimes for IPsec.

Question Question 16

Single Choice | Topic: VPN

The main goal of a VPN is to provide:

- A. Secure communication over public networks
- B. Faster Internet speed
- C. Free public Wi-Fi
- D. Unencrypted remote access

Correct Answer: A

Concept: VPNs create secure encrypted tunnels over untrusted networks.

Question Question 17

Single Choice | Topic: VPN

GRE protocol number is:

- A. 47
- B. 50
- C. 51
- D. 89

Correct Answer: A

Concept: GRE is encapsulated using IP protocol 47.

Question Question 18

Single Choice | Topic: VPN

IPsec ESP provides:

- A. Encryption and authentication
- B. Only routing
- C. Only NAT
- D. Only compression

Correct Answer: A

Concept: ESP provides confidentiality, integrity, and authentication.

Question Question 19

Single Choice | Topic: VPN

IPsec AH provides:

- A. Authentication and integrity without encryption
- B. Encryption only
- C. Compression only
- D. Key exchange

Correct Answer: A

Concept: AH authenticates and protects integrity but does not encrypt.

Question Question 20

Single Choice | Topic: VPN

IKE negotiates:

- A. Security associations and keys for IPsec
- B. DHCP leases
- C. DNS records
- D. MAC addresses

Correct Answer: A

Concept: IKE handles key exchange and SA negotiation.

Question Question 21

Single Choice | Topic: VPN

SSL VPN commonly uses port:

- A. 80
- B. 443
- C. 22
- D. 1701

Correct Answer: B

Concept: SSL VPN uses HTTPS/TCP 443.

Question Question 22

Single Choice | Topic: VPN

L2TP is commonly paired with which protocol for encryption?

- A. IPsec
- B. GRE
- C. OSPF
- D. BGP

Correct Answer: A

Concept: L2TP provides tunneling and is paired with IPsec for encryption.

Question Question 23

Single Choice | Topic: VPN

Which concept is described as: provides secure communication over public networks?

- A. IPsec
- B. VPN
- C. Tunnel mode
- D. AH

Correct Answer: B

Concept: VPN refers to the concept where provides secure communication over public networks.

Question Question 24

Single Choice | Topic: VPN

Which concept is described as: connects entire networks at different locations?

- A. GRE
- B. Web proxy mode
- C. Tunnel mode
- D. Site-to-site VPN

Correct Answer: D**Concept:** Site-to-site VPN refers to the concept where connects entire networks at different locations.

Question Question 25

Single Choice | Topic: VPN

Which concept is described as: connects individual users to a corporate network?

- A. Remote-access VPN
- B. L2TP
- C. L2TP over IPsec
- D. IKEv2

Correct Answer: A**Concept:** Remote-access VPN refers to the concept where connects individual users to a corporate network.

Question Question 26

Single Choice | Topic: VPN

Which concept is described as: is IP protocol 47 and lacks built-in encryption?

- A. IKEv2
- B. GRE
- C. Transport mode
- D. IKE

Correct Answer: B**Concept:** GRE refers to the concept where is IP protocol 47 and lacks built-in encryption.

Question Question 27

Single Choice | Topic: VPN

Which concept is described as: can tunnel multicast and non-IP protocols?

- A. Transport mode
- B. GRE
- C. Site-to-site VPN
- D. Web proxy mode

Correct Answer: B**Concept:** GRE refers to the concept where can tunnel multicast and non-IP protocols.

Question Question 28

Single Choice | Topic: VPN

Which concept is described as: provides confidentiality, integrity, and authentication?

- A. IPsec
- B. Tunnel mode
- C. Remote-access VPN
- D. ESP

Correct Answer: A

Concept: IPsec refers to the concept where provides confidentiality, integrity, and authentication.

Question Question 29

Single Choice | Topic: VPN

Which concept is described as: provides encryption and authentication?

- A. ESP
- B. Transport mode
- C. VPN
- D. GRE

Correct Answer: A

Concept: ESP refers to the concept where provides encryption and authentication.

Question Question 30

Single Choice | Topic: VPN

Which concept is described as: provides authentication without encryption?

- A. AH
- B. Site-to-site VPN
- C. Transport mode
- D. IKEv2

Correct Answer: A

Concept: AH refers to the concept where provides authentication without encryption.

Question Question 31

Single Choice | Topic: VPN

Which concept is described as: negotiates IPsec keys and security associations?

- A. IKE
- B. Transport mode
- C. ESP
- D. Remote-access VPN

Correct Answer: A

Concept: IKE refers to the concept where negotiates IPsec keys and security associations.

Question Question 32

Single Choice | Topic: VPN

Which concept is described as: encrypts the entire original IP packet?

- A. Transport mode
- B. GRE
- C. Tunnel mode
- D. L2TP

Correct Answer: C**Concept:** Tunnel mode refers to the concept where encrypts the entire original IP packet.

Question Question 33

Single Choice | Topic: VPN

Which concept is described as: encrypts only the IP payload?

- A. SSL VPN
- B. Transport mode
- C. L2TP over IPsec
- D. GRE

Correct Answer: B**Concept:** Transport mode refers to the concept where encrypts only the IP payload.

Question Question 34

Single Choice | Topic: VPN

Which concept is described as: uses UDP port 1701 and lacks encryption?

- A. L2TP
- B. IKE
- C. GRE
- D. Remote-access VPN

Correct Answer: A**Concept:** L2TP refers to the concept where uses UDP port 1701 and lacks encryption.

Question Question 35

Single Choice | Topic: VPN

Which concept is described as: combines L2TP tunneling with IPsec encryption?

- A. Transport mode
- B. IKE
- C. L2TP over IPsec
- D. AH

Correct Answer: C**Concept:** L2TP over IPsec refers to the concept where combines L2TP tunneling with IPsec encryption.

Question Question 36

Single Choice | Topic: VPN

Which concept is described as: commonly uses TCP port 443?

- A. L2TP
- B. IPsec
- C. SSL VPN
- D. Transport mode

Correct Answer: C

Concept: SSL VPN refers to the concept where commonly uses TCP port 443.

Question Question 37

Single Choice | Topic: VPN

Which concept is described as: provides browser-based access to web apps?

- A. IPsec
- B. Web proxy mode
- C. IKE
- D. GRE

Correct Answer: B

Concept: Web proxy mode refers to the concept where provides browser-based access to web apps.

Question Question 38

Single Choice | Topic: VPN

Which concept is described as: assigns a virtual IP to the remote host?

- A. Network extension mode
- B. Web proxy mode
- C. Tunnel mode
- D. IKE

Correct Answer: A

Concept: Network extension mode refers to the concept where assigns a virtual IP to the remote host.

Question Question 39

Single Choice | Topic: VPN

Which concept is described as: is faster and more reliable than IKEv1?

- A. GRE
- B. IPsec
- C. IKEv2
- D. L2TP

Correct Answer: C

Concept: IKEv2 refers to the concept where is faster and more reliable than IKEv1.

39 VPN Overview

Question Question 1

Single Choice | Topic: VPN Overview

What does VPN stand for?

- A. Virtual Private Network
- B. Virtual Public Network
- C. Verified Private Node
- D. Virtual Protocol Network

Correct Answer: A

Concept: A VPN extends a private network across a public network using encryption and tunneling.

Question Question 2

Single Choice | Topic: VPN Overview

Which benefit does a VPN provide over the public Internet?

- A. Higher physical speed
- B. Confidentiality and secure remote access
- C. Free bandwidth
- D. No encryption needed

Correct Answer: B

Concept: VPNs provide confidentiality, integrity, and secure access over untrusted networks.

Question Question 3

Single Choice | Topic: VPN Overview

Which VPN type connects individual remote users to a corporate network?

- A. Site-to-site VPN
- B. Remote-access VPN
- C. Intranet VPN
- D. Extranet VPN

Correct Answer: B

Concept: Remote-access VPNs allow individual users to securely connect to a corporate network.

40 GRE VPN

Question Question 1

Single Choice | Topic: GRE VPN

Which protocol does GRE use as its transport protocol?

- A. TCP
- B. UDP
- C. IP protocol 47
- D. IP protocol 50

Correct Answer: C

Concept: GRE encapsulates packets using IP protocol number 47.

Question Question 2

Single Choice | Topic: GRE VPN

Which statement about GRE is correct?

- A. GRE provides encryption by default
- B. GRE is a simple tunneling protocol without built-in encryption
- C. GRE cannot tunnel multicast
- D. GRE uses UDP port 500

Correct Answer: B

Concept: GRE provides encapsulation and tunneling but does not provide encryption or authentication by itself.

41 IPsec VPN

Question Question 1

Single Choice | Topic: IPsec VPN

Which IPsec protocol provides encryption and confidentiality?

- A. AH
- B. ESP
- C. IKE
- D. GRE

Correct Answer: B

Concept: ESP (Encapsulating Security Payload) provides encryption, authentication, and integrity.

Question Question 2

Single Choice | Topic: IPsec VPN

Which IPsec protocol provides authentication and integrity but not encryption?

- A. AH
- B. ESP
- C. IKE
- D. L2TP

Correct Answer: A

Concept: AH (Authentication Header) provides source authentication and integrity but does not encrypt payloads.

Question Question 3

Single Choice | Topic: IPsec VPN

Which protocol is used to establish and manage IPsec security associations?

- A. AH
- B. ESP
- C. IKE
- D. GRE

Correct Answer: C

Concept: IKE (Internet Key Exchange) negotiates keys and Security Associations for IPsec.

Question Question 4

Single Choice | Topic: IPsec VPN

Which IPsec mode encrypts the entire original IP packet and adds a new IP header?

- A. Transport mode
- B. Tunnel mode
- C. Gre mode
- D. Bridge mode

Correct Answer: B**Concept:** Tunnel mode encapsulates and encrypts the entire original packet, adding a new IP header.

Question Question 5

Single Choice | Topic: IPsec VPN

Which IPsec mode encrypts only the payload and is used for host-to-host communications?

- A. Transport mode
- B. Tunnel mode
- C. Bridge mode
- D. Trunk mode

Correct Answer: A**Concept:** Transport mode encrypts only the IP payload, leaving the original IP header intact.

42 L2TP VPN

Question Question 1

Single Choice | Topic: L2TP VPN

Which layer-2 tunneling protocol is often combined with IPsec for encryption?

- A. PPTP
- B. L2TP
- C. GRE
- D. MPLS

Correct Answer: B**Concept:** L2TP provides tunneling and is commonly paired with IPsec to provide encryption.

43 SSL VPN

Question Question 1

Single Choice | Topic: SSL VPN

Which VPN type is commonly accessed through a web browser without a dedicated client?

- A. IPsec VPN
- B. SSL VPN
- C. GRE VPN
- D. L2TP VPN

Correct Answer: B**Concept:** SSL VPN can be accessed via a web browser, making it convenient for remote users.

Question Question 2

Single Choice | Topic: SSL VPN

Which SSL VPN mode provides access to specific web applications through a browser?

- A. Web proxy mode
- B. Network extension mode
- C. File sharing mode
- D. Full tunnel mode

Correct Answer: A**Concept:** Web proxy mode allows browser-based access to internal web applications.

PART II

Multiple-Answer Questions

44 Network Security Concepts

Question Question 1

Multiple Choice | Topic: Network Security Concepts

Which of the following are characteristics of network security? (Choose all that apply)

- A. Confidentiality
- B. Integrity
- C. Availability
- D. Controllability
- E. Non-repudiation

Correct Answer: A,B,C,D,E**Concept:** Network security commonly encompasses confidentiality, integrity, availability, controllability, and non-repudiation.

Question Question 2

Multiple Choice | Topic: Network Security Concepts

Which are stages in the development history of network security? (Choose all that apply)

- A. Communication security period
- B. Information security period
- C. Information assurance period
- D. Cyberspace security period

Correct Answer: A,B,C,D**Concept:** Network security has evolved through communication security, information security, information assurance, and cyberspace security periods.

Question Question 3

Multiple Choice | Topic: Network Security Concepts

Which are core information security properties? (Choose all that apply)

- A. Confidentiality
- B. Integrity
- C. Availability
- D. Controllability
- E. Non-repudiation

Correct Answer: A,B,C,D,E

Concept: Network security includes confidentiality, integrity, availability, controllability, and non-repudiation.

45 Network Security Standards

Question Question 1

Multiple Choice | Topic: Network Security Standards

Which are typical Classified Protection 2.0 lifecycle phases? (Choose all that apply)

- A. Classification
- B. Filing
- C. Construction and rectification
- D. Level assessment
- E. Supervision and inspection

Correct Answer: A,B,C,D,E

Concept: The full lifecycle includes classification, filing, construction/rectification, level assessment, and supervision/inspection.

Question Question 2

Multiple Choice | Topic: Network Security Standards

Which are benefits of implementing ISO 27001? (Choose all that apply)

- A. Systematic risk management
- B. Improved stakeholder confidence
- C. Legal/regulatory compliance
- D. Guaranteed zero attacks

Correct Answer: A,B,C

Concept: ISO 27001 helps manage risk, build confidence, and comply with regulations, but cannot guarantee zero attacks.

Question Question 3

Multiple Choice | Topic: Network Security Standards

Which are typical Classified Protection 2.0 security extension requirements? (Choose all that apply)

- A. Cloud computing
- B. Mobile Internet
- C. Internet of Things
- D. Industrial control systems

Correct Answer: A,B,C,D

Concept: Classified Protection 2.0 extends coverage to cloud, mobile Internet, IoT, and industrial control scenarios.

Question Question 4

Multiple Choice | Topic: Network Security Standards

Which belong to the Classified Protection lifecycle? (Choose all that apply)

- A. Classification
- B. Filing
- C. Construction and rectification
- D. Level assessment
- E. Supervision and inspection

Correct Answer: A,B,C,D,E

Concept: The full lifecycle includes classification, filing, construction, level assessment, and supervision.

Question Question 5

Multiple Choice | Topic: Network Security Standards

Which are ISO 27000 family standards? (Choose all that apply)

- A. ISO 27001
- B. ISO 27002
- C. ISO 27005
- D. ISO 9001

Correct Answer: A,B,C

Concept: ISO 27001, 27002, and 27005 are information security standards; ISO 9001 is quality management.

Question Question 6

Multiple Choice | Topic: Network Security Standards

Which of the following relate to ISO? (Choose all that apply)

- A. ISO 27001
- B. ISO 27002
- C. ISO 27001:2022

Correct Answer: A,B,C

Concept: These concepts all relate to ISO in Network Security Standards.

Question Question 7

Multiple Choice | Topic: Network Security Standards

Which of the following relate to Classified? (Choose all that apply)

- A. Classified Protection 2.0
- B. Classified Protection levels
- C. Classified Protection lifecycle

Correct Answer: A,B,C

Concept: These concepts all relate to Classified in Network Security Standards.

46 Network Fundamentals

Question Question 1

Multiple Choice | Topic: Network Fundamentals

Which layers are included in the TCP/IP equivalent model? (Choose all that apply)

- A. Application
- B. Transport
- C. Network
- D. Data link
- E. Physical

Correct Answer: A,B,C,D,E

Concept: The TCP/IP equivalent model includes application, transport, network, data link, and physical layers.

Question Question 2

Multiple Choice | Topic: Network Fundamentals

Which are connectionless protocols? (Choose all that apply)

- A. UDP
- B. IP
- C. ICMP
- D. TCP

Correct Answer: A,B,C

Concept: UDP, IP, and ICMP are connectionless; TCP is connection-oriented.

Question Question 3

Multiple Choice | Topic: Network Fundamentals

Which fields are in a TCP header? (Choose all that apply)

- A. Source port
- B. Destination port
- C. Sequence number
- D. Acknowledgment number

Correct Answer: A,B,C,D

Concept: TCP headers include source/destination ports, sequence and acknowledgment numbers, flags, window size, etc.

Question Question 4

Multiple Choice | Topic: Network Fundamentals

Which devices operate primarily at Layer 1? (Choose all that apply)

- A. Hub
- B. Repeater
- C. Switch
- D. Router

Correct Answer: A,B

Concept: Hubs and repeaters operate at the physical layer.

Question Question 5

Multiple Choice | Topic: Network Fundamentals

Which protocols use TCP? (Choose all that apply)

- A. HTTP
- B. FTP
- C. SMTP
- D. DNS (sometimes)

Correct Answer: A,B,C,D

Concept: HTTP, FTP, and SMTP use TCP; DNS primarily uses UDP but can use TCP for zone transfers and large responses.

Question Question 6

Multiple Choice | Topic: Network Fundamentals

Which protocols use UDP? (Choose all that apply)

- A. DNS
- B. TFTP
- C. SNMP
- D. DHCP

Correct Answer: A,B,C,D

Concept: DNS, TFTP, SNMP, and DHCP commonly use UDP.

Question Question 7

Multiple Choice | Topic: Network Fundamentals

Which are valid IPv4 private address ranges? (Choose all that apply)

- A. 10.0.0.0/8
- B. 172.16.0.0/12
- C. 192.168.0.0/16
- D. 224.0.0.0/4

Correct Answer: A,B,C

Concept: RFC 1918 private ranges are 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.

Question Question 8

Multiple Choice | Topic: Network Fundamentals

Which are OSI layers? (Choose all that apply)

- A. Physical
- B. Data link
- C. Network
- D. Transport
- E. Application

Correct Answer: A,B,C,D,E

Concept: The OSI model includes these layers plus session and presentation.

Question Question 9

Multiple Choice | Topic: Network Fundamentals

Which are TCP/IP equivalent model layers? (Choose all that apply)

- A. Application
- B. Transport
- C. Network
- D. Data link
- E. Physical

Correct Answer: A,B,C,D,E

Concept: The TCP/IP equivalent model includes all five layers.

Question Question 10

Multiple Choice | Topic: Network Fundamentals

Which protocols use TCP? (Choose all that apply)

- A. HTTP
- B. FTP
- C. SMTP
- D. SSH

Correct Answer: A,B,C,D

Concept: HTTP, FTP, SMTP, and SSH all use TCP.

Question Question 11

Multiple Choice | Topic: Network Fundamentals

Which devices operate at Layer 2? (Choose all that apply)

- A. Switch
- B. Bridge
- C. Router
- D. Hub

Correct Answer: A,B

Concept: Switches and bridges operate at the data link layer.

Question Question 12

Multiple Choice | Topic: Network Fundamentals

Which of the following relate to TCP/IP? (Choose all that apply)

- A. TCP/IP standard model
- B. TCP/IP equivalent model

Correct Answer: A,B

Concept: These concepts all relate to TCP/IP in Network Fundamentals.

47 OSI and TCP/IP Models

Question Question 1

Multiple Choice | Topic: OSI and TCP/IP Models

Which layers are part of the OSI model? (Choose all that apply)

- A. Physical
- B. Data link
- C. Network
- D. Transport
- E. Application

Correct Answer: A,B,C,D,E

Concept: The OSI model includes physical, data link, network, transport, session, presentation, and application layers.

48 Network Protocols

Question Question 1

Multiple Choice | Topic: Network Protocols

Which protocols operate at the application layer? (Choose all that apply)

- A. HTTP
- B. FTP
- C. DNS
- D. SMTP

Correct Answer: A,B,C,D

Concept: HTTP, FTP, DNS, and SMTP are all application-layer protocols.

49 Network Devices

Question Question 1

Multiple Choice | Topic: Network Devices

Which devices can operate at Layer 2? (Choose all that apply)

- A. Switch
- B. Bridge
- C. Router
- D. Hub

Correct Answer: A,B**Concept:** Switches and bridges operate at the data link layer; routers are Layer 3 and hubs are Layer 1.

50 Enterprise Security Threats

Question Question 1

Multiple Choice | Topic: Enterprise Security Threats

Which are examples of malware? (Choose all that apply)

- A. Virus
- B. Worm
- C. Trojan
- D. Ransomware
- E. Spyware

Correct Answer: A,B,C,D,E**Concept:** All listed items are types of malware.

Question Question 2

Multiple Choice | Topic: Enterprise Security Threats

Which are social engineering attacks? (Choose all that apply)

- A. Phishing
- B. Pretexting
- C. Baiting
- D. Tailgating

Correct Answer: A,B,C,D**Concept:** Social engineering manipulates people through phishing, pretexting, baiting, tailgating, etc.

Question Question 3

Multiple Choice | Topic: Enterprise Security Threats

Which protect communication confidentiality? (Choose all that apply)

- A. Encryption
- B. VPN
- C. TLS/SSL
- D. MAC filtering

Correct Answer: A,B,C**Concept:** Encryption, VPNs, and TLS/SSL protect confidentiality; MAC filtering controls device access.

Question Question 4

Multiple Choice | Topic: Enterprise Security Threats

Which are typical security zones? (Choose all that apply)

- A. Trust
- B. Untrust
- C. DMZ
- D. Local

Correct Answer: A,B,C,D

Concept: Huawei firewalls define Trust, Untrust, DMZ, and Local security zones by default.

Question Question 5

Multiple Choice | Topic: Enterprise Security Threats

Which are endpoint security controls? (Choose all that apply)

- A. Antivirus
- B. Host firewall
- C. Application whitelisting
- D. Full-disk encryption

Correct Answer: A,B,C,D

Concept: Endpoint security includes antivirus, host firewalls, whitelisting, and encryption.

Question Question 6

Multiple Choice | Topic: Enterprise Security Threats

Which are malware types? (Choose all that apply)

- A. Virus
- B. Worm
- C. Trojan
- D. Ransomware

Correct Answer: A,B,C,D

Concept: All listed are categories of malware.

Question Question 7

Multiple Choice | Topic: Enterprise Security Threats

Which are denial-of-service attacks? (Choose all that apply)

- A. SYN flood
- B. DDoS
- C. Ping of death
- D. Smurf attack

Correct Answer: A,B,C,D

Concept: These attacks aim to disrupt service availability.

51 Zone Border Security

Question Question 1

Multiple Choice | Topic: Zone Border Security

Which controls are typically applied at zone borders? (Choose all that apply)

- A. Firewall policies
- B. NAT
- C. IPS/IDS
- D. VPN termination

Correct Answer: A,B,C,D**Concept:** Zone borders commonly use firewalls, NAT, IPS/IDS, and VPN gateways for protection.

52 Firewall Basics

Question Question 1

Multiple Choice | Topic: Firewall Basics

Which are Huawei firewall default security zones? (Choose all that apply)

- A. Trust
- B. Untrust
- C. DMZ
- D. Local

Correct Answer: A,B,C,D**Concept:** Huawei firewalls provide Trust, Untrust, DMZ, and Local zones by default.

Question Question 2

Multiple Choice | Topic: Firewall Basics

Which actions can a security policy perform? (Choose all that apply)

- A. Permit
- B. Deny
- C. Log
- D. Redirect

Correct Answer: A,B,C**Concept:** Policies can permit, deny, and log traffic; redirect may be available in specific features.

Question Question 3

Multiple Choice | Topic: Firewall Basics

Which are common firewall deployment modes? (Choose all that apply)

- A. Routing mode
- B. Transparent mode
- C. Hybrid mode
- D. Bridge mode

Correct Answer: A,B,C**Concept:** Firewalls commonly deploy in routing, transparent (bridge), and hybrid modes.

Question Question 4

Multiple Choice | Topic: Firewall Basics

Which conditions can be matched in a security policy? (Choose all that apply)

- A. Source zone
- B. Destination zone
- C. Source IP
- D. Destination port
- E. User

Correct Answer: A,B,C,D,E

Concept: Policies can match zones, IPs, ports, users, applications, services, and time.

Question Question 5

Multiple Choice | Topic: Firewall Basics

Which are default Huawei security zones? (Choose all that apply)

- A. Trust
- B. Untrust
- C. DMZ
- D. Local

Correct Answer: A,B,C,D

Concept: Huawei firewalls provide Trust, Untrust, DMZ, and Local zones by default.

Question Question 6

Multiple Choice | Topic: Firewall Basics

Which can be matched in security policies? (Choose all that apply)

- A. Source zone
- B. Destination zone
- C. Source IP
- D. Destination port
- E. User

Correct Answer: A,B,C,D,E

Concept: Policies can match zones, addresses, ports, users, services, applications, and time.

Question Question 7

Multiple Choice | Topic: Firewall Basics

Which of the following relate to Security? (Choose all that apply)

- A. Security zones
- B. Security policy logging

Correct Answer: A,B

Concept: These concepts all relate to Security in Firewall Basics.

53 Security Zones

Question Question 1

Multiple Choice | Topic: Security Zones

Which are default security zones on Huawei firewalls? (Choose all that apply)

- A. Trust
- B. Untrust
- C. DMZ
- D. Local

Correct Answer: A,B,C,D**Concept:** Huawei firewalls provide four default zones: Trust, Untrust, DMZ, and Local.

54 Security Policies

Question Question 1

Multiple Choice | Topic: Security Policies

Which elements can be referenced in a security policy? (Choose all that apply)

- A. Security zones
- B. IP addresses/address groups
- C. Services
- D. Users and user groups
- E. Time ranges

Correct Answer: A,B,C,D,E**Concept:** Security policies can reference zones, addresses, services, users, applications, and time ranges.

55 Stateful Inspection

Question Question 1

Multiple Choice | Topic: Stateful Inspection

Which are advantages of stateful inspection over simple packet filtering? (Choose all that apply)

- A. Tracks connection state
- B. Allows return traffic automatically
- C. Better performance for established sessions
- D. Can inspect application-layer payloads

Correct Answer: A,B,C**Concept:** Stateful inspection tracks connections, permits return traffic, and improves performance; deep payload inspection is more advanced.

56 NAT

Question Question 1

Multiple Choice | Topic: NAT

Which are private IPv4 address ranges? (Choose all that apply)

- A. 10.0.0.0/8
- B. 172.16.0.0/12
- C. 192.168.0.0/16
- D. 127.0.0.0/8

Correct Answer: A,B,C

Concept: RFC 1918 private ranges are 10/8, 172.16/12, and 192.168/16.

Question Question 2

Multiple Choice | Topic: NAT

Which are valid uses of NAT Server? (Choose all that apply)

- A. Publishing a web server
- B. Publishing an email server
- C. Remote access to an internal host
- D. Hiding internal topology

Correct Answer: A,B,C,D

Concept: NAT Server can publish services, enable remote access, and obscure internal addressing.

Question Question 3

Multiple Choice | Topic: NAT

Which are NAT types? (Choose all that apply)

- A. Source NAT
- B. Destination NAT
- C. Bidirectional NAT
- D. NAT Server

Correct Answer: A,B,C,D

Concept: Huawei firewalls support all four NAT types.

Question Question 4

Multiple Choice | Topic: NAT

Which source NAT modes exist? (Choose all that apply)

- A. NAT No-PAT
- B. NAT
- C. Easy-IP
- D. NAT Server

Correct Answer: A,B,C

Concept: NAT Server is not a source NAT mode.

Question Question 5

Multiple Choice | Topic: NAT

Which of the following relate to NAT? (Choose all that apply)

- A. NAT
- B. NAT Server
- C. NAT No-PAT
- D. NAT ALG

Correct Answer: A,B,C,D

Concept: These concepts all relate to NAT in NAT.

57 NAT Basics

Question Question 1

Multiple Choice | Topic: NAT Basics

Which are types of NAT supported by Huawei firewalls? (Choose all that apply)

- A. Source NAT
- B. Destination NAT
- C. Bidirectional NAT
- D. NAT Server

Correct Answer: A,B,C,D

Concept: Huawei firewalls support source NAT, destination NAT, bidirectional NAT, and NAT server.

58 Source NAT

Question Question 1

Multiple Choice | Topic: Source NAT

Which are common source NAT modes? (Choose all that apply)

- A. NAT No-PAT
- B. NAPT
- C. Easy-IP
- D. NAT Server

Correct Answer: A,B,C

Concept: Source NAT includes NAT No-PAT (one-to-one), NAPT (many-to-one), and Easy-IP (interface-based).

59 NAT ALG

Question Question 1

Multiple Choice | Topic: NAT ALG

Which protocols may require NAT ALG? (Choose all that apply)

- A. FTP
- B. SIP
- C. H.323
- D. DNS

Correct Answer: A,B,C

Concept: Protocols embedding address/port information in payloads (FTP, SIP, H.323) need NAT ALG.

60 Firewall Hot Standby

Question Question 1

Multiple Choice | Topic: Firewall Hot Standby

Which are VRRP components? (Choose all that apply)

- A. Virtual IP
- B. Virtual MAC
- C. Master router
- D. Backup router

Correct Answer: A,B,C,D

Concept: VRRP uses a virtual IP, virtual MAC, master, and backup routers.

Question Question 2

Multiple Choice | Topic: Firewall Hot Standby

Which are HRP synchronization modes? (Choose all that apply)

- A. Real-time
- B. Fast
- C. Batch
- D. Auto

Correct Answer: A,B,C

Concept: HRP supports real-time, fast, and batch synchronization modes.

Question Question 3

Multiple Choice | Topic: Firewall Hot Standby

Which are valid hot-standby modes? (Choose all that apply)

- A. Active/standby
- B. Load sharing
- C. Cluster
- D. Standalone

Correct Answer: A,B

Concept: Huawei firewall hot standby supports active/standby and load-sharing modes.

Question Question 4

Multiple Choice | Topic: Firewall Hot Standby

Which items should be checked when HRP is not synchronizing? (Choose all that apply)

- A. Heartbeat link status
- B. HRP enable status
- C. VGMP priority
- D. Configuration consistency

Correct Answer: A,B,C,D

Concept: Check heartbeat links, HRP status, priorities, and configuration consistency.

Question Question 5

Multiple Choice | Topic: Firewall Hot Standby

Which are HRP synchronization contents? (Choose all that apply)

- A. Session table
- B. NAT entries
- C. Server-map entries
- D. Configuration

Correct Answer: A,B,C,D

Concept: HRP synchronizes all these items between peers.

Question Question 6

Multiple Choice | Topic: Firewall Hot Standby

Which are VRRP states? (Choose all that apply)

- A. Master
- B. Backup
- C. Initialize
- D. Failover

Correct Answer: A,B,C

Concept: VRRP states include master, backup, and initialize.

Question Question 7

Multiple Choice | Topic: Firewall Hot Standby

Which of the following relate to VRRP? (Choose all that apply)

- A. VRRP
- B. VRRP master
- C. VRRP backup

Correct Answer: A,B,C

Concept: These concepts all relate to VRRP in Firewall Hot Standby.

Question Question 8

Multiple Choice | Topic: Firewall Hot Standby

Which of the following relate to HRP? (Choose all that apply)

- A. HRP
- B. HRP session backup

Correct Answer: A,B

Concept: These concepts all relate to HRP in Firewall Hot Standby.

61 VRRP

Question Question 1

Multiple Choice | Topic: VRRP

Which are VRRP roles? (Choose all that apply)

- A. Master
- B. Backup
- C. Initialize
- D. Failover

Correct Answer: A,B,C

Concept: VRRP devices can be in master, backup, or initialize states.

62 HRP

Question Question 1

Multiple Choice | Topic: HRP

Which data does HRP synchronize? (Choose all that apply)

- A. Session table
- B. NAT entries
- C. Server-map entries
- D. Configurations

Correct Answer: A,B,C,D

Concept: HRP synchronizes sessions, NAT entries, server-map entries, and configurations.

63 Intrusion Prevention and Antivirus

Question Question 1

Multiple Choice | Topic: Intrusion Prevention and Antivirus

Which are common attack phases? (Choose all that apply)

- A. Reconnaissance
- B. Exploitation
- C. Lateral movement
- D. Exfiltration

Correct Answer: A,B,C,D

Concept: Cyber kill chain phases include reconnaissance, exploitation, lateral movement, and exfiltration.

Question Question 2

Multiple Choice | Topic: Intrusion Prevention and Antivirus

Which are types of intrusion detection/prevention systems? (Choose all that apply)

- A. NIDS
- B. NIPS
- C. HIDS
- D. HIPS

Correct Answer: A,B,C,D

Concept: Network and host-based IDS/IPS exist for different monitoring scopes.

Question Question 3

Multiple Choice | Topic: Intrusion Prevention and Antivirus

Which factors should be considered when tuning IPS signatures? (Choose all that apply)

- A. False positive rate
- B. False negative rate
- C. Performance impact
- D. Business context

Correct Answer: A,B,C,D

Concept: Effective tuning balances detection accuracy, performance, and business needs.

Question Question 4

Multiple Choice | Topic: Intrusion Prevention and Antivirus

Which are methods to improve malware detection? (Choose all that apply)

- A. Keep signatures updated
- B. Use heuristics
- C. Deploy sandboxing
- D. User education

Correct Answer: A,B,C,D

Concept: Updated signatures, heuristics, sandboxing, and user awareness improve detection.

Question Question 5

Multiple Choice | Topic: Intrusion Prevention and Antivirus

Which are IPS deployment modes? (Choose all that apply)

- A. Inline
- B. Tap/SPAN
- C. Out-of-band
- D. Bridge

Correct Answer: A,B,C**Concept:** IPS can be inline, tapped, or out-of-band.

Question Question 6

Multiple Choice | Topic: Intrusion Prevention and Antivirus

Which are antivirus detection methods? (Choose all that apply)

- A. Signature-based
- B. Heuristic
- C. Sandboxing
- D. Behavioral analysis

Correct Answer: A,B,C,D**Concept:** Modern antivirus uses all these techniques.

Question Question 7

Multiple Choice | Topic: Intrusion Prevention and Antivirus

Which of the following relate to False? (Choose all that apply)

- A. False positive
- B. False negative

Correct Answer: A,B**Concept:** These concepts all relate to False in Intrusion Prevention and Antivirus.

64 Intrusion Prevention

Question Question 1

Multiple Choice | Topic: Intrusion Prevention

Which are common IPS actions? (Choose all that apply)

- A. Block
- B. Permit
- C. Reset connection
- D. Alert/log

Correct Answer: A,C,D**Concept:** IPS can block malicious traffic, reset connections, and alert/log events.

65 Antivirus

Question Question 1

Multiple Choice | Topic: Antivirus

Which are antivirus detection techniques? (Choose all that apply)

- A. Signature-based
- B. Heuristic
- C. Sandboxing
- D. Behavioral analysis

Correct Answer: A,B,C,D**Concept:** Modern antivirus uses signatures, heuristics, sandboxing, and behavioral analysis.

66 AAA and User Authentication

Question Question 1

Multiple Choice | Topic: AAA and User Authentication

Which are AAA functions? (Choose all that apply)

- A. Authentication
- B. Authorization
- C. Accounting
- D. Alerting

Correct Answer: A,B,C**Concept:** AAA provides authentication, authorization, and accounting.

Question Question 2

Multiple Choice | Topic: AAA and User Authentication

Which can be authentication sources on Huawei firewalls? (Choose all that apply)

- A. Local database
- B. RADIUS server
- C. HWTACACS server
- D. LDAP server

Correct Answer: A,B,C,D**Concept:** Firewalls can authenticate against local, RADIUS, HWTACACS, and LDAP sources.

Question Question 3

Multiple Choice | Topic: AAA and User Authentication

Which are differences between RADIUS and TACACS+? (Choose all that apply)

- A. RADIUS uses UDP; TACACS+ uses TCP
- B. RADIUS combines auth/authz; TACACS+ separates them
- C. RADIUS encrypts only password; TACACS+ encrypts entire payload
- D. RADIUS is Cisco proprietary

Correct Answer: A,B,C**Concept:** RADIUS uses UDP and combines auth/authz; TACACS+ uses TCP and separates AAA; only the password is encrypted in RADIUS.

Question Question 4

Multiple Choice | Topic: AAA and User Authentication

Which are AAA components? (Choose all that apply)

- A. Authentication
- B. Authorization
- C. Accounting
- D. Auditing

Correct Answer: A,B,C**Concept:** AAA consists of authentication, authorization, and accounting.

Question Question 5

Multiple Choice | Topic: AAA and User Authentication

Which can authenticate users on a Huawei firewall? (Choose all that apply)

- A. Local database
- B. RADIUS
- C. HWTACACS
- D. LDAP

Correct Answer: A,B,C,D**Concept:** Firewalls support all these authentication sources.

67 AAA Basics

Question Question 1

Multiple Choice | Topic: AAA Basics

Which are components of AAA? (Choose all that apply)

- A. Authentication
- B. Authorization
- C. Accounting
- D. Auditing

Correct Answer: A,B,C**Concept:** AAA consists of authentication, authorization, and accounting.

68 Firewall User Authentication

Question Question 1

Multiple Choice | Topic: Firewall User Authentication

Which are common firewall authentication methods? (Choose all that apply)

- A. Local authentication
- B. RADIUS
- C. LDAP
- D. HWTACACS

Correct Answer: A,B,C,D**Concept:** Firewalls support local, RADIUS, LDAP, and HWTACACS authentication.

69 Cryptography and PKI

Question Question 1

Multiple Choice | Topic: Cryptography and PKI

Which are asymmetric algorithms? (Choose all that apply)

- A. RSA
- B. ECC
- C. DSA
- D. AES

Correct Answer: A,B,C

Concept: RSA, ECC, and DSA are asymmetric; AES is symmetric.

Question Question 2

Multiple Choice | Topic: Cryptography and PKI

Which are block cipher modes? (Choose all that apply)

- A. ECB
- B. CBC
- C. CTR
- D. GCM

Correct Answer: A,B,C,D

Concept: ECB, CBC, CTR, and GCM are all AES block cipher modes.

Question Question 3

Multiple Choice | Topic: Cryptography and PKI

Which are secure hash algorithms? (Choose all that apply)

- A. SHA-256
- B. SHA-3
- C. BLAKE2
- D. MD5

Correct Answer: A,B,C

Concept: SHA-256, SHA-3, and BLAKE2 are considered secure; MD5 is broken.

Question Question 4

Multiple Choice | Topic: Cryptography and PKI

Which are PKI trust models? (Choose all that apply)

- A. Single CA
- B. Hierarchical CA
- C. Cross-certification
- D. Bridge CA

Correct Answer: A,B,C,D

Concept: PKI can use single, hierarchical, cross-certified, or bridge CA models.

Question Question 5

Multiple Choice | Topic: Cryptography and PKI

Which fields are typically found in an X.509 certificate? (Choose all that apply)

- A. Subject
- B. Issuer
- C. Public key
- D. Validity period

Correct Answer: A,B,C,D

Concept: X.509 certificates contain subject, issuer, public key, validity, serial number, and signature.

Question Question 6

Multiple Choice | Topic: Cryptography and PKI

Which are symmetric algorithms? (Choose all that apply)

- A. AES
- B. DES
- C. 3DES
- D. RSA

Correct Answer: A,B,C

Concept: AES, DES, and 3DES are symmetric; RSA is asymmetric.

Question Question 7

Multiple Choice | Topic: Cryptography and PKI

Which are hash algorithms? (Choose all that apply)

- A. MD5
- B. SHA-256
- C. SHA-1
- D. SHA-3

Correct Answer: A,B,C,D

Concept: These are all hash algorithms, though MD5 and SHA-1 are deprecated for security use.

Question Question 8

Multiple Choice | Topic: Cryptography and PKI

Which are PKI components? (Choose all that apply)

- A. CA
- B. RA
- C. Repository
- D. CRL/OCSP

Correct Answer: A,B,C,D

Concept: PKI includes certificate authorities, registration authorities, repositories, and revocation systems.

70 Cryptography Basics

Question Question 1

Multiple Choice | Topic: Cryptography Basics

Which are symmetric encryption algorithms? (Choose all that apply)

- A. AES
- B. DES
- C. 3DES
- D. RSA

Correct Answer: A,B,C

Concept: AES, DES, and 3DES are symmetric; RSA is asymmetric.

71 PKI

Question Question 1

Multiple Choice | Topic: PKI

Which are core PKI components? (Choose all that apply)

- A. Certificate Authority
- B. Registration Authority
- C. Certificate repository
- D. CRL/OCSP

Correct Answer: A,B,C,D

Concept: PKI includes CAs, RAs, repositories, and revocation mechanisms such as CRL and OCSP.

72 VPN

Question Question 1

Multiple Choice | Topic: VPN

Which are benefits of using a VPN? (Choose all that apply)

- A. Confidentiality
- B. Data integrity
- C. Authentication
- D. Secure remote access

Correct Answer: A,B,C,D

Concept: VPNs provide confidentiality, integrity, authentication, and secure remote access.

Question Question 2

Multiple Choice | Topic: VPN

Which are IPsec security services? (Choose all that apply)

- A. Confidentiality
- B. Integrity
- C. Authentication
- D. Anti-replay

Correct Answer: A,B,C,D

Concept: IPsec provides confidentiality, integrity, authentication, and anti-replay protection.

Question Question 3

Multiple Choice | Topic: VPN

Which are phases of IKEv1? (Choose all that apply)

- A. Phase 1
- B. Phase 2
- C. Phase 3
- D. Phase 1.5

Correct Answer: A,B

Concept: IKEv1 has two phases: Phase 1 (IKE SA) and Phase 2 (IPsec SA).

Question Question 4

Multiple Choice | Topic: VPN

Which protocols operate at the IP layer? (Choose all that apply)

- A. AH
- B. ESP
- C. GRE
- D. L2TP

Correct Answer: A,B

Concept: AH and ESP are IP-layer protocols; GRE is also IP-layer but L2TP is not.

Question Question 5

Multiple Choice | Topic: VPN

Which are VPN types? (Choose all that apply)

- A. Site-to-site
- B. Remote-access
- C. Intranet
- D. Extranet

Correct Answer: A,B,C,D

Concept: VPNs can be categorized by these deployment models.

Question Question 6

Multiple Choice | Topic: VPN

Which are IPsec protocols? (Choose all that apply)

- A. AH
- B. ESP
- C. IKE
- D. GRE

Correct Answer: A,B,C

Concept: AH, ESP, and IKE are IPsec protocols; GRE is separate.

Question Question 7

Multiple Choice | Topic: VPN

Which are SSL VPN access modes? (Choose all that apply)

- A. Web proxy
- B. File sharing
- C. Network extension
- D. Port forwarding

Correct Answer: A,B,C,D

Concept: SSL VPN commonly supports all these modes.

Question Question 8

Multiple Choice | Topic: VPN

Which of the following relate to GRE? (Choose all that apply)

- A. GRE
- B. GRE

Correct Answer: A,B

Concept: These concepts all relate to GRE in VPN.

Question Question 9

Multiple Choice | Topic: VPN

Which of the following relate to L2TP? (Choose all that apply)

- A. L2TP
- B. L2TP over IPsec

Correct Answer: A,B

Concept: These concepts all relate to L2TP in VPN.

73 VPN Overview

Question Question 1

Multiple Choice | Topic: VPN Overview

Which are common VPN types? (Choose all that apply)

- A. Site-to-site VPN
- B. Remote-access VPN
- C. Intranet VPN
- D. Extranet VPN

Correct Answer: A,B,C,D

Concept: VPNs can be site-to-site, remote-access, intranet, or extranet based on deployment.

74 IPsec VPN

Question Question 1

Multiple Choice | Topic: IPsec VPN

Which are IPsec protocols? (Choose all that apply)

- A. AH
- B. ESP
- C. IKE
- D. GRE

Correct Answer: A,B,C

Concept: AH, ESP, and IKE are part of IPsec; GRE is a separate tunneling protocol.

75 SSL VPN

Question Question 1

Multiple Choice | Topic: SSL VPN

Which are common SSL VPN access modes? (Choose all that apply)

- A. Web proxy
- B. File sharing
- C. Network extension
- D. Port forwarding

Correct Answer: A,B,C,D

Concept: SSL VPN can offer web proxy, file sharing, network extension, and port forwarding.

PART III

True / False Questions

76 Network Security Concepts

Question Question 1

True / False | Topic: Network Security Concepts

Network security in the narrow sense only refers to national cybersecurity laws.

- A. True
- B. False

Correct Answer: False

Concept: Narrow-sense network security covers vendor solutions and enterprise network protection, not only national laws.

Question Question 2

True / False | Topic: Network Security Concepts

The CIA triad includes confidentiality, integrity, and availability.

- A. True
- B. False

Correct Answer: True

Concept: Confidentiality, integrity, and availability are the three foundational goals of information security.

Question Question 3

True / False | Topic: Network Security Concepts

Building network security helps enterprises reduce losses from attacks.

- A. True
- B. False

Correct Answer: True

Concept: Security controls reduce the impact of breaches, downtime, and data loss, thereby lowering financial losses.

Question Question 4

True / False | Topic: Network Security Concepts

Non-repudiation ensures that senders or receivers cannot deny their actions.

- A. True
- B. False

Correct Answer: True

Concept: Non-repudiation provides evidence to prevent denial of participation.

Question Question 5

True / False | Topic: Network Security Concepts

Availability means anyone can access the system at any time.

- A. True
- B. False

Correct Answer: False

Concept: Availability means authorized users can access information when required, not unlimited access.

Question Question 6

True / False | Topic: Network Security Concepts

Zero trust means everything inside the network is trusted by default.

- A. True
- B. False

Correct Answer: False

Concept: Zero trust never assumes trust based on network location and verifies every request.

Question Question 7

True / False | Topic: Network Security Concepts

Integrity ensures that data is not tampered with during transmission.

- A. True
- B. False

Correct Answer: True**Concept:** Integrity protects against unauthorized modification or destruction of data.

Question Question 8

True / False | Topic: Network Security Concepts

Network security broadly refers to cyberspace security at a national level.

- A. True
- B. False

Correct Answer: True**Concept:** Broad network security covers national cyberspace protection.

Question Question 9

True / False | Topic: Network Security Concepts

The CIA triad excludes availability.

- A. True
- B. False

Correct Answer: False**Concept:** The CIA triad includes confidentiality, integrity, and availability.

Question Question 10

True / False | Topic: Network Security Concepts

Risk is the combination of threat likelihood and impact.

- A. True
- B. False

Correct Answer: True**Concept:** Risk assesses probability and consequence of security incidents.

Question Question 11

True / False | Topic: Network Security Concepts

Broad network security means that is synonymous with national cyberspace security.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Broad network security.

Question Question 12

True / False | Topic: Network Security Concepts

Broad network security is unrelated to is synonymous with national cyberspace security.

- A. True
- B. False

Correct Answer: False**Concept:** Broad network security is directly related to is synonymous with national cyberspace security.

Question Question 13

True / False | Topic: Network Security Concepts

Narrow network security means that focuses on enterprise network protection solutions.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Narrow network security.

Question Question 14

True / False | Topic: Network Security Concepts

Narrow network security is unrelated to focuses on enterprise network protection solutions.

- A. True
- B. False

Correct Answer: False**Concept:** Narrow network security is directly related to focuses on enterprise network protection solutions.

Question Question 15

True / False | Topic: Network Security Concepts

CIA triad means that consists of confidentiality, integrity, and availability.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes CIA triad.

Question Question 16

True / False | Topic: Network Security Concepts

CIA triad is unrelated to consists of confidentiality, integrity, and availability.

- A. True
- B. False

Correct Answer: False**Concept:** CIA triad is directly related to consists of confidentiality, integrity, and availability.

Question Question 17

True / False | Topic: Network Security Concepts

Information assurance period means that added controllability and non-repudiation.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Information assurance period.

Question Question 18

True / False | Topic: Network Security Concepts

Information assurance period is unrelated to added controllability and non-repudiation.

- A. True
- B. False

Correct Answer: False**Concept:** Information assurance period is directly related to added controllability and non-repudiation.

Question Question 19

True / False | Topic: Network Security Concepts

Cyberspace security means that protects facilities, data, users, and operations.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Cyberspace security.

Question Question 20

True / False | Topic: Network Security Concepts

Cyberspace security is unrelated to protects facilities, data, users, and operations.

- A. True
- B. False

Correct Answer: False**Concept:** Cyberspace security is directly related to protects facilities, data, users, and operations.

Question Question 21

True / False | Topic: Network Security Concepts

Confidentiality means that ensures information is accessible only to authorized users.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Confidentiality.

Question Question 22

True / False | Topic: Network Security Concepts

Confidentiality is unrelated to ensures information is accessible only to authorized users.

- A. True
- B. False

Correct Answer: False**Concept:** Confidentiality is directly related to ensures information is accessible only to authorized users.

Question Question 23

True / False | Topic: Network Security Concepts

Integrity means that ensures data is not tampered with during transmission or storage.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Integrity.

Question Question 24

True / False | Topic: Network Security Concepts

Integrity is unrelated to ensures data is not tampered with during transmission or storage.

- A. True
- B. False

Correct Answer: False**Concept:** Integrity is directly related to ensures data is not tampered with during transmission or storage.

Question Question 25

True / False | Topic: Network Security Concepts

Availability means that ensures authorized users can access information when needed.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Availability.

Question Question 26

True / False | Topic: Network Security Concepts

Availability is unrelated to ensures authorized users can access information when needed.

- A. True
- B. False

Correct Answer: False**Concept:** Availability is directly related to ensures authorized users can access information when needed.

Question Question 27

True / False | Topic: Network Security Concepts

Controllability means that enables monitoring and control of information and systems.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Controllability.

Question Question 28

True / False | Topic: Network Security Concepts

Controllability is unrelated to enables monitoring and control of information and systems.

- A. True
- B. False

Correct Answer: False**Concept:** Controllability is directly related to enables monitoring and control of information and systems.

Question Question 29

True / False | Topic: Network Security Concepts

Non-repudiation means that prevents senders or receivers from denying actions.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Non-repudiation.

Question Question 30

True / False | Topic: Network Security Concepts

Non-repudiation is unrelated to prevents senders or receivers from denying actions.

- A. True
- B. False

Correct Answer: False**Concept:** Non-repudiation is directly related to prevents senders or receivers from denying actions.

Question Question 31

True / False | Topic: Network Security Concepts

Risk means that is the combination of threat likelihood and impact.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Risk.

Question Question 32

True / False | Topic: Network Security Concepts

Risk is unrelated to is the combination of threat likelihood and impact.

- A. True
- B. False

Correct Answer: False**Concept:** Risk is directly related to is the combination of threat likelihood and impact.

Question Question 33

True / False | Topic: Network Security Concepts

Vulnerability means that is a weakness that can be exploited by threats.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Vulnerability.

Question Question 34

True / False | Topic: Network Security Concepts

Vulnerability is unrelated to is a weakness that can be exploited by threats.

- A. True
- B. False

Correct Answer: False**Concept:** Vulnerability is directly related to is a weakness that can be exploited by threats.

Question Question 35

True / False | Topic: Network Security Concepts

Threat means that is a potential danger to an asset.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Threat.

Question Question 36

True / False | Topic: Network Security Concepts

Threat is unrelated to is a potential danger to an asset.

- A. True
- B. False

Correct Answer: False**Concept:** Threat is directly related to is a potential danger to an asset.

Question Question 37

True / False | Topic: Network Security Concepts

Exploit means that is a method that takes advantage of a vulnerability.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Exploit.

Question Question 38

True / False | Topic: Network Security Concepts

Exploit is unrelated to is a method that takes advantage of a vulnerability.

- A. True
- B. False

Correct Answer: False**Concept:** Exploit is directly related to is a method that takes advantage of a vulnerability.

Question Question 39

True / False | Topic: Network Security Concepts

Future security trend means that includes zero-trust architecture.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Future security trend.

Question Question 40

True / False | Topic: Network Security Concepts

Future security trend is unrelated to includes zero-trust architecture.

- A. True
- B. False

Correct Answer: False**Concept:** Future security trend is directly related to includes zero-trust architecture.

Question Question 41

True / False | Topic: Network Security Concepts

AI and machine learning means that help detect unknown threats and anomalies.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes AI and machine learning.

Question Question 42

True / False | Topic: Network Security Concepts

AI and machine learning is unrelated to help detect unknown threats and anomalies.

- A. True
- B. False

Correct Answer: False**Concept:** AI and machine learning is directly related to help detect unknown threats and anomalies.

77 Network Security Standards

Question Question 1

True / False | Topic: Network Security Standards

ISO 27002 can be used to certify an enterprise information security system.

- A. True
- B. False

Correct Answer: False**Concept:** ISO 27001 is the certification standard; ISO 27002 is a code of practice with controls.

Question Question 2

True / False | Topic: Network Security Standards

Classified Protection 2.0 defines protection levels from Level 1 to Level 5.

- A. True
- B. False

Correct Answer: True**Concept:** Classified Protection 2.0 uses five levels, with higher levels requiring stronger controls and oversight.

Question Question 3

True / False | Topic: Network Security Standards

Network operators can always determine the final protection level on their own without review.

- A. True
- B. False

Correct Answer: False**Concept:** For Level 2 and above, expert review and approval are required; only Level 1 can be determined by the operator.

Question Question 4

True / False | Topic: Network Security Standards

ISO 27001 provides a list of security controls but is not certifiable.

- A. True
- B. False

Correct Answer: False**Concept:** ISO 27001 is the certifiable standard specifying ISMS requirements; ISO 27002 provides controls.

Question Question 5

True / False | Topic: Network Security Standards

Classified Protection 2.0 requires security and informatization to be planned together.

- A. True
- B. False

Correct Answer: True**Concept:** The framework emphasizes synchronized planning, construction, and use of security.

Question Question 6

True / False | Topic: Network Security Standards

Level assessment is part of the Classified Protection lifecycle.

- A. True
- B. False

Correct Answer: True**Concept:** After construction, a level assessment verifies that protection measures meet the required level.

Question Question 7

True / False | Topic: Network Security Standards

ISO 27001 is certifiable.

- A. True
- B. False

Correct Answer: True**Concept:** Organizations can be certified against ISO 27001 requirements.

Question Question 8

True / False | Topic: Network Security Standards

Classified Protection 2.0 has only three levels.

- A. True
- B. False

Correct Answer: False**Concept:** Classified Protection 2.0 defines five levels.

Question Question 9

True / False | Topic: Network Security Standards

ISO 27001 means that specifies certifiable ISMS requirements.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes ISO 27001.

Question Question 10

True / False | Topic: Network Security Standards

ISO 27001 is unrelated to specifies certifiable ISMS requirements.

- A. True
- B. False

Correct Answer: False

Concept: ISO 27001 is directly related to specifies certifiable ISMS requirements.

Question Question 11

True / False | Topic: Network Security Standards

ISO 27002 means that provides a code of practice for security controls.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes ISO 27002.

Question Question 12

True / False | Topic: Network Security Standards

ISO 27002 is unrelated to provides a code of practice for security controls.

- A. True
- B. False

Correct Answer: False

Concept: ISO 27002 is directly related to provides a code of practice for security controls.

Question Question 13

True / False | Topic: Network Security Standards

PDCA means that is the continuous improvement methodology used in ISO 27001.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes PDCA.

Question Question 14

True / False | Topic: Network Security Standards

PDCA is unrelated to is the continuous improvement methodology used in ISO 27001.

- A. True
- B. False

Correct Answer: False

Concept: PDCA is directly related to is the continuous improvement methodology used in ISO 27001.

Question Question 15

True / False | Topic: Network Security Standards

Classified Protection 2.0 means that is a Chinese national cybersecurity framework.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Classified Protection 2.0.

Question Question 16

True / False | Topic: Network Security Standards

Classified Protection 2.0 is unrelated to is a Chinese national cybersecurity framework.

- A. True
- B. False

Correct Answer: False**Concept:** Classified Protection 2.0 is directly related to is a Chinese national cybersecurity framework.

Question Question 17

True / False | Topic: Network Security Standards

Classified Protection levels means that range from Level 1 to Level 5.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Classified Protection levels.

Question Question 18

True / False | Topic: Network Security Standards

Classified Protection levels is unrelated to range from Level 1 to Level 5.

- A. True
- B. False

Correct Answer: False**Concept:** Classified Protection levels is directly related to range from Level 1 to Level 5.

Question Question 19

True / False | Topic: Network Security Standards

Level 2 and above means that require expert review and approval.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Level 2 and above.

Question Question 20

True / False | Topic: Network Security Standards

Level 2 and above is unrelated to require expert review and approval.

- A. True
- B. False

Correct Answer: False

Concept: Level 2 and above is directly related to require expert review and approval.

Question Question 21

True / False | Topic: Network Security Standards

Classified Protection lifecycle means that includes classification, filing, construction, assessment, and supervision.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Classified Protection lifecycle.

Question Question 22

True / False | Topic: Network Security Standards

Classified Protection lifecycle is unrelated to includes classification, filing, construction, assessment, and supervision.

- A. True
- B. False

Correct Answer: False

Concept: Classified Protection lifecycle is directly related to includes classification, filing, construction, assessment, and supervision.

Question Question 23

True / False | Topic: Network Security Standards

Security synchronization means that means planning security with information system construction.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Security synchronization.

Question Question 24

True / False | Topic: Network Security Standards

Security synchronization is unrelated to means planning security with information system construction.

- A. True
- B. False

Correct Answer: False

Concept: Security synchronization is directly related to means planning security with information system construction.

Question Question 25

True / False | Topic: Network Security Standards

ISO 27001:2022 means that organizes controls into four themes.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes ISO 27001:2022.

Question Question 26

True / False | Topic: Network Security Standards

ISO 27001:2022 is unrelated to organizes controls into four themes.

- A. True
- B. False

Correct Answer: False**Concept:** ISO 27001:2022 is directly related to organizes controls into four themes.

Question Question 27

True / False | Topic: Network Security Standards

Cybersecurity Law means that underpins Classified Protection in China.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Cybersecurity Law.

Question Question 28

True / False | Topic: Network Security Standards

Cybersecurity Law is unrelated to underpins Classified Protection in China.

- A. True
- B. False

Correct Answer: False**Concept:** Cybersecurity Law is directly related to underpins Classified Protection in China.

78 Network Fundamentals

Question Question 1

True / False | Topic: Network Fundamentals

The OSI model was developed before the TCP/IP model.

- A. True
- B. False

Correct Answer: False**Concept:** TCP/IP was developed earlier and became the de facto standard; OSI was later as a reference model.

Question Question 2

True / False | Topic: Network Fundamentals

DNS always uses TCP port 53.

- A. True
- B. False

Correct Answer: False

Concept: DNS primarily uses UDP port 53; TCP port 53 is used for zone transfers and large responses.

Question Question 3

True / False | Topic: Network Fundamentals

A switch forwards packets based on IP addresses.

- A. True
- B. False

Correct Answer: False

Concept: A switch forwards frames based on MAC addresses; routers forward packets based on IP addresses.

Question Question 4

True / False | Topic: Network Fundamentals

IPv4 addresses are 32 bits long.

- A. True
- B. False

Correct Answer: True

Concept: IPv4 addresses are 32-bit binary numbers, usually written in dotted-decimal notation.

Question Question 5

True / False | Topic: Network Fundamentals

ICMP is a transport-layer protocol.

- A. True
- B. False

Correct Answer: False

Concept: ICMP is a network-layer protocol used for diagnostics and error reporting.

Question Question 6

True / False | Topic: Network Fundamentals

TCP provides connection-oriented reliable delivery.

- A. True
- B. False

Correct Answer: True

Concept: TCP establishes connections and ensures reliable, ordered delivery.

Question Question 7

True / False | Topic: Network Fundamentals

A hub creates separate collision domains per port.

- A. True
- B. False

Correct Answer: False

Concept: A hub shares a single collision domain; switches create separate collision domains.

Question Question 8

True / False | Topic: Network Fundamentals

IPv6 addresses are 128 bits long.

- A. True
- B. False

Correct Answer: True

Concept: IPv6 addresses are 128-bit binary numbers.

Question Question 9

True / False | Topic: Network Fundamentals

OSI model means that has seven layers.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes OSI model.

Question Question 10

True / False | Topic: Network Fundamentals

OSI model is unrelated to has seven layers.

- A. True
- B. False

Correct Answer: False

Concept: OSI model is directly related to has seven layers.

Question Question 11

True / False | Topic: Network Fundamentals

TCP/IP standard model means that has four layers.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes TCP/IP standard model.

Question Question 12

True / False | Topic: Network Fundamentals

TCP/IP standard model is unrelated to has four layers.

- A. True
- B. False

Correct Answer: False**Concept:** TCP/IP standard model is directly related to has four layers.

Question Question 13

True / False | Topic: Network Fundamentals

TCP/IP equivalent model means that has five layers.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes TCP/IP equivalent model.

Question Question 14

True / False | Topic: Network Fundamentals

TCP/IP equivalent model is unrelated to has five layers.

- A. True
- B. False

Correct Answer: False**Concept:** TCP/IP equivalent model is directly related to has five layers.

Question Question 15

True / False | Topic: Network Fundamentals

Physical layer means that transmits raw bits over a medium.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Physical layer.

Question Question 16

True / False | Topic: Network Fundamentals

Physical layer is unrelated to transmits raw bits over a medium.

- A. True
- B. False

Correct Answer: False**Concept:** Physical layer is directly related to transmits raw bits over a medium.

Question Question 17

True / False | Topic: Network Fundamentals

Data link layer means that handles framing and MAC addressing.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Data link layer.

Question Question 18

True / False | Topic: Network Fundamentals

Data link layer is unrelated to handles framing and MAC addressing.

- A. True
- B. False

Correct Answer: False**Concept:** Data link layer is directly related to handles framing and MAC addressing.

Question Question 19

True / False | Topic: Network Fundamentals

Network layer means that handles logical addressing and routing.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Network layer.

Question Question 20

True / False | Topic: Network Fundamentals

Network layer is unrelated to handles logical addressing and routing.

- A. True
- B. False

Correct Answer: False**Concept:** Network layer is directly related to handles logical addressing and routing.

Question Question 21

True / False | Topic: Network Fundamentals

Transport layer means that provides end-to-end communication and reliability.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Transport layer.

Question Question 22

True / False | Topic: Network Fundamentals

Transport layer is unrelated to provides end-to-end communication and reliability.

- A. True
- B. False

Correct Answer: False**Concept:** Transport layer is directly related to provides end-to-end communication and reliability.

Question Question 23

True / False | Topic: Network Fundamentals

Session layer means that manages application dialogs.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Session layer.

Question Question 24

True / False | Topic: Network Fundamentals

Session layer is unrelated to manages application dialogs.

- A. True
- B. False

Correct Answer: False**Concept:** Session layer is directly related to manages application dialogs.

Question Question 25

True / False | Topic: Network Fundamentals

Presentation layer means that handles encryption and data format conversion.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Presentation layer.

Question Question 26

True / False | Topic: Network Fundamentals

Presentation layer is unrelated to handles encryption and data format conversion.

- A. True
- B. False

Correct Answer: False**Concept:** Presentation layer is directly related to handles encryption and data format conversion.

Question Question 27

True / False | Topic: Network Fundamentals

Application layer means that provides network services to applications.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Application layer.

Question Question 28

True / False | Topic: Network Fundamentals

Application layer is unrelated to provides network services to applications.

- A. True
- B. False

Correct Answer: False**Concept:** Application layer is directly related to provides network services to applications.

Question Question 29

True / False | Topic: Network Fundamentals

TCP means that is connection-oriented and reliable.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes TCP.

Question Question 30

True / False | Topic: Network Fundamentals

TCP is unrelated to is connection-oriented and reliable.

- A. True
- B. False

Correct Answer: False**Concept:** TCP is directly related to is connection-oriented and reliable.

Question Question 31

True / False | Topic: Network Fundamentals

UDP means that is connectionless and has low overhead.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes UDP.

Question Question 32

True / False | Topic: Network Fundamentals

UDP is unrelated to is connectionless and has low overhead.

- A. True
- B. False

Correct Answer: False**Concept:** UDP is directly related to is connectionless and has low overhead.

Question Question 33

True / False | Topic: Network Fundamentals

FTP means that uses TCP ports 20 and 21.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes FTP.

Question Question 34

True / False | Topic: Network Fundamentals

FTP is unrelated to uses TCP ports 20 and 21.

- A. True
- B. False

Correct Answer: False**Concept:** FTP is directly related to uses TCP ports 20 and 21.

Question Question 35

True / False | Topic: Network Fundamentals

HTTP means that uses TCP port 80.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes HTTP.

Question Question 36

True / False | Topic: Network Fundamentals

HTTP is unrelated to uses TCP port 80.

- A. True
- B. False

Correct Answer: False**Concept:** HTTP is directly related to uses TCP port 80.

Question Question 37

True / False | Topic: Network Fundamentals

HTTPS means that uses TCP port 443.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes HTTPS.

Question Question 38

True / False | Topic: Network Fundamentals

HTTPS is unrelated to uses TCP port 443.

- A. True
- B. False

Correct Answer: False**Concept:** HTTPS is directly related to uses TCP port 443.

Question Question 39

True / False | Topic: Network Fundamentals

SSH means that uses TCP port 22.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes SSH.

Question Question 40

True / False | Topic: Network Fundamentals

SSH is unrelated to uses TCP port 22.

- A. True
- B. False

Correct Answer: False**Concept:** SSH is directly related to uses TCP port 22.

Question Question 41

True / False | Topic: Network Fundamentals

Telnet means that uses TCP port 23 and is insecure.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Telnet.

Question Question 42

True / False | Topic: Network Fundamentals

Telnet is unrelated to uses TCP port 23 and is insecure.

- A. True
- B. False

Correct Answer: False**Concept:** Telnet is directly related to uses TCP port 23 and is insecure.

Question Question 43

True / False | Topic: Network Fundamentals

DNS means that primarily uses UDP port 53.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes DNS.

Question Question 44

True / False | Topic: Network Fundamentals

DNS is unrelated to primarily uses UDP port 53.

- A. True
- B. False

Correct Answer: False**Concept:** DNS is directly related to primarily uses UDP port 53.

Question Question 45

True / False | Topic: Network Fundamentals

SMTP means that uses TCP port 25.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes SMTP.

Question Question 46

True / False | Topic: Network Fundamentals

SMTP is unrelated to uses TCP port 25.

- A. True
- B. False

Correct Answer: False**Concept:** SMTP is directly related to uses TCP port 25.

Question Question 47

True / False | Topic: Network Fundamentals

DHCP means that dynamically assigns IP addresses.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes DHCP.

Question Question 48

True / False | Topic: Network Fundamentals

DHCP is unrelated to dynamically assigns IP addresses.

- A. True
- B. False

Correct Answer: False**Concept:** DHCP is directly related to dynamically assigns IP addresses.

Question Question 49

True / False | Topic: Network Fundamentals

ARP means that maps IP addresses to MAC addresses.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes ARP.

Question Question 50

True / False | Topic: Network Fundamentals

ARP is unrelated to maps IP addresses to MAC addresses.

- A. True
- B. False

Correct Answer: False**Concept:** ARP is directly related to maps IP addresses to MAC addresses.

Question Question 51

True / False | Topic: Network Fundamentals

ICMP means that provides error reporting and diagnostics.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes ICMP.

Question Question 52

True / False | Topic: Network Fundamentals

ICMP is unrelated to provides error reporting and diagnostics.

- A. True
- B. False

Correct Answer: False**Concept:** ICMP is directly related to provides error reporting and diagnostics.

Question Question 53

True / False | Topic: Network Fundamentals

IPv4 address means that is 32 bits long.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes IPv4 address.

Question Question 54

True / False | Topic: Network Fundamentals

IPv4 address is unrelated to is 32 bits long.

- A. True
- B. False

Correct Answer: False**Concept:** IPv4 address is directly related to is 32 bits long.

Question Question 55

True / False | Topic: Network Fundamentals

IPv6 address means that is 128 bits long.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes IPv6 address.

Question Question 56

True / False | Topic: Network Fundamentals

IPv6 address is unrelated to is 128 bits long.

- A. True
- B. False

Correct Answer: False**Concept:** IPv6 address is directly related to is 128 bits long.

Question Question 57

True / False | Topic: Network Fundamentals

MAC address means that is a Layer 2 hardware address.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes MAC address.

Question Question 58

True / False | Topic: Network Fundamentals

MAC address is unrelated to is a Layer 2 hardware address.

- A. True
- B. False

Correct Answer: False**Concept:** MAC address is directly related to is a Layer 2 hardware address.

Question Question 59

True / False | Topic: Network Fundamentals

Switch means that operates at Layer 2 and forwards frames by MAC.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Switch.

Question Question 60

True / False | Topic: Network Fundamentals

Switch is unrelated to operates at Layer 2 and forwards frames by MAC.

- A. True
- B. False

Correct Answer: False**Concept:** Switch is directly related to operates at Layer 2 and forwards frames by MAC.

Question Question 61

True / False | Topic: Network Fundamentals

Router means that operates at Layer 3 and forwards packets by IP.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Router.

Question Question 62

True / False | Topic: Network Fundamentals

Router is unrelated to operates at Layer 3 and forwards packets by IP.

- A. True
- B. False

Correct Answer: False**Concept:** Router is directly related to operates at Layer 3 and forwards packets by IP.

Question Question 63

True / False | Topic: Network Fundamentals

Hub means that operates at Layer 1 and repeats signals.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Hub.

Question Question 64

True / False | Topic: Network Fundamentals

Hub is unrelated to operates at Layer 1 and repeats signals.

- A. True
- B. False

Correct Answer: False**Concept:** Hub is directly related to operates at Layer 1 and repeats signals.

Question Question 65

True / False | Topic: Network Fundamentals

VLAN means that logically segments a network into broadcast domains.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes VLAN.

Question Question 66

True / False | Topic: Network Fundamentals

VLAN is unrelated to logically segments a network into broadcast domains.

- A. True
- B. False

Correct Answer: False**Concept:** VLAN is directly related to logically segments a network into broadcast domains.

Question Question 67

True / False | Topic: Network Fundamentals

STP means that prevents Layer 2 loops.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes STP.

Question Question 68

True / False | Topic: Network Fundamentals

STP is unrelated to prevents Layer 2 loops.

- A. True
- B. False

Correct Answer: False**Concept:** STP is directly related to prevents Layer 2 loops.

Question Question 69

True / False | Topic: Network Fundamentals

RFC 1918 means that defines private IPv4 address ranges.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes RFC 1918.

Question Question 70

True / False | Topic: Network Fundamentals

RFC 1918 is unrelated to defines private IPv4 address ranges.

- A. True
- B. False

Correct Answer: False**Concept:** RFC 1918 is directly related to defines private IPv4 address ranges.

79 OSI and TCP/IP Models

Question Question 1

True / False | Topic: OSI and TCP/IP Models

The TCP/IP model combines the OSI session and presentation layers into the application layer.

- A. True
- B. False

Correct Answer: True**Concept:** The TCP/IP model does not have separate session and presentation layers; their functions are included in the application layer.

80 Network Protocols

Question Question 1

True / False | Topic: Network Protocols

UDP guarantees ordered delivery of packets.

- A. True
- B. False

Correct Answer: False**Concept:** UDP is connectionless and does not guarantee delivery, ordering, or duplicate protection.

81 Network Devices

Question Question 1

True / False | Topic: Network Devices

A router forwards frames based on MAC addresses.

- A. True
- B. False

Correct Answer: False**Concept:** Routers forward packets based on IP addresses; switches forward frames based on MAC addresses.

Question Question 2

True / False | Topic: Network Devices

A switch creates a separate collision domain for each port.

- A. True
- B. False

Correct Answer: True**Concept:** Each switch port is its own collision domain, reducing collisions compared to a hub.

82 Enterprise Security Threats

Question Question 1

True / False | Topic: Enterprise Security Threats

A worm requires a host program to spread.

- A. True
- B. False

Correct Answer: False**Concept:** Worms are standalone malware that can self-replicate; viruses require host programs.

Question Question 2

True / False | Topic: Enterprise Security Threats

Phishing attacks target human psychology rather than software vulnerabilities.

- A. True
- B. False

Correct Answer: True**Concept:** Phishing uses deception and social engineering to trick users.

Question Question 3

True / False | Topic: Enterprise Security Threats

Using HTTP instead of HTTPS improves confidentiality.

- A. True
- B. False

Correct Answer: False**Concept:** HTTP sends data in plaintext; HTTPS encrypts data to protect confidentiality.

Question Question 4

True / False | Topic: Enterprise Security Threats

The DMZ zone is typically more trusted than the Untrust zone but less trusted than the Trust zone.

- A. True
- B. False

Correct Answer: True**Concept:** DMZ has an intermediate trust level for public-facing servers.

Question Question 5

True / False | Topic: Enterprise Security Threats

Patch management helps eliminate known vulnerabilities on hosts.

- A. True
- B. False

Correct Answer: True**Concept:** Applying patches closes security holes that attackers could exploit.

Question Question 6

True / False | Topic: Enterprise Security Threats

All security threats originate from external attackers.

- A. True
- B. False

Correct Answer: False**Concept:** Threats can be external or internal, accidental or malicious.

Question Question 7

True / False | Topic: Enterprise Security Threats

Phishing is a type of social engineering.

- A. True
- B. False

Correct Answer: True**Concept:** Phishing manipulates users into revealing information.

Question Question 8

True / False | Topic: Enterprise Security Threats

A firewall at the network border can enforce zone-based policies.

- A. True
- B. False

Correct Answer: True**Concept:** Firewalls enforce policies between security zones.

Question Question 9

True / False | Topic: Enterprise Security Threats

Least privilege reduces the impact of compromised accounts.

- A. True
- B. False

Correct Answer: True**Concept:** Limiting access limits damage from compromised credentials.

Question Question 10

True / False | Topic: Enterprise Security Threats

Passive attack means that monitors traffic without altering it.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Passive attack.

Question Question 11

True / False | Topic: Enterprise Security Threats

Passive attack is unrelated to monitors traffic without altering it.

- A. True
- B. False

Correct Answer: False**Concept:** Passive attack is directly related to monitors traffic without altering it.

Question Question 12

True / False | Topic: Enterprise Security Threats

Active attack means that modifies or disrupts systems or data.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Active attack.

Question Question 13

True / False | Topic: Enterprise Security Threats

Active attack is unrelated to modifies or disrupts systems or data.

- A. True
- B. False

Correct Answer: False**Concept:** Active attack is directly related to modifies or disrupts systems or data.

Question Question 14

True / False | Topic: Enterprise Security Threats

Malware means that includes viruses, worms, Trojans, and ransomware.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Malware.

Question Question 15

True / False | Topic: Enterprise Security Threats

Malware is unrelated to includes viruses, worms, Trojans, and ransomware.

- A. True
- B. False

Correct Answer: False**Concept:** Malware is directly related to includes viruses, worms, Trojans, and ransomware.

Question Question 16

True / False | Topic: Enterprise Security Threats

Virus means that requires a host program to spread.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Virus.

Question Question 17

True / False | Topic: Enterprise Security Threats

Virus is unrelated to requires a host program to spread.

- A. True
- B. False

Correct Answer: False**Concept:** Virus is directly related to requires a host program to spread.

Question Question 18

True / False | Topic: Enterprise Security Threats

Worm means that can self-replicate across networks.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Worm.

Question Question 19

True / False | Topic: Enterprise Security Threats

Worm is unrelated to can self-replicate across networks.

- A. True
- B. False

Correct Answer: False**Concept:** Worm is directly related to can self-replicate across networks.

Question Question 20

True / False | Topic: Enterprise Security Threats

Trojan means that disguises itself as legitimate software.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Trojan.

Question Question 21

True / False | Topic: Enterprise Security Threats

Trojan is unrelated to disguises itself as legitimate software.

- A. True
- B. False

Correct Answer: False**Concept:** Trojan is directly related to disguises itself as legitimate software.

Question Question 22

True / False | Topic: Enterprise Security Threats

Ransomware means that encrypts data and demands payment.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Ransomware.

Question Question 23

True / False | Topic: Enterprise Security Threats

Ransomware is unrelated to encrypts data and demands payment.

- A. True
- B. False

Correct Answer: False**Concept:** Ransomware is directly related to encrypts data and demands payment.

Question Question 24

True / False | Topic: Enterprise Security Threats

Phishing means that uses deception to steal information.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Phishing.

Question Question 25

True / False | Topic: Enterprise Security Threats

Phishing is unrelated to uses deception to steal information.

- A. True
- B. False

Correct Answer: False**Concept:** Phishing is directly related to uses deception to steal information.

Question Question 26

True / False | Topic: Enterprise Security Threats

Social engineering means that manipulates people into breaking security rules.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Social engineering.

Question Question 27

True / False | Topic: Enterprise Security Threats

Social engineering is unrelated to manipulates people into breaking security rules.

- A. True
- B. False

Correct Answer: False

Concept: Social engineering is directly related to manipulates people into breaking security rules.

Question Question 28

True / False | Topic: Enterprise Security Threats

DoS attack means that aims to make a service unavailable.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes DoS attack.

Question Question 29

True / False | Topic: Enterprise Security Threats

DoS attack is unrelated to aims to make a service unavailable.

- A. True
- B. False

Correct Answer: False

Concept: DoS attack is directly related to aims to make a service unavailable.

Question Question 30

True / False | Topic: Enterprise Security Threats

DDoS attack means that uses many distributed sources to flood a target.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes DDoS attack.

Question Question 31

True / False | Topic: Enterprise Security Threats

DDoS attack is unrelated to uses many distributed sources to flood a target.

- A. True
- B. False

Correct Answer: False

Concept: DDoS attack is directly related to uses many distributed sources to flood a target.

Question Question 32

True / False | Topic: Enterprise Security Threats

Man-in-the-middle attack means that intercepts communication between two parties.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Man-in-the-middle attack.

Question Question 33

True / False | Topic: Enterprise Security Threats

Man-in-the-middle attack is unrelated to intercepts communication between two parties.

- A. True
- B. False

Correct Answer: False**Concept:** Man-in-the-middle attack is directly related to intercepts communication between two parties.

Question Question 34

True / False | Topic: Enterprise Security Threats

ARP spoofing means that forges ARP replies to redirect traffic.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes ARP spoofing.

Question Question 35

True / False | Topic: Enterprise Security Threats

ARP spoofing is unrelated to forges ARP replies to redirect traffic.

- A. True
- B. False

Correct Answer: False**Concept:** ARP spoofing is directly related to forges ARP replies to redirect traffic.

Question Question 36

True / False | Topic: Enterprise Security Threats

SYN flood means that sends many half-open TCP connections.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes SYN flood.

Question Question 37

True / False | Topic: Enterprise Security Threats

SYN flood is unrelated to sends many half-open TCP connections.

- A. True
- B. False

Correct Answer: False**Concept:** SYN flood is directly related to sends many half-open TCP connections.

Question Question 38

True / False | Topic: Enterprise Security Threats

Ping of death means that sends oversized ICMP packets.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Ping of death.

Question Question 39

True / False | Topic: Enterprise Security Threats

Ping of death is unrelated to sends oversized ICMP packets.

- A. True
- B. False

Correct Answer: False**Concept:** Ping of death is directly related to sends oversized ICMP packets.

Question Question 40

True / False | Topic: Enterprise Security Threats

Smurf attack means that uses broadcast pings to amplify traffic.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Smurf attack.

Question Question 41

True / False | Topic: Enterprise Security Threats

Smurf attack is unrelated to uses broadcast pings to amplify traffic.

- A. True
- B. False

Correct Answer: False**Concept:** Smurf attack is directly related to uses broadcast pings to amplify traffic.

Question Question 42

True / False | Topic: Enterprise Security Threats

Least privilege means that limits user access to the minimum necessary.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Least privilege.

Question Question 43

True / False | Topic: Enterprise Security Threats

Least privilege is unrelated to limits user access to the minimum necessary.

- A. True
- B. False

Correct Answer: False**Concept:** Least privilege is directly related to limits user access to the minimum necessary.

Question Question 44

True / False | Topic: Enterprise Security Threats

Patch management means that eliminates known vulnerabilities.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Patch management.

Question Question 45

True / False | Topic: Enterprise Security Threats

Patch management is unrelated to eliminates known vulnerabilities.

- A. True
- B. False

Correct Answer: False**Concept:** Patch management is directly related to eliminates known vulnerabilities.

Question Question 46

True / False | Topic: Enterprise Security Threats

SIEM means that centralizes log collection and analysis.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes SIEM.

Question Question 47

True / False | Topic: Enterprise Security Threats

SIEM is unrelated to centralizes log collection and analysis.

- A. True
- B. False

Correct Answer: False**Concept:** SIEM is directly related to centralizes log collection and analysis.

Question Question 48

True / False | Topic: Enterprise Security Threats

DMZ means that hosts public-facing servers between Internet and internal network.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes DMZ.

Question Question 49

True / False | Topic: Enterprise Security Threats

DMZ is unrelated to hosts public-facing servers between Internet and internal network.

- A. True
- B. False

Correct Answer: False**Concept:** DMZ is directly related to hosts public-facing servers between Internet and internal network.

Question Question 50

True / False | Topic: Enterprise Security Threats

Insider threat means that originates from within the organization.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Insider threat.

Question Question 51

True / False | Topic: Enterprise Security Threats

Insider threat is unrelated to originates from within the organization.

- A. True
- B. False

Correct Answer: False**Concept:** Insider threat is directly related to originates from within the organization.

83 Communication Network Security

Question Question 1

True / False | Topic: Communication Network Security

Encryption is an effective way to protect data confidentiality over public networks.

- A. True
- B. False

Correct Answer: True**Concept:** Encryption ensures that intercepted data cannot be read without the proper key.

84 Zone Border Security

Question Question 1

True / False | Topic: Zone Border Security

A firewall at the zone border can enforce security policies between trusted and untrusted networks.

- A. True
- B. False

Correct Answer: True**Concept:** Firewalls inspect and control traffic crossing security zone boundaries based on policy.

85 Firewall Basics

Question Question 1

True / False | Topic: Firewall Basics

Security policies on Huawei firewalls are matched from bottom to top by default.

- A. True
- B. False

Correct Answer: False**Concept:** Policies are matched from top to bottom based on configured order/priority.

Question Question 2

True / False | Topic: Firewall Basics

A packet-filtering firewall tracks the state of every connection.

- A. True
- B. False

Correct Answer: False**Concept:** Only stateful firewalls maintain connection state; packet filters inspect packets individually.

Question Question 3

True / False | Topic: Firewall Basics

ASPF can generate temporary server-map entries for dynamic protocols.

- A. True
- B. False

Correct Answer: True

Concept: ASPF creates server-map entries to permit negotiated secondary connections.

Question Question 4

True / False | Topic: Firewall Basics

Next-generation firewalls can identify applications regardless of port.

- A. True
- B. False

Correct Answer: True

Concept: NGFWs use application-layer inspection to identify applications beyond simple port numbers.

Question Question 5

True / False | Topic: Firewall Basics

Traffic between interfaces in the same security zone is always allowed.

- A. True
- B. False

Correct Answer: False

Concept: Even intra-zone traffic can be controlled by policies, though default behavior may vary by configuration.

Question Question 6

True / False | Topic: Firewall Basics

A deny action in a security policy can generate logs.

- A. True
- B. False

Correct Answer: True

Concept: Both permit and deny actions can be configured to log traffic.

Question Question 7

True / False | Topic: Firewall Basics

The default Huawei firewall action is permit.

- A. True
- B. False

Correct Answer: False

Concept: The default action is deny unless explicitly permitted.

Question Question 8

True / False | Topic: Firewall Basics

ASPF inspects application-layer negotiation.

- A. True
- B. False

Correct Answer: True

Concept: ASPF handles dynamic protocols such as FTP.

Question Question 9

True / False | Topic: Firewall Basics

Stateful firewalls use session tables to track connections.

- A. True
- B. False

Correct Answer: True

Concept: Session tables store connection state for return traffic.

Question Question 10

True / False | Topic: Firewall Basics

Trust zone priority means that is 85 by default.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Trust zone priority.

Question Question 11

True / False | Topic: Firewall Basics

Trust zone priority is unrelated to is 85 by default.

- A. True
- B. False

Correct Answer: False

Concept: Trust zone priority is directly related to is 85 by default.

Question Question 12

True / False | Topic: Firewall Basics

DMZ zone priority means that is 50 by default.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes DMZ zone priority.

Question Question 13

True / False | Topic: Firewall Basics

DMZ zone priority is unrelated to is 50 by default.

- A. True
- B. False

Correct Answer: False

Concept: DMZ zone priority is directly related to is 50 by default.

Question Question 14

True / False | Topic: Firewall Basics

Untrust zone priority means that is 5 by default.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Untrust zone priority.

Question Question 15

True / False | Topic: Firewall Basics

Untrust zone priority is unrelated to is 5 by default.

- A. True
- B. False

Correct Answer: False

Concept: Untrust zone priority is directly related to is 5 by default.

Question Question 16

True / False | Topic: Firewall Basics

Local zone priority means that is 100 by default.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Local zone priority.

Question Question 17

True / False | Topic: Firewall Basics

Local zone priority is unrelated to is 100 by default.

- A. True
- B. False

Correct Answer: False

Concept: Local zone priority is directly related to is 100 by default.

Question Question 18

True / False | Topic: Firewall Basics

Default security policy action means that is deny.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Default security policy action.

Question Question 19

True / False | Topic: Firewall Basics

Default security policy action is unrelated to is deny.

- A. True
- B. False

Correct Answer: False

Concept: Default security policy action is directly related to is deny.

Question Question 20

True / False | Topic: Firewall Basics

Stateful firewall means that tracks connections in a session table.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Stateful firewall.

Question Question 21

True / False | Topic: Firewall Basics

Stateful firewall is unrelated to tracks connections in a session table.

- A. True
- B. False

Correct Answer: False

Concept: Stateful firewall is directly related to tracks connections in a session table.

Question Question 22

True / False | Topic: Firewall Basics

Packet-filtering firewall means that inspects packets individually without state.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Packet-filtering firewall.

Question Question 23

True / False | Topic: Firewall Basics

Packet-filtering firewall is unrelated to inspects packets individually without state.

- A. True
- B. False

Correct Answer: False

Concept: Packet-filtering firewall is directly related to inspects packets individually without state.

Question Question 24

True / False | Topic: Firewall Basics

ASPF means that inspects multi-channel protocols like FTP.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes ASPF.

Question Question 25

True / False | Topic: Firewall Basics

ASPF is unrelated to inspects multi-channel protocols like FTP.

- A. True
- B. False

Correct Answer: False

Concept: ASPF is directly related to inspects multi-channel protocols like FTP.

Question Question 26

True / False | Topic: Firewall Basics

Next-generation firewall means that includes application awareness and IPS.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Next-generation firewall.

Question Question 27

True / False | Topic: Firewall Basics

Next-generation firewall is unrelated to includes application awareness and IPS.

- A. True
- B. False

Correct Answer: False

Concept: Next-generation firewall is directly related to includes application awareness and IPS.

Question Question 28

True / False | Topic: Firewall Basics

Security zones means that group interfaces with similar trust levels.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Security zones.

Question Question 29

True / False | Topic: Firewall Basics

Security zones is unrelated to group interfaces with similar trust levels.

- A. True
- B. False

Correct Answer: False**Concept:** Security zones is directly related to group interfaces with similar trust levels.

Question Question 30

True / False | Topic: Firewall Basics

Intra-zone traffic means that can be controlled by policies.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Intra-zone traffic.

Question Question 31

True / False | Topic: Firewall Basics

Intra-zone traffic is unrelated to can be controlled by policies.

- A. True
- B. False

Correct Answer: False**Concept:** Intra-zone traffic is directly related to can be controlled by policies.

Question Question 32

True / False | Topic: Firewall Basics

Policy hit count means that helps verify which rules are used.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Policy hit count.

Question Question 33

True / False | Topic: Firewall Basics

Policy hit count is unrelated to helps verify which rules are used.

- A. True
- B. False

Correct Answer: False

Concept: Policy hit count is directly related to helps verify which rules are used.

Question Question 34

True / False | Topic: Firewall Basics

Security policy logging means that can be enabled for permit and deny actions.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Security policy logging.

Question Question 35

True / False | Topic: Firewall Basics

Security policy logging is unrelated to can be enabled for permit and deny actions.

- A. True
- B. False

Correct Answer: False

Concept: Security policy logging is directly related to can be enabled for permit and deny actions.

Question Question 36

True / False | Topic: Firewall Basics

Routing mode means that is a firewall deployment mode.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Routing mode.

Question Question 37

True / False | Topic: Firewall Basics

Routing mode is unrelated to is a firewall deployment mode.

- A. True
- B. False

Correct Answer: False

Concept: Routing mode is directly related to is a firewall deployment mode.

Question Question 38

True / False | Topic: Firewall Basics

Transparent mode means that is a firewall deployment mode.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Transparent mode.

Question Question 39

True / False | Topic: Firewall Basics

Transparent mode is unrelated to is a firewall deployment mode.

- A. True
- B. False

Correct Answer: False**Concept:** Transparent mode is directly related to is a firewall deployment mode.

Question Question 40

True / False | Topic: Firewall Basics

Hybrid mode means that is a firewall deployment mode.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Hybrid mode.

Question Question 41

True / False | Topic: Firewall Basics

Hybrid mode is unrelated to is a firewall deployment mode.

- A. True
- B. False

Correct Answer: False**Concept:** Hybrid mode is directly related to is a firewall deployment mode.

86 Security Zones

Question Question 1

True / False | Topic: Security Zones

The Local zone represents traffic destined to or originated from the firewall itself.

- A. True
- B. False

Correct Answer: True**Concept:** Local zone traffic involves the firewall services and management interfaces.

87 Security Policies

Question Question 1

True / False | Topic: Security Policies

By default, Huawei firewalls permit all traffic unless a policy explicitly denies it.

- A. True
- B. False

Correct Answer: False**Concept:** The default action is to deny traffic unless a security policy explicitly permits it.

88 Stateful Inspection

Question Question 1

True / False | Topic: Stateful Inspection

A stateful firewall can allow return traffic for an established session without a separate policy.

- A. True
- B. False

Correct Answer: True**Concept:** Return packets matching an existing session table entry are permitted automatically.

89 ASPF

Question Question 1

True / False | Topic: ASPF

ASPF is used to support protocols that open secondary or dynamic connections.

- A. True
- B. False

Correct Answer: True**Concept:** ASPF inspects application-layer negotiation to dynamically permit related channels.

90 NAT

Question Question 1

True / False | Topic: NAT

NAT can hide internal network topology from external observers.

- A. True
- B. False

Correct Answer: True**Concept:** NAT translates private addresses, making internal topology invisible from outside.

Question Question 2

True / False | Topic: NAT

Easy-IP uses a fixed public IP address pool.

- A. True
- B. False

Correct Answer: False

Concept: Easy-IP uses the firewall egress interface IP address dynamically.

Question Question 3

True / False | Topic: NAT

NAT ALG is required for HTTP because HTTP embeds IP addresses in payloads.

- A. True
- B. False

Correct Answer: False

Concept: HTTP generally does not embed IP addresses in payloads, so ALG is not required.

Question Question 4

True / False | Topic: NAT

NAPT conserves public IPv4 addresses by using port multiplexing.

- A. True
- B. False

Correct Answer: True

Concept: NAPT maps many private addresses to one public IP using unique port numbers.

Question Question 5

True / False | Topic: NAT

FTP data channel failure can be caused by disabled NAT ALG.

- A. True
- B. False

Correct Answer: True

Concept: Without ALG, the firewall cannot identify and permit the negotiated FTP data channel.

Question Question 6

True / False | Topic: NAT

NAPT translates both IP addresses and port numbers.

- A. True
- B. False

Correct Answer: True

Concept: NAPT uses port multiplexing for many-to-one translation.

Question Question 7

True / False | Topic: NAT

NAT Server is used for outbound Internet access.

- A. True
- B. False

Correct Answer: False

Concept: NAT Server publishes inbound services.

Question Question 8

True / False | Topic: NAT

Static NAT preserves a one-to-one public-private mapping.

- A. True
- B. False

Correct Answer: True

Concept: Static NAT maps one public IP to one private IP consistently.

Question Question 9

True / False | Topic: NAT

NAT means that translates IP addresses between private and public.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes NAT.

Question Question 10

True / False | Topic: NAT

NAT is unrelated to translates IP addresses between private and public.

- A. True
- B. False

Correct Answer: False

Concept: NAT is directly related to translates IP addresses between private and public.

Question Question 11

True / False | Topic: NAT

Source NAT means that translates the source IP of outbound packets.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Source NAT.

Question Question 12

True / False | Topic: NAT

Source NAT is unrelated to translates the source IP of outbound packets.

- A. True
- B. False

Correct Answer: False**Concept:** Source NAT is directly related to translates the source IP of outbound packets.

Question Question 13

True / False | Topic: NAT

Destination NAT means that translates the destination IP of inbound packets.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Destination NAT.

Question Question 14

True / False | Topic: NAT

Destination NAT is unrelated to translates the destination IP of inbound packets.

- A. True
- B. False

Correct Answer: False**Concept:** Destination NAT is directly related to translates the destination IP of inbound packets.

Question Question 15

True / False | Topic: NAT

NAT Server means that publishes internal services to the Internet.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes NAT Server.

Question Question 16

True / False | Topic: NAT

NAT Server is unrelated to publishes internal services to the Internet.

- A. True
- B. False

Correct Answer: False**Concept:** NAT Server is directly related to publishes internal services to the Internet.

Question Question 17

True / False | Topic: NAT

Bidirectional NAT means that translates both source and destination addresses.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Bidirectional NAT.

Question Question 18

True / False | Topic: NAT

Bidirectional NAT is unrelated to translates both source and destination addresses.

- A. True
- B. False

Correct Answer: False**Concept:** Bidirectional NAT is directly related to translates both source and destination addresses.

Question Question 19

True / False | Topic: NAT

NAPT means that translates many private addresses to one public IP using ports.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes NAPT.

Question Question 20

True / False | Topic: NAT

NAPT is unrelated to translates many private addresses to one public IP using ports.

- A. True
- B. False

Correct Answer: False**Concept:** NAPT is directly related to translates many private addresses to one public IP using ports.

Question Question 21

True / False | Topic: NAT

NAT No-PAT means that performs one-to-one address translation preserving ports.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes NAT No-PAT.

Question Question 22

True / False | Topic: NAT

NAT No-PAT is unrelated to performs one-to-one address translation preserving ports.

- A. True
- B. False

Correct Answer: False

Concept: NAT No-PAT is directly related to performs one-to-one address translation preserving ports.

Question Question 23

True / False | Topic: NAT

Easy-IP means that uses the outbound interface IP for translation.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Easy-IP.

Question Question 24

True / False | Topic: NAT

Easy-IP is unrelated to uses the outbound interface IP for translation.

- A. True
- B. False

Correct Answer: False

Concept: Easy-IP is directly related to uses the outbound interface IP for translation.

Question Question 25

True / False | Topic: NAT

NAT ALG means that rewrites addresses inside application payloads.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes NAT ALG.

Question Question 26

True / False | Topic: NAT

NAT ALG is unrelated to rewrites addresses inside application payloads.

- A. True
- B. False

Correct Answer: False

Concept: NAT ALG is directly related to rewrites addresses inside application payloads.

Question Question 27

True / False | Topic: NAT

RFC 1918 means that defines private IPv4 address ranges.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes RFC 1918.

Question Question 28

True / False | Topic: NAT

RFC 1918 is unrelated to defines private IPv4 address ranges.

- A. True
- B. False

Correct Answer: False

Concept: RFC 1918 is directly related to defines private IPv4 address ranges.

Question Question 29

True / False | Topic: NAT

Hairpin NAT means that lets internal users access servers via public NAT address.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Hairpin NAT.

Question Question 30

True / False | Topic: NAT

Hairpin NAT is unrelated to lets internal users access servers via public NAT address.

- A. True
- B. False

Correct Answer: False

Concept: Hairpin NAT is directly related to lets internal users access servers via public NAT address.

Question Question 31

True / False | Topic: NAT

Static NAT means that provides a persistent one-to-one mapping.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Static NAT.

Question Question 32

True / False | Topic: NAT

Static NAT is unrelated to provides a persistent one-to-one mapping.

- A. True
- B. False

Correct Answer: False**Concept:** Static NAT is directly related to provides a persistent one-to-one mapping.

91 NAT Basics

Question Question 1

True / False | Topic: NAT Basics

NAT can conserve public IPv4 addresses.

- A. True
- B. False

Correct Answer: True**Concept:** NAT allows many private hosts to share a smaller number of public IP addresses.

92 Source NAT

Question Question 1

True / False | Topic: Source NAT

NAPT translates both IP addresses and port numbers.

- A. True
- B. False

Correct Answer: True**Concept:** NAPT multiplexes many private addresses onto one public IP by translating ports.

93 Destination NAT

Question Question 1

True / False | Topic: Destination NAT

Destination NAT is typically used for outbound Internet access.

- A. True
- B. False

Correct Answer: False**Concept:** Destination NAT is used for inbound traffic to expose internal services.

94 NAT Server

Question Question 1

True / False | Topic: NAT Server

NAT Server maps public IP addresses to private server addresses.

- A. True
- B. False

Correct Answer: True**Concept:** NAT Server publishes internal services by mapping public IPs/ports to private servers.

95 NAT ALG

Question Question 1

True / False | Topic: NAT ALG

NAT ALG is needed when protocols embed IP addresses inside application payloads.

- A. True
- B. False

Correct Answer: True**Concept:** ALG inspects application-layer data to update embedded addresses/ports.

96 Firewall Hot Standby

Question Question 1

True / False | Topic: Firewall Hot Standby

In active/standby mode, both firewalls actively forward the same traffic.

- A. True
- B. False

Correct Answer: False**Concept:** Only the active firewall forwards traffic; the standby takes over upon failure.

Question Question 2

True / False | Topic: Firewall Hot Standby

VRRP provides default gateway redundancy for end hosts.

- A. True
- B. False

Correct Answer: True**Concept:** VRRP allows multiple routers to act as a single virtual gateway.

Question Question 3

True / False | Topic: Firewall Hot Standby

In VRRP, only the router with the physical IP matching the virtual IP can become master.

- A. True
- B. False

Correct Answer: False**Concept:** Any router with sufficient priority can become master, not just the IP owner.

Question Question 4

True / False | Topic: Firewall Hot Standby

VGMP prevents individual VRRP groups from becoming inconsistent during failover.

- A. True
- B. False

Correct Answer: True**Concept:** VGMP manages multiple VRRP groups as a single entity.

Question Question 5

True / False | Topic: Firewall Hot Standby

HRP can synchronize NAT sessions between active and standby firewalls.

- A. True
- B. False

Correct Answer: True**Concept:** HRP keeps NAT/session state consistent across hot-standby peers.

Question Question 6

True / False | Topic: Firewall Hot Standby

Load-sharing mode improves resource utilization compared to active/standby mode.

- A. True
- B. False

Correct Answer: True**Concept:** Multiple firewalls forward traffic simultaneously in load-sharing mode.

Question Question 7

True / False | Topic: Firewall Hot Standby

The HRP heartbeat link can share bandwidth with regular user traffic without risk.

- A. True
- B. False

Correct Answer: False**Concept:** Sharing heartbeat and user traffic can cause false failover due to congestion.

Question Question 8

True / False | Topic: Firewall Hot Standby

VRRP provides a virtual gateway IP.

- A. True
- B. False

Correct Answer: True**Concept:** VRRP allows multiple routers to share a virtual IP.

Question Question 9

True / False | Topic: Firewall Hot Standby

HRP synchronizes only configurations.

- A. True
- B. False

Correct Answer: False**Concept:** HRP synchronizes configurations and runtime state.

Question Question 10

True / False | Topic: Firewall Hot Standby

Load-sharing mode allows multiple firewalls to forward traffic.

- A. True
- B. False

Correct Answer: True**Concept:** Multiple active firewalls share the traffic load.

Question Question 11

True / False | Topic: Firewall Hot Standby

VRRP means that provides gateway redundancy with a virtual IP.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes VRRP.

Question Question 12

True / False | Topic: Firewall Hot Standby

VRRP is unrelated to provides gateway redundancy with a virtual IP.

- A. True
- B. False

Correct Answer: False**Concept:** VRRP is directly related to provides gateway redundancy with a virtual IP.

Question Question 13

True / False | Topic: Firewall Hot Standby

VRRP master means that forwards traffic for the virtual IP.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes VRRP master.

Question Question 14

True / False | Topic: Firewall Hot Standby

VRRP master is unrelated to forwards traffic for the virtual IP.

- A. True
- B. False

Correct Answer: False**Concept:** VRRP master is directly related to forwards traffic for the virtual IP.

Question Question 15

True / False | Topic: Firewall Hot Standby

VRRP backup means that takes over if the master fails.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes VRRP backup.

Question Question 16

True / False | Topic: Firewall Hot Standby

VRRP backup is unrelated to takes over if the master fails.

- A. True
- B. False

Correct Answer: False**Concept:** VRRP backup is directly related to takes over if the master fails.

Question Question 17

True / False | Topic: Firewall Hot Standby

VGMP means that manages VRRP group states consistently.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes VGMP.

Question Question 18

True / False | Topic: Firewall Hot Standby

VGMP is unrelated to manages VRRP group states consistently.

- A. True
- B. False

Correct Answer: False**Concept:** VGMP is directly related to manages VRRP group states consistently.

Question Question 19

True / False | Topic: Firewall Hot Standby

HRP means that synchronizes sessions and configurations.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes HRP.

Question Question 20

True / False | Topic: Firewall Hot Standby

HRP is unrelated to synchronizes sessions and configurations.

- A. True
- B. False

Correct Answer: False**Concept:** HRP is directly related to synchronizes sessions and configurations.

Question Question 21

True / False | Topic: Firewall Hot Standby

Active/standby mode means that has one firewall forwarding traffic.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Active/standby mode.

Question Question 22

True / False | Topic: Firewall Hot Standby

Active/standby mode is unrelated to has one firewall forwarding traffic.

- A. True
- B. False

Correct Answer: False**Concept:** Active/standby mode is directly related to has one firewall forwarding traffic.

Question Question 23

True / False | Topic: Firewall Hot Standby

Load-sharing mode means that has multiple firewalls forwarding traffic.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Load-sharing mode.

Question Question 24

True / False | Topic: Firewall Hot Standby

Load-sharing mode is unrelated to has multiple firewalls forwarding traffic.

- A. True
- B. False

Correct Answer: False**Concept:** Load-sharing mode is directly related to has multiple firewalls forwarding traffic.

Question Question 25

True / False | Topic: Firewall Hot Standby

Heartbeat link means that should be dedicated and reliable.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Heartbeat link.

Question Question 26

True / False | Topic: Firewall Hot Standby

Heartbeat link is unrelated to should be dedicated and reliable.

- A. True
- B. False

Correct Answer: False**Concept:** Heartbeat link is directly related to should be dedicated and reliable.

Question Question 27

True / False | Topic: Firewall Hot Standby

HRP session backup means that ensures connections survive failover.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes HRP session backup.

Question Question 28

True / False | Topic: Firewall Hot Standby

HRP session backup is unrelated to ensures connections survive failover.

- A. True
- B. False

Correct Answer: False**Concept:** HRP session backup is directly related to ensures connections survive failover.

Question Question 29

True / False | Topic: Firewall Hot Standby

Preempt delay means that reduces rapid master-backup flapping.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Preempt delay.

Question Question 30

True / False | Topic: Firewall Hot Standby

Preempt delay is unrelated to reduces rapid master-backup flapping.

- A. True
- B. False

Correct Answer: False**Concept:** Preempt delay is directly related to reduces rapid master-backup flapping.

97 VRRP

Question Question 1

True / False | Topic: VRRP

VRRP uses a virtual IP address shared between multiple physical routers.

- A. True
- B. False

Correct Answer: True**Concept:** VRRP allows multiple routers to present a single virtual gateway IP to hosts.

98 VGMP

Question Question 1

True / False | Topic: VGMP

VGMP ensures that VRRP groups on a firewall switch state consistently.

- A. True
- B. False

Correct Answer: True**Concept:** VGMP prevents split-brain scenarios by managing all VRRP groups as a single entity.

99 HRP

Question Question 1

True / False | Topic: HRP

HRP only synchronizes configuration files, not session tables.

- A. True
- B. False

Correct Answer: False

Concept: HRP synchronizes both configurations and runtime state such as session tables.

100 Intrusion Prevention and Antivirus

Question Question 1

True / False | Topic: Intrusion Prevention and Antivirus

IPS is typically deployed inline to block threats.

- A. True
- B. False

Correct Answer: True

Concept: Inline deployment allows IPS to actively block malicious traffic.

Question Question 2

True / False | Topic: Intrusion Prevention and Antivirus

Anomaly-based detection requires a database of known attack signatures.

- A. True
- B. False

Correct Answer: False

Concept: Anomaly detection uses baselines of normal behavior, not attack signatures.

Question Question 3

True / False | Topic: Intrusion Prevention and Antivirus

Heuristic analysis can detect previously unknown malware.

- A. True
- B. False

Correct Answer: True

Concept: Heuristics identify suspicious behavior even without a known signature.

Question Question 4

True / False | Topic: Intrusion Prevention and Antivirus

Signature-based antivirus is effective against all zero-day threats.

- A. True
- B. False

Correct Answer: False

Concept: Zero-day threats lack signatures, so signature-only detection cannot catch them.

Question Question 5

True / False | Topic: Intrusion Prevention and Antivirus

An IDS can block malicious traffic inline.

- A. True
- B. False

Correct Answer: False

Concept: IDS typically detects and alerts; IPS blocks inline.

Question Question 6

True / False | Topic: Intrusion Prevention and Antivirus

Heuristic detection can identify unknown threats.

- A. True
- B. False

Correct Answer: True

Concept: Heuristics analyze behavior to find unknown malware.

Question Question 7

True / False | Topic: Intrusion Prevention and Antivirus

Sandboxing executes files in an isolated environment.

- A. True
- B. False

Correct Answer: True

Concept: Sandboxing observes behavior without risking production systems.

Question Question 8

True / False | Topic: Intrusion Prevention and Antivirus

IDS means that detects intrusions and generates alerts.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes IDS.

Question Question 9

True / False | Topic: Intrusion Prevention and Antivirus

IDS is unrelated to detects intrusions and generates alerts.

- A. True
- B. False

Correct Answer: False

Concept: IDS is directly related to detects intrusions and generates alerts.

Question Question 10

True / False | Topic: Intrusion Prevention and Antivirus

IPS means that detects and blocks intrusions inline.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes IPS.

Question Question 11

True / False | Topic: Intrusion Prevention and Antivirus

IPS is unrelated to detects and blocks intrusions inline.

- A. True
- B. False

Correct Answer: False**Concept:** IPS is directly related to detects and blocks intrusions inline.

Question Question 12

True / False | Topic: Intrusion Prevention and Antivirus

Signature detection means that matches known attack patterns.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Signature detection.

Question Question 13

True / False | Topic: Intrusion Prevention and Antivirus

Signature detection is unrelated to matches known attack patterns.

- A. True
- B. False

Correct Answer: False**Concept:** Signature detection is directly related to matches known attack patterns.

Question Question 14

True / False | Topic: Intrusion Prevention and Antivirus

Anomaly detection means that flags deviations from normal behavior.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Anomaly detection.

Question Question 15

True / False | Topic: Intrusion Prevention and Antivirus

Anomaly detection is unrelated to flags deviations from normal behavior.

- A. True
- B. False

Correct Answer: False**Concept:** Anomaly detection is directly related to flags deviations from normal behavior.

Question Question 16

True / False | Topic: Intrusion Prevention and Antivirus

False positive means that incorrectly blocks legitimate traffic.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes False positive.

Question Question 17

True / False | Topic: Intrusion Prevention and Antivirus

False positive is unrelated to incorrectly blocks legitimate traffic.

- A. True
- B. False

Correct Answer: False**Concept:** False positive is directly related to incorrectly blocks legitimate traffic.

Question Question 18

True / False | Topic: Intrusion Prevention and Antivirus

False negative means that misses an actual attack.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes False negative.

Question Question 19

True / False | Topic: Intrusion Prevention and Antivirus

False negative is unrelated to misses an actual attack.

- A. True
- B. False

Correct Answer: False**Concept:** False negative is directly related to misses an actual attack.

Question Question 20

True / False | Topic: Intrusion Prevention and Antivirus

Sandboxing means that observes file behavior in an isolated environment.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Sandboxing.

Question Question 21

True / False | Topic: Intrusion Prevention and Antivirus

Sandboxing is unrelated to observes file behavior in an isolated environment.

- A. True
- B. False

Correct Answer: False**Concept:** Sandboxing is directly related to observes file behavior in an isolated environment.

Question Question 22

True / False | Topic: Intrusion Prevention and Antivirus

Heuristic analysis means that detects malware by examining behavior.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Heuristic analysis.

Question Question 23

True / False | Topic: Intrusion Prevention and Antivirus

Heuristic analysis is unrelated to detects malware by examining behavior.

- A. True
- B. False

Correct Answer: False**Concept:** Heuristic analysis is directly related to detects malware by examining behavior.

Question Question 24

True / False | Topic: Intrusion Prevention and Antivirus

Whitelisting means that allows only approved applications.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Whitelisting.

Question Question 25

True / False | Topic: Intrusion Prevention and Antivirus

Whitelisting is unrelated to allows only approved applications.

- A. True
- B. False

Correct Answer: False**Concept:** Whitelisting is directly related to allows only approved applications.

Question Question 26

True / False | Topic: Intrusion Prevention and Antivirus

Antivirus signatures means that must be updated regularly.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Antivirus signatures.

Question Question 27

True / False | Topic: Intrusion Prevention and Antivirus

Antivirus signatures is unrelated to must be updated regularly.

- A. True
- B. False

Correct Answer: False**Concept:** Antivirus signatures is directly related to must be updated regularly.

Question Question 28

True / False | Topic: Intrusion Prevention and Antivirus

NIDS means that monitors network traffic for intrusions.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes NIDS.

Question Question 29

True / False | Topic: Intrusion Prevention and Antivirus

NIDS is unrelated to monitors network traffic for intrusions.

- A. True
- B. False

Correct Answer: False**Concept:** NIDS is directly related to monitors network traffic for intrusions.

Question Question 30

True / False | Topic: Intrusion Prevention and Antivirus

HIDS means that monitors host activity for intrusions.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes HIDS.

Question Question 31

True / False | Topic: Intrusion Prevention and Antivirus

HIDS is unrelated to monitors host activity for intrusions.

- A. True
- B. False

Correct Answer: False**Concept:** HIDS is directly related to monitors host activity for intrusions.

Question Question 32

True / False | Topic: Intrusion Prevention and Antivirus

Reconnaissance means that is an early attack phase.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Reconnaissance.

Question Question 33

True / False | Topic: Intrusion Prevention and Antivirus

Reconnaissance is unrelated to is an early attack phase.

- A. True
- B. False

Correct Answer: False**Concept:** Reconnaissance is directly related to is an early attack phase.

Question Question 34

True / False | Topic: Intrusion Prevention and Antivirus

Exploitation means that gains unauthorized access using vulnerabilities.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Exploitation.

Question Question 35

True / False | Topic: Intrusion Prevention and Antivirus

Exploitation is unrelated to gains unauthorized access using vulnerabilities.

- A. True
- B. False

Correct Answer: False**Concept:** Exploitation is directly related to gains unauthorized access using vulnerabilities.

101 Intrusion Prevention

Question Question 1

True / False | Topic: Intrusion Prevention

An IDS is typically deployed inline and can block attacks.

- A. True
- B. False

Correct Answer: False**Concept:** IDS is usually passive/out-of-band and only detects and alerts; IPS blocks inline.

Question Question 2

True / False | Topic: Intrusion Prevention

Signature-based IPS can detect zero-day attacks effectively.

- A. True
- B. False

Correct Answer: False**Concept:** Signatures require known attack patterns; zero-day attacks are unknown and often missed.

102 Antivirus

Question Question 1

True / False | Topic: Antivirus

Antivirus signature databases must be updated regularly to detect new malware.

- A. True
- B. False

Correct Answer: True**Concept:** New malware variants require updated signatures and detection engines.

Question Question 2

True / False | Topic: Antivirus

Sandboxing can detect unknown malware by observing its runtime behavior.

- A. True
- B. False

Correct Answer: True**Concept:** Sandboxing reveals malicious behavior of unknown samples in an isolated environment.

103 AAA and User Authentication

Question Question 1

True / False | Topic: AAA and User Authentication

Authentication verifies who a user is.

- A. True
- B. False

Correct Answer: True

Concept: Authentication confirms identity before granting access.

Question Question 2

True / False | Topic: AAA and User Authentication

TACACS+ separates authentication, authorization, and accounting.

- A. True
- B. False

Correct Answer: True

Concept: TACACS+ provides independent AAA functions.

Question Question 3

True / False | Topic: AAA and User Authentication

RADIUS encrypts the entire authentication request.

- A. True
- B. False

Correct Answer: False

Concept: RADIUS encrypts only the password attribute.

Question Question 4

True / False | Topic: AAA and User Authentication

Local authentication does not require an external server.

- A. True
- B. False

Correct Answer: True

Concept: Local authentication uses the firewall own user database.

Question Question 5

True / False | Topic: AAA and User Authentication

Authorization happens after authentication.

- A. True
- B. False

Correct Answer: True

Concept: Authorization defines what an authenticated user may do.

Question Question 6

True / False | Topic: AAA and User Authentication

TACACS+ uses UDP.

- A. True
- B. False

Correct Answer: False

Concept: TACACS+ uses TCP.

Question Question 7

True / False | Topic: AAA and User Authentication

Portal authentication requires a web browser.

- A. True
- B. False

Correct Answer: True

Concept: Portal presents a web login page to users.

Question Question 8

True / False | Topic: AAA and User Authentication

Authentication means that verifies user identity.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Authentication.

Question Question 9

True / False | Topic: AAA and User Authentication

Authentication is unrelated to verifies user identity.

- A. True
- B. False

Correct Answer: False

Concept: Authentication is directly related to verifies user identity.

Question Question 10

True / False | Topic: AAA and User Authentication

Authorization means that determines what a user can access.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Authorization.

Question Question 11

True / False | Topic: AAA and User Authentication

Authorization is unrelated to determines what a user can access.

- A. True
- B. False

Correct Answer: False**Concept:** Authorization is directly related to determines what a user can access.

Question Question 12

True / False | Topic: AAA and User Authentication

Accounting means that records user activities and resource usage.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Accounting.

Question Question 13

True / False | Topic: AAA and User Authentication

Accounting is unrelated to records user activities and resource usage.

- A. True
- B. False

Correct Answer: False**Concept:** Accounting is directly related to records user activities and resource usage.

Question Question 14

True / False | Topic: AAA and User Authentication

RADIUS means that uses UDP ports 1812 and 1813.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes RADIUS.

Question Question 15

True / False | Topic: AAA and User Authentication

RADIUS is unrelated to uses UDP ports 1812 and 1813.

- A. True
- B. False

Correct Answer: False**Concept:** RADIUS is directly related to uses UDP ports 1812 and 1813.

Question Question 16

True / False | Topic: AAA and User Authentication

TACACS+ means that uses TCP port 49.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes TACACS+.

Question Question 17

True / False | Topic: AAA and User Authentication

TACACS+ is unrelated to uses TCP port 49.

- A. True
- B. False

Correct Answer: False**Concept:** TACACS+ is directly related to uses TCP port 49.

Question Question 18

True / False | Topic: AAA and User Authentication

Portal authentication means that uses a web login page.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Portal authentication.

Question Question 19

True / False | Topic: AAA and User Authentication

Portal authentication is unrelated to uses a web login page.

- A. True
- B. False

Correct Answer: False**Concept:** Portal authentication is directly related to uses a web login page.

Question Question 20

True / False | Topic: AAA and User Authentication

802.1X means that provides port-based network access control.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes 802.1X.

Question Question 21

True / False | Topic: AAA and User Authentication

802.1X is unrelated to provides port-based network access control.

- A. True
- B. False

Correct Answer: False**Concept:** 802.1X is directly related to provides port-based network access control.

Question Question 22

True / False | Topic: AAA and User Authentication

Local authentication means that uses the device user database.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Local authentication.

Question Question 23

True / False | Topic: AAA and User Authentication

Local authentication is unrelated to uses the device user database.

- A. True
- B. False

Correct Answer: False**Concept:** Local authentication is directly related to uses the device user database.

Question Question 24

True / False | Topic: AAA and User Authentication

LDAP means that queries a directory server for authentication.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes LDAP.

Question Question 25

True / False | Topic: AAA and User Authentication

LDAP is unrelated to queries a directory server for authentication.

- A. True
- B. False

Correct Answer: False**Concept:** LDAP is directly related to queries a directory server for authentication.

Question Question 26

True / False | Topic: AAA and User Authentication

User group means that applies policies to multiple users collectively.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes User group.

Question Question 27

True / False | Topic: AAA and User Authentication

User group is unrelated to applies policies to multiple users collectively.

- A. True
- B. False

Correct Answer: False**Concept:** User group is directly related to applies policies to multiple users collectively.

Question Question 28

True / False | Topic: AAA and User Authentication

SSO means that allows one authentication for multiple services.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes SSO.

Question Question 29

True / False | Topic: AAA and User Authentication

SSO is unrelated to allows one authentication for multiple services.

- A. True
- B. False

Correct Answer: False**Concept:** SSO is directly related to allows one authentication for multiple services.

104 AAA Basics

Question Question 1

True / False | Topic: AAA Basics

Authorization determines whether an authenticated user is allowed to perform an action.

- A. True
- B. False

Correct Answer: True**Concept:** Authorization enforces access control decisions based on identity and policy.

Question Question 2

True / False | Topic: AAA Basics

Accounting in AAA records user activities and resource usage.

- A. True
- B. False

Correct Answer: True**Concept:** Accounting provides logs for auditing, billing, and forensic analysis.

105 Firewall User Authentication

Question Question 1

True / False | Topic: Firewall User Authentication

Portal authentication is commonly used for guest access and web-based login.

- A. True
- B. False

Correct Answer: True**Concept:** Portal authentication provides a convenient web-based login experience.

Question Question 2

True / False | Topic: Firewall User Authentication

RADIUS encrypts the entire authentication packet including the username.

- A. True
- B. False

Correct Answer: False**Concept:** RADIUS encrypts only the password field; the rest of the packet is not encrypted.

106 Cryptography and PKI

Question Question 1

True / False | Topic: Cryptography and PKI

AES is a symmetric encryption algorithm.

- A. True
- B. False

Correct Answer: True**Concept:** AES uses the same key for encryption and decryption.

Question Question 2

True / False | Topic: Cryptography and PKI

RSA is faster than AES for bulk data encryption.

- A. True
- B. False

Correct Answer: False**Concept:** RSA is slower and typically used for key exchange or signatures, not bulk encryption.

Question Question 3

True / False | Topic: Cryptography and PKI

SHA-256 produces a 256-bit digest.

- A. True
- B. False

Correct Answer: True**Concept:** SHA-256 generates a fixed 256-bit output.

Question Question 4

True / False | Topic: Cryptography and PKI

Hash functions are reversible.

- A. True
- B. False

Correct Answer: False**Concept:** Hash functions are designed to be one-way.

Question Question 5

True / False | Topic: Cryptography and PKI

A digital signature provides non-repudiation.

- A. True
- B. False

Correct Answer: True**Concept:** Digital signatures prove the signer created the message and cannot easily deny it.

Question Question 6

True / False | Topic: Cryptography and PKI

A root CA certificate is typically signed by another CA.

- A. True
- B. False

Correct Answer: False**Concept:** Root CA certificates are self-signed and act as trust anchors.

Question Question 7

True / False | Topic: Cryptography and PKI

OCSP provides real-time certificate revocation checking.

- A. True
- B. False

Correct Answer: True**Concept:** OCSP allows clients to query revocation status online.

Question Question 8

True / False | Topic: Cryptography and PKI

A hash function output can be reversed to find the input.

- A. True
- B. False

Correct Answer: False

Concept: Hash functions are one-way.

Question Question 9

True / False | Topic: Cryptography and PKI

A digital signature verifies integrity and authenticity.

- A. True
- B. False

Correct Answer: True

Concept: Signatures prove the message came from the signer and was not altered.

Question Question 10

True / False | Topic: Cryptography and PKI

RSA is a symmetric encryption algorithm.

- A. True
- B. False

Correct Answer: False

Concept: RSA is asymmetric.

Question Question 11

True / False | Topic: Cryptography and PKI

Symmetric encryption means that uses the same key for encryption and decryption.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Symmetric encryption.

Question Question 12

True / False | Topic: Cryptography and PKI

Symmetric encryption is unrelated to uses the same key for encryption and decryption.

- A. True
- B. False

Correct Answer: False

Concept: Symmetric encryption is directly related to uses the same key for encryption and decryption.

Question Question 13

True / False | Topic: Cryptography and PKI

Asymmetric encryption means that uses public and private key pairs.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Asymmetric encryption.

Question Question 14

True / False | Topic: Cryptography and PKI

Asymmetric encryption is unrelated to uses public and private key pairs.

- A. True
- B. False

Correct Answer: False**Concept:** Asymmetric encryption is directly related to uses public and private key pairs.

Question Question 15

True / False | Topic: Cryptography and PKI

AES means that is a symmetric encryption standard.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes AES.

Question Question 16

True / False | Topic: Cryptography and PKI

AES is unrelated to is a symmetric encryption standard.

- A. True
- B. False

Correct Answer: False**Concept:** AES is directly related to is a symmetric encryption standard.

Question Question 17

True / False | Topic: Cryptography and PKI

RSA means that is an asymmetric encryption algorithm.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes RSA.

Question Question 18

True / False | Topic: Cryptography and PKI

RSA is unrelated to is an asymmetric encryption algorithm.

- A. True
- B. False

Correct Answer: False**Concept:** RSA is directly related to is an asymmetric encryption algorithm.

Question Question 19

True / False | Topic: Cryptography and PKI

DES means that uses a 56-bit effective key.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes DES.

Question Question 20

True / False | Topic: Cryptography and PKI

DES is unrelated to uses a 56-bit effective key.

- A. True
- B. False

Correct Answer: False**Concept:** DES is directly related to uses a 56-bit effective key.

Question Question 21

True / False | Topic: Cryptography and PKI

Hash function means that produces a fixed-length digest.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Hash function.

Question Question 22

True / False | Topic: Cryptography and PKI

Hash function is unrelated to produces a fixed-length digest.

- A. True
- B. False

Correct Answer: False**Concept:** Hash function is directly related to produces a fixed-length digest.

Question Question 23

True / False | Topic: Cryptography and PKI

MD5 means that is considered broken due to collision attacks.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes MD5.

Question Question 24

True / False | Topic: Cryptography and PKI

MD5 is unrelated to is considered broken due to collision attacks.

- A. True
- B. False

Correct Answer: False**Concept:** MD5 is directly related to is considered broken due to collision attacks.

Question Question 25

True / False | Topic: Cryptography and PKI

SHA-256 means that produces a 256-bit digest.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes SHA-256.

Question Question 26

True / False | Topic: Cryptography and PKI

SHA-256 is unrelated to produces a 256-bit digest.

- A. True
- B. False

Correct Answer: False**Concept:** SHA-256 is directly related to produces a 256-bit digest.

Question Question 27

True / False | Topic: Cryptography and PKI

HMAC means that provides message authentication using a hash and key.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes HMAC.

Question Question 28

True / False | Topic: Cryptography and PKI

HMAC is unrelated to provides message authentication using a hash and key.

- A. True
- B. False

Correct Answer: False**Concept:** HMAC is directly related to provides message authentication using a hash and key.

Question Question 29

True / False | Topic: Cryptography and PKI

Digital signature means that verifies authenticity and integrity.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Digital signature.

Question Question 30

True / False | Topic: Cryptography and PKI

Digital signature is unrelated to verifies authenticity and integrity.

- A. True
- B. False

Correct Answer: False**Concept:** Digital signature is directly related to verifies authenticity and integrity.

Question Question 31

True / False | Topic: Cryptography and PKI

Public key means that is used to verify a digital signature.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Public key.

Question Question 32

True / False | Topic: Cryptography and PKI

Public key is unrelated to is used to verify a digital signature.

- A. True
- B. False

Correct Answer: False**Concept:** Public key is directly related to is used to verify a digital signature.

Question Question 33

True / False | Topic: Cryptography and PKI

Private key means that is used to create a digital signature.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Private key.

Question Question 34

True / False | Topic: Cryptography and PKI

Private key is unrelated to is used to create a digital signature.

- A. True
- B. False

Correct Answer: False**Concept:** Private key is directly related to is used to create a digital signature.

Question Question 35

True / False | Topic: Cryptography and PKI

PKI means that manages public keys and certificates.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes PKI.

Question Question 36

True / False | Topic: Cryptography and PKI

PKI is unrelated to manages public keys and certificates.

- A. True
- B. False

Correct Answer: False**Concept:** PKI is directly related to manages public keys and certificates.

Question Question 37

True / False | Topic: Cryptography and PKI

CA means that issues and signs digital certificates.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes CA.

Question Question 38

True / False | Topic: Cryptography and PKI

CA is unrelated to issues and signs digital certificates.

- A. True
- B. False

Correct Answer: False

Concept: CA is directly related to issues and signs digital certificates.

Question Question 39

True / False | Topic: Cryptography and PKI

RA means that validates identity before certificate issuance.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes RA.

Question Question 40

True / False | Topic: Cryptography and PKI

RA is unrelated to validates identity before certificate issuance.

- A. True
- B. False

Correct Answer: False

Concept: RA is directly related to validates identity before certificate issuance.

Question Question 41

True / False | Topic: Cryptography and PKI

CRL means that lists revoked certificates.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes CRL.

Question Question 42

True / False | Topic: Cryptography and PKI

CRL is unrelated to lists revoked certificates.

- A. True
- B. False

Correct Answer: False

Concept: CRL is directly related to lists revoked certificates.

Question Question 43

True / False | Topic: Cryptography and PKI

OCSP means that provides online certificate status checking.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes OCSP.

Question Question 44

True / False | Topic: Cryptography and PKI

OCSP is unrelated to provides online certificate status checking.

- A. True
- B. False

Correct Answer: False**Concept:** OCSP is directly related to provides online certificate status checking.

Question Question 45

True / False | Topic: Cryptography and PKI

X.509 means that is the standard certificate format.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes X.509.

Question Question 46

True / False | Topic: Cryptography and PKI

X.509 is unrelated to is the standard certificate format.

- A. True
- B. False

Correct Answer: False**Concept:** X.509 is directly related to is the standard certificate format.

Question Question 47

True / False | Topic: Cryptography and PKI

Root CA certificate means that is self-signed and acts as a trust anchor.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Root CA certificate.

Question Question 48

True / False | Topic: Cryptography and PKI

Root CA certificate is unrelated to is self-signed and acts as a trust anchor.

- A. True
- B. False

Correct Answer: False**Concept:** Root CA certificate is directly related to is self-signed and acts as a trust anchor.

Question Question 49

True / False | Topic: Cryptography and PKI

GCM means that provides authenticated encryption.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes GCM.

Question Question 50

True / False | Topic: Cryptography and PKI

GCM is unrelated to provides authenticated encryption.

- A. True
- B. False

Correct Answer: False**Concept:** GCM is directly related to provides authenticated encryption.

Question Question 51

True / False | Topic: Cryptography and PKI

ECB means that should be avoided because it reveals plaintext patterns.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes ECB.

Question Question 52

True / False | Topic: Cryptography and PKI

ECB is unrelated to should be avoided because it reveals plaintext patterns.

- A. True
- B. False

Correct Answer: False**Concept:** ECB is directly related to should be avoided because it reveals plaintext patterns.

107 Cryptography Basics

Question Question 1

True / False | Topic: Cryptography Basics

Symmetric encryption is generally faster than asymmetric encryption.

- A. True
- B. False

Correct Answer: True**Concept:** Symmetric algorithms are faster and suitable for bulk data encryption.

Question Question 2

True / False | Topic: Cryptography Basics

Asymmetric encryption is ideal for encrypting large amounts of data directly.

- A. True
- B. False

Correct Answer: False**Concept:** Asymmetric encryption is slower; it is often used for key exchange and signatures, while symmetric encryption handles bulk data.

108 Hash Algorithms

Question Question 1

True / False | Topic: Hash Algorithms

A hash function can be reversed to recover the original input.

- A. True
- B. False

Correct Answer: False**Concept:** Hash functions are one-way; the original input cannot be derived from the digest.

109 PKI

Question Question 1

True / False | Topic: PKI

A digital certificate binds a public key to an identity.

- A. True
- B. False

Correct Answer: True**Concept:** Certificates contain a public key and identity information, signed by a trusted CA.

Question Question 2

True / False | Topic: PKI

A CRL lists certificates that are still valid.

- A. True
- B. False

Correct Answer: False**Concept:** A CRL lists certificates that have been revoked and should no longer be trusted.

110 VPN

Question Question 1

True / False | Topic: VPN

A VPN allows secure communication over an untrusted public network.

- A. True
- B. False

Correct Answer: True**Concept:** VPNs create encrypted tunnels over public networks.

Question Question 2

True / False | Topic: VPN

GRE can tunnel multicast traffic.

- A. True
- B. False

Correct Answer: True**Concept:** GRE supports multicast tunneling, which is useful for routing protocols.

Question Question 3

True / False | Topic: VPN

IPsec transport mode encrypts the entire original IP packet.

- A. True
- B. False

Correct Answer: False**Concept:** Transport mode encrypts only the payload; tunnel mode encrypts the entire packet.

Question Question 4

True / False | Topic: VPN

IKE is responsible for negotiating IPsec keys and security associations.

- A. True
- B. False

Correct Answer: True**Concept:** IKE automates key exchange and SA negotiation.

Question Question 5

True / False | Topic: VPN

L2TP provides encryption by itself.

- A. True
- B. False

Correct Answer: False

Concept: L2TP does not encrypt traffic; it is usually combined with IPsec.

Question Question 6

True / False | Topic: VPN

SSL VPN can be accessed through a standard web browser.

- A. True
- B. False

Correct Answer: True

Concept: SSL VPN supports browser-based access via HTTPS.

Question Question 7

True / False | Topic: VPN

GRE provides encryption by default.

- A. True
- B. False

Correct Answer: False

Concept: GRE encapsulates but does not encrypt traffic.

Question Question 8

True / False | Topic: VPN

SSL VPN commonly uses TCP port 443.

- A. True
- B. False

Correct Answer: True

Concept: SSL VPN uses HTTPS/TCP 443.

Question Question 9

True / False | Topic: VPN

IPsec tunnel mode encrypts the entire original IP packet.

- A. True
- B. False

Correct Answer: True

Concept: Tunnel mode encapsulates the whole original packet.

Question Question 10

True / False | Topic: VPN

VPN means that provides secure communication over public networks.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes VPN.

Question Question 11

True / False | Topic: VPN

VPN is unrelated to provides secure communication over public networks.

- A. True
- B. False

Correct Answer: False**Concept:** VPN is directly related to provides secure communication over public networks.

Question Question 12

True / False | Topic: VPN

Site-to-site VPN means that connects entire networks at different locations.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Site-to-site VPN.

Question Question 13

True / False | Topic: VPN

Site-to-site VPN is unrelated to connects entire networks at different locations.

- A. True
- B. False

Correct Answer: False**Concept:** Site-to-site VPN is directly related to connects entire networks at different locations.

Question Question 14

True / False | Topic: VPN

Remote-access VPN means that connects individual users to a corporate network.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Remote-access VPN.

Question Question 15

True / False | Topic: VPN

Remote-access VPN is unrelated to connects individual users to a corporate network.

- A. True
- B. False

Correct Answer: False

Concept: Remote-access VPN is directly related to connects individual users to a corporate network.

Question Question 16

True / False | Topic: VPN

GRE means that is IP protocol 47 and lacks built-in encryption.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes GRE.

Question Question 17

True / False | Topic: VPN

GRE is unrelated to is IP protocol 47 and lacks built-in encryption.

- A. True
- B. False

Correct Answer: False

Concept: GRE is directly related to is IP protocol 47 and lacks built-in encryption.

Question Question 18

True / False | Topic: VPN

GRE means that can tunnel multicast and non-IP protocols.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes GRE.

Question Question 19

True / False | Topic: VPN

GRE is unrelated to can tunnel multicast and non-IP protocols.

- A. True
- B. False

Correct Answer: False

Concept: GRE is directly related to can tunnel multicast and non-IP protocols.

Question Question 20

True / False | Topic: VPN

IPsec means that provides confidentiality, integrity, and authentication.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes IPsec.

Question Question 21

True / False | Topic: VPN

IPsec is unrelated to provides confidentiality, integrity, and authentication.

- A. True
- B. False

Correct Answer: False**Concept:** IPsec is directly related to provides confidentiality, integrity, and authentication.

Question Question 22

True / False | Topic: VPN

ESP means that provides encryption and authentication.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes ESP.

Question Question 23

True / False | Topic: VPN

ESP is unrelated to provides encryption and authentication.

- A. True
- B. False

Correct Answer: False**Concept:** ESP is directly related to provides encryption and authentication.

Question Question 24

True / False | Topic: VPN

AH means that provides authentication without encryption.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes AH.

Question Question 25

True / False | Topic: VPN

AH is unrelated to provides authentication without encryption.

- A. True
- B. False

Correct Answer: False

Concept: AH is directly related to provides authentication without encryption.

Question Question 26

True / False | Topic: VPN

IKE means that negotiates IPsec keys and security associations.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes IKE.

Question Question 27

True / False | Topic: VPN

IKE is unrelated to negotiates IPsec keys and security associations.

- A. True
- B. False

Correct Answer: False

Concept: IKE is directly related to negotiates IPsec keys and security associations.

Question Question 28

True / False | Topic: VPN

Tunnel mode means that encrypts the entire original IP packet.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Tunnel mode.

Question Question 29

True / False | Topic: VPN

Tunnel mode is unrelated to encrypts the entire original IP packet.

- A. True
- B. False

Correct Answer: False

Concept: Tunnel mode is directly related to encrypts the entire original IP packet.

Question Question 30

True / False | Topic: VPN

Transport mode means that encrypts only the IP payload.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Transport mode.

Question Question 31

True / False | Topic: VPN

Transport mode is unrelated to encrypts only the IP payload.

- A. True
- B. False

Correct Answer: False

Concept: Transport mode is directly related to encrypts only the IP payload.

Question Question 32

True / False | Topic: VPN

L2TP means that uses UDP port 1701 and lacks encryption.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes L2TP.

Question Question 33

True / False | Topic: VPN

L2TP is unrelated to uses UDP port 1701 and lacks encryption.

- A. True
- B. False

Correct Answer: False

Concept: L2TP is directly related to uses UDP port 1701 and lacks encryption.

Question Question 34

True / False | Topic: VPN

L2TP over IPsec means that combines L2TP tunneling with IPsec encryption.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes L2TP over IPsec.

Question Question 35

True / False | Topic: VPN

L2TP over IPsec is unrelated to combines L2TP tunneling with IPsec encryption.

- A. True
- B. False

Correct Answer: False

Concept: L2TP over IPsec is directly related to combines L2TP tunneling with IPsec encryption.

Question Question 36

True / False | Topic: VPN

SSL VPN means that commonly uses TCP port 443.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes SSL VPN.

Question Question 37

True / False | Topic: VPN

SSL VPN is unrelated to commonly uses TCP port 443.

- A. True
- B. False

Correct Answer: False

Concept: SSL VPN is directly related to commonly uses TCP port 443.

Question Question 38

True / False | Topic: VPN

Web proxy mode means that provides browser-based access to web apps.

- A. True
- B. False

Correct Answer: True

Concept: This statement correctly describes Web proxy mode.

Question Question 39

True / False | Topic: VPN

Web proxy mode is unrelated to provides browser-based access to web apps.

- A. True
- B. False

Correct Answer: False

Concept: Web proxy mode is directly related to provides browser-based access to web apps.

Question Question 40

True / False | Topic: VPN

Network extension mode means that assigns a virtual IP to the remote host.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes Network extension mode.

Question Question 41

True / False | Topic: VPN

Network extension mode is unrelated to assigns a virtual IP to the remote host.

- A. True
- B. False

Correct Answer: False**Concept:** Network extension mode is directly related to assigns a virtual IP to the remote host.

Question Question 42

True / False | Topic: VPN

IKEv2 means that is faster and more reliable than IKEv1.

- A. True
- B. False

Correct Answer: True**Concept:** This statement correctly describes IKEv2.

Question Question 43

True / False | Topic: VPN

IKEv2 is unrelated to is faster and more reliable than IKEv1.

- A. True
- B. False

Correct Answer: False**Concept:** IKEv2 is directly related to is faster and more reliable than IKEv1.

111 GRE VPN

Question Question 1

True / False | Topic: GRE VPN

GRE provides strong encryption by default.

- A. True
- B. False

Correct Answer: False**Concept:** GRE only encapsulates traffic; encryption must be added separately, often with IPsec.

112 IPsec VPN

Question Question 1

True / False | Topic: IPsec VPN

ESP provides both encryption and authentication.

- A. True
- B. False

Correct Answer: True**Concept:** ESP can encrypt payloads and authenticate packets.

Question Question 2

True / False | Topic: IPsec VPN

AH provides payload encryption.

- A. True
- B. False

Correct Answer: False**Concept:** AH provides authentication and integrity but does not encrypt the payload.

113 L2TP VPN

Question Question 1

True / False | Topic: L2TP VPN

L2TP alone does not provide encryption.

- A. True
- B. False

Correct Answer: True**Concept:** L2TP provides tunneling but relies on IPsec or other mechanisms for encryption.

114 SSL VPN

Question Question 1

True / False | Topic: SSL VPN

SSL VPN typically uses TCP port 443.

- A. True
- B. False

Correct Answer: True**Concept:** SSL/TLS VPNs commonly use HTTPS/TCP 443, which is usually allowed through firewalls.